Consciousness Trilogy

Geometry and the finite first-person organization

Complete Collected Volume

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Introduction

These papers ask how geometry and the organization of a reader constrain one another. A Cut has a precise modular meaning within its stated class. Higher-dimensional ball data test geometric consistency against a prepared physical source. A finite self-model faces limits that depend on its permissions, records and drift.

The mathematical structures and the experience question retain different proof obligations. A successful self-calibration task is evidence for functional reflection, not a theorem of subjective experience. The experience bridge is kept as an explicit hypothesis with possible interventions rather than removed from the theory or hidden inside the definition of a Cut.

Reading map

C-I *Geometry Relative to a Cut: Selection, Generation and Modular Inseparability.* A local seed generates a standard Cut under the stated modular dynamics. Finite stress-flow tasks have a paid linear-current realization in the neutral carrier; maintaining that access under selecting noise remains a separate question.

C-II *Higher-Dimensional Consistency of Geometry and Modular Flow: Physical Source Identification and a Second-Order Extension.* Ball data test geometric consistency using a physical source. The second-order extension retains extrinsic terms and the preparation and boundary data that change with geometry.

C-III *Finite Self-Models and the Explanatory Gap: What a Continuing Observer Can Know About Itself.* Self-description has task-specific limits. Finite records, drift and chosen permissions determine risk; a proposed bridge to experience requires its own intervention tests.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.

Geometry Relative to a Cut: Selection, Generation and Modular Inseparability

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Abstract

Can a geometric world and the structure of a perspective be specified independently? Standard half-sided modular inclusions give an exact setting in which to ask this question. The triple of two algebras and a vacuum determines a strongly additive chiral net, and the net determines the triple (Guido–Longo–Wiesbrock). Their geometric and modular data cannot be varied independently, although this symmetric correspondence establishes neither ontological priority nor experience.

We construct the inclusion rather than assuming it for a class of local seeds. In the vacuum representation of Vir_c, let $f(x) = xg(\log x)$, where g is a positive smooth bump, strictly increasing near its left endpoint and strictly decreasing near its right endpoint. The past modular orbit of the bounded observable $O_f = (2/\pi)\arctan T(f)$ generates a proper standard Cut. Its algebra contains a full tail half-line and lies inside a proper half-line of the original net. The proof uses geometric conjugation, smooth closure and positive translates of a localized bracket kernel; no independent forward-translation permission is needed. The full two-sided modular orbit consequently generates the entire Virasoro half-line algebra. For arbitrary nonzero stress seeds the earlier dilation-plus-forward-translation generation theorem remains valid.

In the SU(2) level-one net, two specified random rotations select exactly the $c = 1$ Virasoro subnet. Generation and noise selection therefore meet in one model, but at different levels: the selected subnet is not vacuum-cyclic on the ambient Hilbert space, whereas the Cut is standard in that subnet's own vacuum space. For local unitary excitations we also obtain the exact relative-entropy loss under Cut displacement, $2\pi aE$.

These results remove a separate Cut and a separate translation permission from the seed construction. The carrier, vacuum, modular evolution and ideal spectral and strong closures remain inputs. Finite-word tasks have a physical implementation in the neutral current model: local linear couplings, a retained helper and a paid autonomous clock approximate the required stress flows uniformly on bounded mean-energy inputs. Coherent control intervals and growing precision budgets are required. An exact Cut is not a finite observer, and modular flow is not thereby consciousness.

1 Introduction

Arguments for an explanatory gap between physics and experience [8] share a structure. A physical description is given: a space, a causal structure, matter fields in a state. Then one asks

why anything is experienced from a particular place within it. The gap appears because the description was complete before the question of a perspective arose. We call this the *independence assumption*: the physical structure can be specified without specifying a perspective, and the perspective is then added to, or located in, what was specified.

The assumption is an assumption about *specification*, and it can be examined in a limited way in an exact setting: ask whether the data that would constitute a perspective are an addition to the physical specification or are already fixed by it. An equivalence of descriptions does not settle which comes first, and it does not show that any algebra is a perspective; what it does settle is whether the perspective data can be varied independently of the physical ones. This paper examines that in the one setting where both sides are exact: von Neumann algebras of observables with a distinguished state.

Setting. *Geometry* enters through a conformal net on the circle: a family of algebras $\mathcal{A}(I)$ indexed by intervals, with the translations and dilations that move them. A *perspective* is represented by an accessible algebra A, a subalgebra of a global algebra M with a state ω. The state defines a one-parameter group of automorphisms of M, the modular flow σ_s. For the half-line algebra of a conformal net the modular flow is the dilation (Bisognano–Wichmann [5], Brunetti–Guido–Longo [3]), and in general the modular flow of an algebra in a state is the time that the state itself defines for that algebra (Connes–Rovelli [9]). We use the modular structure of an accessible algebra as the structural proxy for the temporal structure of a perspective. This is a proxy, not an identification; Section 8 says what it does and does not support.

Results.

(i) *Cuts and geometry* (Section 3). A proper half-sided modular inclusion $N \subset M$ forces a one-directional strict nesting of the modular orbit of N and is generated back into M by that orbit (Lemmas 3.1, 3.2); in the factor setting it lives only in type III_1 (Proposition 3.3). For any subalgebra $A \subset M$ whose past modular orbit is cyclic and proper, the Cut and its translations are reconstructed from (M, Ω, A), and the full net when the inclusion is also standard (Proposition 3.4). Geometry and the modular data of a standard Cut determine each other (Corollary 3.5), so they cannot be varied independently; the correspondence is symmetric and does not say which is prior.

(ii) *Selection* (Section 4). If a compact group acts on a net by internal symmetries and the unrecorded part of the environment applies it at random, the noise semigroup has the fixed-point algebra as its exact fixed algebra (Proposition 4.3). For SU(2) level one with $SO(3)$ this is the Virasoro net with $c = 1$ (Rehren), and two fixed rotations about non-parallel axes suffice to select it (Theorem 4.4). Several non-commuting noises select the intersection of their fixed algebras (Theorem 4.5). With the $\mathbb{Z}_2$ noise of the $U(1)$ current net the fixed algebra is strictly larger than the Virasoro net (Proposition 4.8).

(iii) *Generation* (Section 5). For positive endpoint-monotone logarithmic bumps, a single bounded stress seed and its past modular orbit generate a proper standard Cut (Theorem 5.7); the full modular orbit generates the Virasoro half-line algebra (Corollary 5.8). For every nonzero stress seed, without the bump restriction, dilations plus forward translations suffice (Theorem 5.1), while forward translations alone do not (Proposition 5.2). In SU(2) level one this generated algebra equals the noise-selected algebra (Theorem 5.4). The generated Cut is taken in its own vacuum representation, not in the larger ambient representation (Remark 5.5).

(iv) *Response* (Section 6). For local unitary excitations the relative entropy on the algebra of a displaced Cut is an exact expression in the null energy, and the loss under displacement is 2π times the null energy times the displacement (Theorem 6.1). Restriction to a smaller half-line therefore loses information even when the whole excitation remains accessible.

What enters the construction. The carrier algebra, vacuum and modular dynamics are specified. The seed construction then supplies a Cut and its reconstructed translations; it does not derive the carrier or its modular dynamics from a microscopic detector. Continuous parameters, spectral calculus and strong limits are used throughout. Section 7 connects these ideal structures to finite task permissions without identifying the two.

Modular decoherence and an ambient-algebra shell rule do not alone derive a proper half-sided inclusion; Appendix D records their boundaries. The constructive route here uses local stress fields and geometric covariance. The original question remains: which physical history produces a usable boundary, and what does that boundary contribute to a perspective?

2 Preliminaries

2.1 Modular theory and half-sided modular inclusions

Let M be a von Neumann algebra on a Hilbert space H and $\Omega \in H$ a vector that is cyclic and separating for M. Write Δ_M, J_M for the modular operator and conjugation of (M, Ω) and $\sigma_s^M(x) = \Delta_M^{is} x \Delta_M^{-is}$. Then $\sigma_s^M(M) = M$ and $\Delta_M^{is}\Omega = \Omega$.

Definition 2.1 (Half-sided modular inclusion)**.** Let $N \subset M$ be von Neumann algebras for which Ω is cyclic and separating. The inclusion is *+hsm* if $\sigma_s^M(N) \subset N$ for all $s \leq 0$ (equivalently $\Delta_M^{-it} N \Delta_M^{it} \subset N$ for $t \geq 0$). It is *standard* if Ω is also cyclic for $M \cap N'$. It is *proper* if $N \neq M$. We call a proper +hsm inclusion a *Cut*.

Theorem 2.2 (Wiesbrock; Araki–Zsidó [2, Thm. 2.1])**.** *Let $N \subset M$ be +hsm. Then $P := \overline{(\log \Delta_N - \log \Delta_M)/2\pi}$ is a positive self-adjoint operator, and with $U(s) = e^{isP}$:*

(1) $\Delta_M^{-it} U(s) \Delta_M^{it} = U(e^{2\pi t} s) = \Delta_N^{-it} U(s) \Delta_N^{it}$;

(2) $J_M U(s) J_M = U(-s) = J_N U(s) J_N$;

(3) $\Delta_N^{it} = U(1) \Delta_M^{it} U(1)^*$ *and* $J_N = U(1) J_M U(1)^*$;

(4) $N = U(1) M U(1)^*$;

(5) $U(s) M U(s)^* \subset M$ *for* $s \geq 0$, *and* $\mathrm{Ad}\, U(s)$, $s \geq 0$, *is a strongly continuous semigroup of endomorphisms of* M;

(6) $U(s)\Omega = \Omega$, *so the vacuum state is invariant under* $\mathrm{Ad}\, U(s)$.

Theorem 2.3 ([2, Thm. 2.2(4)], [14, Cor. 1.8])**.** *If $N \subset M$ is +hsm and p is the largest central projection of M with $Mp = Np$, then $M(1-p)$ and $N(1-p)$ are of type III$_1$ whenever $p \neq 1$. In particular, if M is a factor and $N \neq M$, both are type III$_1$ factors.*

Theorem 2.4 (Guido–Longo–Wiesbrock [14, Cor. 1.7–1.9])**.** *There is a one-to-one correspondence between isomorphism classes of* standard *+hsm inclusions $(N \subset M, \Omega)$ and isomorphism classes of strongly additive local conformal nets on S^1; under it $M = \mathcal{A}(0,\infty)$, $N = \mathcal{A}(1,\infty)$, Ω is the vacuum, U is the translation group, and Δ_M^{it} acts as dilation. For a conformal net that is not strongly additive, the hsm reconstruction returns the dual net $\mathcal{A}^d(a,b) = \mathcal{A}(a,\infty) \cap \mathcal{A}(b,\infty)'$, which equals $\mathcal{A}$ if and only if $\mathcal{A}$ is strongly additive.*

2.2 Virasoro and current nets

A conformal net $\mathcal{A}$ is *diffeomorphism covariant* with central charge $c > 0$ if the vacuum representation carries a positive-energy projective representation V of $\mathrm{Diff}_+(S^1)$, with $V(\varphi) \in \mathcal{A}(I)$ when φ is supported in I, and stress tensor T with $V(\mathrm{Exp}\, f) = e^{iT(f)}$ up to a phase (Goodman–Wallach [13], Carpi–Weiner [10]). Write $\mathrm{Vir}_c(I) = \{T(f) : \mathrm{supp} f \subset I\}''$. For real $f \in C_c^\infty$, $T(f)$ is essentially self-adjoint on finite-energy vectors, and if $f_n \to f$ in $\|f\|_{3/2} = \sum_n |\hat{f}_n|(1+|n|^{3/2})$ with supports in a fixed compact set, then $T(f_n) \to T(f)$ in the strong resolvent sense [10, Thm. 4.4, Prop. 4.5]. For the half-line $M = \mathrm{Vir}_c(0,\infty)$ the modular group of the vacuum is the dilation group, $\sigma_s = \mathrm{Ad}\, V(\delta_{e^{-2\pi s}})$ with $\delta_\lambda(x) = \lambda x$ (Bisognano–Wichmann [5]; Brunetti–Guido–Longo [3]), and $T_a = \mathrm{Ad}\, U(a)$ with $U(a) = V(\tau_a)$ implements translations. The transformation law is $V(\varphi)T(x)V(\varphi)^* = \varphi'(x)^2 T(\varphi(x)) - \frac{c}{24\pi}\mathrm{Sch}(\varphi)(x)$. Virasoro nets with $c \le 1$ are strongly additive and for $c > 1$ they are not [7, §2.4]; the fixed-point subnet of a strongly additive net under a compact group is strongly additive [25].

2.3 Relative entropy of coherent states

Let $f \in C_c^\infty((a_0,\infty),\mathbb{R})$ with $a_0 > 0$, $\rho = \mathrm{Exp}\, f$, $U = V(\rho) = e^{iT(f)}$, $\omega_+(A) = \omega(U^*AU)$ and $\omega_-(A) = \omega(UAU^*)$. Put $w = \rho''/\rho'$. Hollands [15, Thm. 1] proves, for the half-line algebra $\mathrm{Vir}_c(0,\infty)$,

$$S(\omega\|\omega_+) = \frac{c}{24}\int_0^\infty x\,(\partial_x \log \rho'(x))^2\, dx. \tag{1}$$

3 Cuts and geometry

Let $(N \subset M, \Omega)$ be $+$hsm with translation group U from Theorem 2.2. Put $N_a = U(a)MU(a)^*$ for $a \ge 0$, so $N_0 = M$, $N_1 = N$.

Lemma 3.1 (The modular orbit of a Cut generates M)**.** $\bigvee_{t\in\mathbb{R}} \sigma_t^M(N) = M$.

Proof. By Theorem 2.2(1),(4) and $\sigma_t(M) = M$, $\sigma_t(N) = \Delta_M^{it}U(1)MU(1)^*\Delta_M^{-it} = U(e^{-2\pi t})MU(e^{-2\pi t})^* = N_{e^{-2\pi t}}$. For $m \in M$ put $m_a = U(a)mU(a)^* \in N_a$. As $t \to \infty$, $a = e^{-2\pi t} \to 0$ and $m_a \to m$ strongly, since U is strongly continuous. Each m_a lies in $\sigma_t(N)$ for $a = e^{-2\pi t}$, and the algebra generated by the $\sigma_t(N)$ is strongly closed, so it contains m. The other inclusion is trivial. □

Lemma 3.2 (One-directional strict nesting)**.** *If $N \subsetneq M$ is $+$hsm, then $\sigma_t(N) \subsetneq N$ for $t < 0$ and $\sigma_t(N) \not\subset N$ for $t > 0$.*

Proof. For $a \ge b \ge 0$, $N_a = U(b)[U(a-b)MU(a-b)^*]U(b)^* \subset U(b)MU(b)^* = N_b$ by Theorem 2.2(5), so $a \mapsto N_a$ is decreasing. Suppose $N_a = N_b$ for some $a > b \ge 0$. Conjugating by $U(b)^*$ gives $N_d = M$ with $d = a - b > 0$. Choose t with $e^{-2\pi t}d = 1$: then $N = N_1 = \sigma_t(N_d) = \sigma_t(M) = M$, a contradiction. Hence $a \mapsto N_a$ is strictly decreasing. Since $\sigma_t(N) = N_{e^{-2\pi t}}$, the claims follow. □

The nesting is a property of the subalgebras. It is not entropy production or an arrow of time: the modular group itself is invertible and acts by automorphisms. Lemma 3.2 says that a proper Cut cannot be stable in both directions.

Proposition 3.3 (A Cut lives in type III_1)**.** *If M is a factor and $N \subsetneq M$ is $+$hsm, then M and N are type III_1 factors. No proper $+$hsm inclusion exists in a type I factor.*

This is Theorem 2.3. It is a statement about the type of the algebras, not about observers: a finite-dimensional system has no exact Cut, but an observer of finite capacity can still couple to a system described by a type III_1 algebra. Finite instruments and exact Cuts therefore require different descriptions; Section 7 gives their task-level interface.

Proposition 3.4 (Accessible algebras and geometry)**.** *Let $A \subset M$ be a subalgebra and $N_A^- = \big(\bigcup_{s\le 0}\sigma_s(A)\big)'' \subset M$. Assume that Ω is cyclic for N_A^- and $N_A^- \neq M$. Then:*

(a) *$(N_A^- \subset M, \Omega)$ is a Cut, with translation group U_A and $N_A^- = U_A(1) M U_A(1)^*$;*

(b) *$\Delta^{it}_{N_A^-} = U_A(1)\Delta_M^{it}U_A(1)^*$, and U_A is recovered from the two modular operators as $P = \dfrac{\cdot}{(\log\Delta_{N_A^-} - \log\Delta_M)/2\pi}$;*

(c) *if the inclusion is standard, there is a unique strongly additive local conformal net $\mathcal{A}_A$ with $\mathcal{A}_A(0,\infty) = M$, $\mathcal{A}_A(1,\infty) = N_A^-$, translations U_A and dilations σ;*

(d) *$\bigvee_t \sigma_t(A) = M$;*

(e) *if $A \subset A' \subset M$ and both satisfy the hypotheses, then $N_A^- \subset N_{A'}^-$.*

Proof. $\sigma_s(N_A^-) \subset N_A^-$ for $s \le 0$ by construction, and Ω is separating for $N_A^- \subset M$, so (a) is Theorem 2.2(4), and (b) is (1),(3). (c) is Theorem 2.4. (d): $\bigvee_t \sigma_t(A) \supset \bigvee_t \sigma_t(N_A^-) = M$ by Lemma 3.1, and the left side is contained in M. (e) is monotonicity of the generated algebra. □

Corollary 3.5 (Geometry and modular data determine each other)**.** *Within the class of standard Cuts, the geometry (the isomorphism class of the strongly additive net) and the modular data (the isomorphism class of the triple $(N \subset M, \Omega)$) determine each other.*

This is Theorem 2.4. The input is the complete standard triple, not a bare flow: Δ_M and U together with M and Ω fix N and Δ_N, and conversely the two modular operators fix U. The triple replaces the list "net, vacuum, translations, dilations"; it does not explain where M, Ω or the Cut come from.

Remark 3.6 (Cuts on one orbit are isomorphic). The Cuts N_a and N_b of one net, $a, b > 0$, are exchanged by σ_t and give isomorphic triples. Distinguishing geometries therefore needs an invariant of the triple (for instance the spectrum of the conformal Hamiltonian of the reconstructed net), not merely a different subalgebra. Foundation I supplies one in this net: the vacuum character of the hull determines the stabilizer of the seed up to conjugacy within that model, so seeds with non-conjugate stabilizers give different vacuum characters and hence non-isomorphic triples [20]. That invariant classifies Foundation I's ambient-translation Cuts $U(1)\,\mathrm{Hull}(S)U(1)^* \subset \mathrm{Hull}(S)$. It does not classify the past-orbit inclusion $N_{O_f}^- \subset M$ by identifying its reconstructed U_A with the ambient U. Conjecture 5.10(b) concerns precisely whether that orbit is an ordinary translated half-line.

Remark 3.7 (Relative position of regions). For nested intervals $(x_2, x_3) \subset (x_1, x_4)$ the cross-ratio $b = \frac{(x_2-x_1)(x_4-x_3)}{(x_3-x_1)(x_4-x_2)} \in (0,1)$ is the unique modulus under $PSL(2,\mathbb{R})$; in the normal form the outer interval is $(0,\infty)$ and the inner one is $(b,1)$. Any statement about the relative position of nested regions in the reconstructed net is a statement about b.

4 Selection of the accessible algebra by noise

4.1 Setting

Assumption 4.1. $\mathcal{A}$ is an irreducible, local, Möbius covariant net on S^1 with vacuum (H, Ω). A compact group G acts by automorphisms α_g, preserving each $\mathcal{A}(I)$, implemented by a strongly

continuous unitary representation $u(g)$ with $u(g)\Omega = \Omega$ that commutes with the Möbius (and, when present, the diffeomorphism) representation. We write $\omega = \langle \Omega, \cdot\, \Omega\rangle$ and $M = \mathcal{A}(0,\infty)$.

For a closed subgroup $H \subset G$ put $Q_H(x) = \int_H \alpha_h(x)\, dh$ with normalised Haar measure.

Lemma 4.2. *Q_H is a normal, faithful, ω-preserving conditional expectation of M onto the fixed algebra M^H; it is M^H-bimodular and commutes with the modular group and with translations.*

Proof. For a vector functional $\rho = \langle \xi, \cdot\, \eta\rangle$ the map $h \mapsto \rho\circ\alpha_h$ is norm continuous in the predual, by strong continuity of u; every normal functional is a norm limit of finite sums of vector functionals, so the same holds for all normal ρ. The Bochner integral $\int_H \rho\circ\alpha_h\, dh$ is therefore a normal functional, and it equals $\rho\circ Q_H$; hence Q_H is normal. Haar invariance gives $Q_H^2 = Q_H$ and $\mathrm{Ran}\, Q_H = M^H$; bimodularity is $\alpha_h(n_1 x n_2) = n_1\alpha_h(x)n_2$ for $n_i \in M^H$. Since $\omega\circ\alpha_h = \omega$, $\omega\circ Q_H = \omega$, and if $Q_H(x^*x) = 0$ then $\omega(x^*x) = 0$, so $x = 0$ (faithfulness). Commutation with σ and translations follows because $u(g)$ commutes with their unitary groups. □

4.2 Noise semigroups

Fix $\kappa > 0$ and put $D_t^H = e^{-\kappa t}\,\mathrm{id} + (1 - e^{-\kappa t})Q_H$. This is the average channel of a Poisson process of rate κ in which each jump applies α_h with h Haar-distributed in H: the probability of no jump in time t is $e^{-\kappa t}$, and a product of at least one Haar element is again Haar.

Proposition 4.3. *D_t^H is a norm-continuous semigroup of normal, unital, completely positive, ω-preserving maps on M, and*

(1) $\mathrm{Fix}(D^H) = M^H$*; $D_t^H - Q_H = e^{-\kappa t}(\mathrm{id} - Q_H)$, so $\|D_t^H - Q_H\| \le 2e^{-\kappa t}$;*

(2) *$H_1 \subset H_2 \Rightarrow M^{H_2} \subset M^{H_1}$; κ changes only the rate of approach;*

(3) *if $M^H \neq M$ and $0 \neq y = y^* \in \ker Q_H$, then for $0 < t < \infty$ and $q = e^{-\kappa t}$ we have $D_t(y^2) - D_t(y)^2 = (1-q)Q_H(y^2) + (q - q^2)y^2 \ge 0$ and $\neq 0$, so D_t is a non-multiplicative channel although it preserves ω.*

Proof. The semigroup law and complete positivity are those of a convex combination of id and Q_H with $Q_H^2 = Q_H$. (1): if $D_t(x) = x$ for some $t > 0$ then $(1 - e^{-\kappa t})(x - Q_H x) = 0$. (2) is clear. (3): choose $x = x^* \notin M^H$ and $y = x - Q_H x$; then $Q_H y = 0$, $D_t(y) = qy$, and $D_t(y^2) = qy^2 + (1-q)Q_H(y^2)$. Since Q_H is faithful, $Q_H(y^2) \neq 0$. □

Theorem 4.4 (Two fixed rotations suffice)**.** *Let $G = SO(3)$ act as in Assumption 4.1. Let $g_1, \dots, g_m \in SO(3)$ generate a subgroup whose closure is $SO(3)$, let $p_j > 0$, $\sum_j p_j = 1$, and put $R = \sum_j p_j \alpha_{g_j}$, $\tilde{D}_t = \exp[\kappa t(R - \mathrm{id})]$. Then $\tilde{D}$ is a norm-continuous semigroup of normal, unital, completely positive, ω-preserving maps with $\mathrm{Fix}(\tilde{D}) = M^{SO(3)}$. The hypothesis holds, in particular, (i) for $m = 2$ when g_1, g_2 are rotations about two non-parallel axes by angles whose ratios to 2π are irrational (any $p_1 \in (0,1)$), and (ii) for the symmetric four-operation choice $R = \frac{1}{4}(\alpha_{g_1} + \alpha_{g_1^{-1}} + \alpha_{g_2} + \alpha_{g_2^{-1}})$ built from such a pair.*

Proof. The generator is bounded, so $\mathrm{Fix}(\tilde{D}) = \mathrm{Fix}(R)$. Write α_j for the automorphisms and u_j for their implementers, which fix Ω. Then $\|(\alpha_j(x) - x)\Omega\|^2 = 2\|x\Omega\|^2 - 2\mathrm{Re}\langle \alpha_j(x)\Omega, x\Omega\rangle$, so $\sum_j p_j\|(\alpha_j(x) - x)\Omega\|^2 = 2\|x\Omega\|^2 - 2\mathrm{Re}\langle R(x)\Omega, x\Omega\rangle$. If $R(x) = x$, the left side vanishes, so $\alpha_j(x)\Omega = x\Omega$ for each j, and $\alpha_j(x) = x$ because Ω is separating. The set $\{g : \alpha_g(x) = x\} = \{g : u(g)x\Omega = x\Omega\}$ is a closed subgroup (strong continuity of u) that contains every g_j, hence the closure of the group they generate. For (i): each g_j generates a dense subgroup of the circle

T_j of rotations about its axis n_j, so the closure H of $\langle g_1, g_2\rangle$ is a closed, hence Lie, subgroup containing T_1 and T_2. Its Lie algebra contains $n_1, n_2 \in \mathfrak{so}(3) \cong \mathbb{R}^3$ (bracket = cross product), hence $n_1 \times n_2 \neq 0$; the three vectors span $\mathbb{R}^3$, so H contains the identity component of $SO(3)$, which is $SO(3)$. For (ii) the generated group is the same. Then x is fixed by all of $SO(3)$; the converse inclusion is clear. □

Two is not claimed to be the minimal number of operations in every mechanism.

Theorem 4.5 (Non-commuting noises select the intersection)**.** *Let $H_1, \dots, H_m \subset G$ be closed subgroups, $\kappa_i > 0$, $L = \sum_i \kappa_i (Q_{H_i} - \mathrm{id})$, $E_t = e^{tL}$. The Q_{H_i} need not commute. Then E_t is a norm-continuous semigroup of normal, unital, completely positive, ω-preserving maps, with*

$$\mathrm{Fix}(E) = \bigcap_i M^{H_i} = M^{\overline{\langle H_1,\dots,H_m\rangle}}, \qquad -\langle x\Omega, Lx\Omega\rangle = \sum_i \kappa_i \|(x - Q_{H_i}x)\Omega\|^2,$$

and E_t converges strongly on $L^2(M,\omega)$ to the Haar average over the closed group generated.

Proof. With $\lambda = \sum \kappa_i$ and $\bar{R} = \sum \kappa_i Q_{H_i}/\lambda$, $E_t = e^{-\lambda t}\sum_n (\lambda t)^n \bar{R}^n/n!$ is a Poisson mixture of powers of a state-preserving unital completely positive map. On the GNS space each Q_{H_i} is the orthogonal projection $P_i = \int_{H_i} u(h)\, dh$; hence L is bounded, self-adjoint and nonpositive there, and the displayed identity holds. $Lx = 0$ forces $Q_{H_i}x = x$ for every i, because Ω is separating. The spectral theorem gives strong convergence to the projection onto $\ker L = \bigcap_i \mathrm{Ran} P_i$, which is the Haar average over the closed group generated. □

No uniform spectral gap or operator-norm rate is claimed in infinite dimension.

4.3 SU(2) level one and the U(1) contrast

Let $\mathcal{A}_{SU(2)_1}$ be the vacuum net of the loop group $LSU(2)$ at level one; $SO(3)$ acts properly by the adjoint action, and Vir_1 denotes the Virasoro subnet of the Sugawara stress tensor, $c = 1$.

Theorem 4.6 (Rehren [17])**.** $\mathcal{A}_{SU(2)_1}(I)^{SO(3)} = \mathrm{Vir}_1(I)$ *for every interval I.*

The proof in [17] decomposes the vacuum representation by isospin: its character is $\sum_j \chi_j(z)\, t^{j^2}(1 - t^{2j+1})\prod_n (1 - t^n)^{-1}$, the $SO(3)$-invariant part is the $j = 0$ term, which is the character of the irreducible Virasoro vacuum representation at $c = 1$ (Segal [18], Kac–Raina [16]), so the invariant vacuum space equals the Virasoro vacuum space. Haag duality of the Virasoro net [6] and locality then give the chain $\mathrm{Vir}(I) \subset \mathcal{A}(I)^G \subset (\mathcal{A}(I^c)^G)' \subset \mathrm{Vir}(I^c)' = \mathrm{Vir}(I)$ on that space, and local faithfulness of the representation lifts the equality to $\mathcal{A}$. We recomputed the Virasoro vacuum module in the charge-zero Fock space with exact rational arithmetic (Appendix B): its dimensions at levels $0, \dots, 9$ are $1, 0, 1, 1, 2, 2, 4, 4, 7, 8 = p(n) - p(n-1)$, as the character requires. Since Vir_1 is strongly additive, the reconstruction of Section 3 returns Vir_1 itself and not a dual net.

Corollary 4.7. *In $\mathcal{A}_{SU(2)_1}$ the exact fixed algebra of the Haar-jump noise $D^{SO(3)}$, of the semigroup $\tilde{D}$ built from finitely many fixed rotations that generate $SO(3)$ (two suffice), and of the pair of non-commuting axis noises H_z, H_x is $\mathrm{Vir}_1(0,\infty)$, a proper type III_1 subfactor of M.*

Properness: α is non-trivial on $\mathcal{A}$ and trivial on Vir_1.

Proposition 4.8 (The $\mathbb{Z}_2$ noise of the $U(1)$ current does not select the Virasoro net)**.** *Let $\mathcal{A}_{U(1)}$ be the net of the $U(1)$ current J, $\gamma : J \mapsto -J$, $F = \frac{1}{2}(\mathrm{id} + \gamma)$. Then $\mathrm{Vir}_1(I) \subsetneq \mathcal{A}_{U(1)}(I)^{\mathbb{Z}_2} \subsetneq \mathcal{A}_{U(1)}(I)$.*

Proof. At level 4 the even Fock space is spanned by $J_{-1}^4\Omega, J_{-3}J_{-1}\Omega, J_{-2}^2\Omega$, dimension 3, whereas the Virasoro vacuum module at level 4 has dimension 2, spanned by $L_{-4}\Omega = (J_{-3}J_{-1} + \frac{1}{2}J_{-2}^2)\Omega$ and $L_{-2}^2\Omega = (\frac{1}{4}J_{-1}^4 + J_{-3}J_{-1})\Omega$, with Gram matrix $\left(\begin{smallmatrix} 5 & 3 \\ 3 & 9/2 \end{smallmatrix}\right)$. The vector $u = \frac{1}{2}J_{-1}^4\Omega - J_{-3}J_{-1}\Omega + \frac{3}{4}J_{-2}^2\Omega$ is even, orthogonal to both, satisfies $L_m u = 0$ for $m = 1, \ldots, 4$, and has $\|u\|^2 = 27/2$ (Appendix B). If $\mathrm{Vir}_1(I) = \mathcal{A}(I)^{\mathbb{Z}_2}$, then by Reeh–Schlieder their vacuum-cyclic subspaces coincide: $F(\mathcal{A}(I))\Omega$ is dense in the even subspace and $\mathrm{Vir}_1(I)\Omega$ is dense in the Virasoro vacuum space; but u lies in the former and not the latter. The second inclusion is strict because γ is non-trivial. □

So the $\mathbb{Z}_2$ noise of the $U(1)$ current protects the stress tensor (and the whole even algebra) but does not select Vir_1; a noise that selects Vir_1 in that net must be specified differently, for instance by the conditional expectation onto Vir_1 itself. The $SU(2)$ construction has the advantage that the noise is specified by internal rotations, not by the target algebra.

Remark 4.9 (Fixed algebra, attracting state, correctable algebra). The three notions are of different types. The semigroups above preserve the vacuum state and act as the identity on the fixed algebra, so information in the fixed algebra is passively preserved and can be recovered by the identity map. A reset channel $E_t(A) = e^{-\gamma t}A + (1 - e^{-\gamma t})\mathrm{tr}(\rho_* A)\mathbf{1}$ has an attracting state ρ_* but trivial fixed algebra, and actively protected information is described by correctable algebras (Bény–Kempf–Kribs [4]). We do not identify a fixed algebra with an attracting state, and we do not claim that the full correctable structure of these semigroups equals their fixed algebra; exact fixed algebras are, moreover, discontinuous in the dynamics (Section 7).

5 Access: one observable generates the selected algebra

Let $\mathcal{A}$ be diffeomorphism covariant with $c > 0$, $M = \mathcal{A}(0, \infty)$, $B = \mathrm{Vir}_c(0, \infty)$. For $f \in C_c^\infty((0, \infty), \mathbb{R})$, $f \neq 0$, the *seed* is $O_f = \frac{2}{\pi}\arctan T(f) \in B$, a bounded self-adjoint observable. An *admissible* algebra is a unital von Neumann algebra $N \subset M$ with $\sigma_s(N) = N$ for all $s \in \mathbb{R}$ and $T_a(N) \subset N$ for $a \geq 0$. The *hull* $\mathrm{Hull}_{\sigma,T}(O_f)$ is the intersection of all admissible N containing O_f.

Theorem 5.1 (Generation)**.** $\mathrm{Hull}_{\sigma,T}(O_f) = \mathrm{Vir}_c(0, \infty)$ *for every nonzero* $f \in C_c^\infty((0, \infty), \mathbb{R})$.

Appendix A proves this for arbitrary nonzero f. The stronger modular-only result below applies to a specified bump class, not to all f.

Proposition 5.2 (Forward translations alone do not suffice)**.** *If* $\mathrm{supp} f \subset (a, b)$ *with* $a > 0$*, then the smallest algebra containing* O_f *and stable under* T_s*,* $s \geq 0$*, lies in* $\mathrm{Vir}_c(a, \infty) \subsetneq B$.

Proof. All forward translates of O_f lie in $\mathrm{Vir}_c(a, \infty)$, which is stable under forward translations. If $\mathrm{Vir}_c(a, \infty) = B$ then $\mathrm{Vir}_c(0, a) \subset B \cap B'$ by locality, which is trivial as B is a factor; but $\mathrm{Vir}_c(0, a)$ is not trivial. □

For the logarithmic bump class of Theorem 5.7, dilations alone suffice by Corollary 5.8. The unrestricted stress-seed question remains open. Foundation I gives a separate modular-only generation theorem for nonzero smeared current seeds [20].

A seed here denotes an ideal observable, not a grant of every function of it as an actual measurement or gate. An injective bounded function has the same spectral algebra as its generator; its implementation has a separate price. Foundation I distinguishes an effect, its spectral hull and the full physical instrument. HAFF-E supplies a concrete contrast: a partially polarized

memory produces an unsharp effect generating the same pointer algebra at every nonzero polarization, while its one-read error varies continuously [23]. The same distinction applies to O_f and to the ideal flows used below.

Example 5.3 (Exact fixedness does not freeze the modular flow)**.** In $M_2\bar{\otimes}M_2$ with state $\mathrm{diag}(.8,.2)\otimes\frac{1}{2}I$, let the noise be id$\otimes$ depolarisation and the observable $X\otimes 1$. The observable is fixed by the noise, but $\sigma_t(X\otimes 1) = (\cos(t\ln 4)X - \sin(t\ln 4)Y)\otimes 1$ generates the non-commutative algebra $M_2\otimes 1$. Exact invariance under the noise does not by itself prevent the modular flow from moving a fixed observable through a non-commutative algebra.

Theorem 5.4 (Access closure equals noise-selected algebra)**.** *In $\mathcal{A}_{SU(2)_1}$, let O_f be built from the Sugawara stress tensor. Then*

$$\mathrm{Hull}_{\sigma,T}(O_f) = \mathrm{Vir}_1(0,\infty) = M^{SO(3)} = \mathrm{Fix}(D^{SO(3)}) = \mathrm{Fix}(\tilde{D}).$$

Proof. The first equality is Theorem 5.1 applied inside M: the argument uses only unitaries $e^{iuT(h)} \in B \subset M$. The second is Theorem 4.6 (the half-line is a Möbius image of an interval), and the last two are Corollary 4.7. □

Two different structural definitions meet here: the algebra generated from one observable by dilations and forward translations, and the algebra left invariant by an explicit random-rotation noise. The equality is specific to this net and group; it fails for the $\mathbb{Z}_2$ noise of the $U(1)$ current (Proposition 4.8).

Remark 5.5 (Two different inclusions). The algebra selected by noise is $B = M^{SO(3)} = \mathrm{Vir}_1(0,\infty) \subset M = \mathcal{A}_{SU(2)_1}(0,\infty)$. By Lemma 4.2 there is an ω-preserving faithful normal conditional expectation of M onto B, so by Takesaki's theorem [24] $\sigma_s^M(B) = B$ for all $s\in\mathbb{R}$ and $\sigma^M|_B$ is the modular group of B in its own vacuum representation, so stability of B under σ in one direction holds trivially. What fails is cyclicity. For $x\in M$, $\int u(g)x\Omega\,dg = Q_{SO(3)}(x)\Omega$ because $u(g)\Omega=\Omega$; so the projection onto the $SO(3)$-invariant subspace $H^{SO(3)}$ maps $M\Omega$ onto $B\Omega$, and since $M\Omega$ is dense, $\overline{B\Omega} = H^{SO(3)}$, a proper subspace because the vacuum representation of $\mathcal{A}_{SU(2)_1}$ contains non-trivial isospin sectors. Ω is therefore not cyclic for B on the ambient Hilbert space, and $B\subset M$ is not a Cut in the sense of Definition 2.1 (which asks Ω to be cyclic and separating for the subalgebra; a proper subalgebra that is σ-invariant and has Ω cyclic would contradict Lemma 3.2). The Cuts of the geometric construction are the half-line subalgebras $N_a = \mathrm{Vir}_1(a,\infty)\subset B$, taken in the vacuum-cyclic representation of B (a factor, so the restriction to that subspace is faithful). The two structures meet in one technical model: noise fixes B, and B carries the net whose Cuts are the N_a. The noise selects B without producing its vacuum or modular dynamics. Theorem 5.7, applied in $H^{SO(3)}$, then constructs a standard Cut from a local seed and that modular dynamics, and reconstructs its translations. These need not be assumed as a separate generation permission. The underlying Virasoro carrier and state are still inherited from the ambient model.

5.1 A Cut generated by one past modular orbit

Here $M = \mathrm{Vir}_c(0,\infty)$ acts on its own vacuum-cyclic space. Write

$$N_{O_f}^- = W^*(\sigma_s(O_f) : s\le 0), \qquad H_\sigma(O_f) = W^*(\sigma_s(O_f) : s\in\mathbb{R}).$$

The first algebra is automatically half-sided invariant; it still needs cyclicity and properness before it is a Cut.

Lemma 5.6 (Smooth flow closure)**.** *For a unital von Neumann algebra N, the real compactly supported vector fields whose stress flows $e^{itT(v)}$ belong to N for every $t \in \mathbb{R}$ are closed under real linear combinations, under C^∞ limits with supports in a fixed compact set, and under the ordinary vector-field bracket $[v, w] = vw' - v'w$.*

Proof. These statements can be checked in any smooth coordinate chart covering the compact supports. Finite-energy vectors form a common core on which $T(v) + T(w) = T(v + w)$; essential self-adjointness and the Trotter formula give linear closure. Fixed-compact C^∞ convergence implies convergence of the circle fields in the Fourier norm $\sum_n |\hat{v}_n|(1 + |n|^{3/2})$. Carpi–Weiner's strong-resolvent result then gives strong convergence of their exponential unitaries [10, Thm. 4.4, Prop. 4.5].

For the bracket, put $\varphi_t = \mathrm{Exp}(tv)$. Up to scalar phases, $V(\varphi_t) = e^{itT(v)} \in N$. Geometric covariance gives

$$V(\varphi_t)e^{iuT(w)}V(\varphi_t)^* = e^{iu\lambda(t,w)}e^{iuT((\varphi_t)_* w)}, \qquad \lambda(t, w) \in \mathbb{R}.$$

The Schwarzian term contributes only the displayed scalar; projective phases do not change membership in a unital algebra. Thus $(\varphi_t)_* w$ has all its flows in N. Linear closure applies to $(w - (\varphi_t)_* w)/t$, and the smooth pushforward formula gives

$$\frac{d}{dt}(\varphi_t)_* w\Big|_{t=0} = v'w - vw' = -[v, w].$$

For sufficiently small $|t|$ all supports stay in one fixed compact set. The difference quotient converges in every C^k norm, so the preceding strong-resolvent closure proves the bracket claim. No commutator of unbounded operators on an unspecified domain is used. □

Theorem 5.7 (A single-orbit standard Cut)**.** *Choose $0 < a < b$, $\alpha = \log a$, $\beta = \log b$, and a real $g \in C_c^\infty(\mathbb{R})$ with support $[\alpha, \beta]$. Suppose $g > 0$ on (α, β), and g is strictly increasing and strictly decreasing on some interior neighborhoods of its left and right endpoints, respectively. Put $f(x) = xg(\log x)$ and $O_f = (2/\pi) \arctan T(f)$. Then*

$$\mathrm{Vir}_c(b, \infty) \subset N_{O_f}^- \subset \mathrm{Vir}_c(a, \infty) \subsetneq M. \tag{2}$$

The triple $(N_{O_f}^- \subset M, \Omega)$ is a proper standard +hsm inclusion. Its translations and strongly additive net are recovered from its modular operators.

Proof. In $u = \log x$, the vector field $f(x)\partial_x$ is $g(u)\partial_u$. A dilation by λ pushes g to $g(u - \log \lambda)$. Past modular times therefore give exactly $g_c(u) = g(u - c)$ with $c = -2\pi s \geq 0$. Injectivity of arctan gives all stress flows of g_c in $N = N_{O_f}^-$, as in Appendix A, Step 1. All these fields have support to the right of α, proving the upper inclusion in (2). For $r \leq 0$, $\sigma_r(N) \subset N$; equivalently, already obtained fields can be shifted further right by any $c \geq 0$.

Let $d = \beta - \alpha$ and choose $\varepsilon > 0$ smaller than both endpoint neighborhoods. Lemma 5.6 supplies

$$k_\varepsilon = [g_0, g_{d-\varepsilon}], \qquad \mathrm{supp}\, k_\varepsilon \subset [\beta - \varepsilon, \beta].$$

On the overlap, the first bump is nonincreasing and the second is nondecreasing, both positive. Hence $k_\varepsilon \geq 0$. Strict monotonicity need not mean a pointwise strict derivative. It does, however, give a positive integral: the nonconstant increasing shifted bump has a subinterval where its derivative is positive, and the other bump is positive there. Thus

$$m_\varepsilon = \int k_\varepsilon(u)\, du > 0, \qquad K_\varepsilon = k_\varepsilon / m_\varepsilon$$

is a unit-mass positive kernel whose stress flows and all right translates' flows belong to N.

For $h \in C_c^\infty((\beta,\infty),\mathbb{R})$, set

$$h_\varepsilon(u) = \int_{c\geq 0} h(c+\beta)K_\varepsilon(u-c)\,dc = \int K_\varepsilon(w)h(u-w+\beta)\,dw. \tag{3}$$

The second equality extends the integral without adding negative shifts: the support of $h(c+\beta)$ already has $c > 0$. If its distance from β is $\eta > 0$, then for $\varepsilon < \eta/2$ the supports of h_ε lie in a fixed compact subset of (β,∞). Since $w \in [\beta-\varepsilon,\beta]$ and the mass is one, for every $j \geq 0$,

$$\|h_\varepsilon^{(j)} - h^{(j)}\|_\infty \leq \varepsilon \|h^{(j+1)}\|_\infty.$$

The integral is a C^∞ limit of finite real Riemann sums of permitted right-translated kernels, all with fixed compact support. Lemma 5.6 first gives the flows of h_ε, then those of h, in N. Returning through $x = e^u$, these are all compact stress fields on (b,∞). They generate $\mathrm{Vir}_c(b,\infty)$, giving the lower inclusion.

Reeh–Schlieder makes Ω cyclic for the lower half-line algebra, hence for N; it is separating for $N \subset M$. Locality gives $\mathrm{Vir}_c(0,a) \subset M \cap N'$, so the relative commutant is also vacuum-cyclic. The upper half-line is proper: otherwise the nontrivial $\mathrm{Vir}_c(0,a)$ would lie in the center of the factor M. The past-orbit definition supplies half-sided invariance. Theorems 2.2 and 2.4 reconstruct translations and the net. □

Corollary 5.8 (Modular flow alone generates the full half-line)**.** *For the seeds in Theorem 5.7,*

$$H_\sigma(O_f) = \mathrm{Vir}_c(0,\infty).$$

Proof. The two-sided hull contains $N_{O_f}^-$, hence $\mathrm{Vir}_c(b,\infty)$, and is invariant under every dilation. It contains $\mathrm{Vir}_c(\lambda b,\infty)$ for every $\lambda > 0$. Every compact stress field on $(0,\infty)$ lies in one of these half-lines, so their generated algebra is M. □

For example, $g(u) = \exp[-1/((u-\alpha)(\beta-u))]$ inside its support and zero outside satisfies the hypotheses. Positivity can be reversed globally without changing the spectral algebra. The theorem is not an all-seed classification: sign-changing and more irregular endpoint behavior are not covered by the positive-kernel argument.

The lower inclusion is enough to prove a Cut, but does not identify $N_{O_f}^-$ with, or distinguish it from, an ordinary translated half-line. For $c > 1$, the standard reconstruction is strongly additive even though the original Virasoro net is not; one must not infer that the reconstructed full net equals the original net from the shared half-line algebra alone. For $c = 1$ the carrier is strongly additive, but whether this seed changes the isomorphism class of the reconstructed triple still requires an invariant.

Proposition 5.9 (Cyclicity supplies the remaining Cut conditions)**.** *For arbitrary nonzero f supported in a compact subset of (a,b), $a > 0$, in the same vacuum representation, cyclicity of $N_{O_f}^-$ implies that $N_{O_f}^- \subset M$ is a proper standard +hsm inclusion.*

Proof. The past orbit lies in $\mathrm{Vir}_c(a,\infty) \subsetneq M$, and $\mathrm{Vir}_c(0,a) \subset M \cap (N_{O_f}^-)'$. Locality and Reeh–Schlieder give standardness, while separatingness is inherited from M. □

Conjecture 5.10 (General seeds and the position of the Cut)**.** *(a) For every nonzero real compact stress seed on $(0,\infty)$, the past modular orbit is vacuum-cyclic. (b) Whenever the past orbit is cyclic, $N_{O_f}^-$ is not $\mathrm{Vir}_c(a',\infty)$ for any $a' > 0$.*

Part (a) is proved above for the endpoint-monotone positive log-bump class. Beyond that class, bracket closure remains available but a localized kernel of controlled nonzero mass has not been obtained in general. Part (b) is independent of cyclicity. Near the left support endpoint, finite bracket expressions built from right translates are multiples of the initial field. This germ restriction does not survive automatically through strong operator closure, so it cannot prove (b). A non-isomorphic geometry would require a further invariant of the full standard triple. The positive construction narrows the formation problem without assuming its remaining general conclusions.

6 Response of the selected algebra to excitation

Let $f \in C_c^\infty((a_0,\infty),\mathbb{R})$, $a_0 > 0$, $\rho = \mathrm{Exp} f$, $w = \rho''/\rho'$, and $0 \le a < a_0$. Put $N_a = \mathrm{Vir}_c(a,\infty)$.

Theorem 6.1 (Cut displacement)**.** *With $U = V(\rho)$, $\omega_\pm$ as in Section 2.3,*

$$D_a := S_{N_a}(\omega\|\omega_+) = S_{N_a}(\omega_-\|\omega) = \frac{c}{24}\int (x-a)\, w(x)^2\, dx.$$

Consequently

$$-\partial_a D_a = 2\pi E, \qquad D_0 - D_a = 2\pi a E, \qquad E = \frac{c}{48\pi}\int w^2\, dx = \int \langle T(x)\rangle_{\omega_-}\, dx,$$

where $\langle T\rangle_{\omega_-} = -\frac{c}{24\pi}\mathrm{Sch}(\rho)$ is the stress expectation in ω_-.

Proof. Since $\omega_+ \circ \mathrm{Ad}\, U = \omega$ and $\omega \circ \mathrm{Ad}\, U = \omega_-$, unitary covariance of the relative entropy gives $S(\omega\|\omega_+) = S(\omega_-\|\omega)$ on any algebra containing U, in particular N_a. Translation covariance moves the algebra N_a to N_0 and ρ to $\tilde\rho(y) = \rho(y+a) - a$, whose generating field $\tilde f(y) = f(y+a)$ is supported in $y > a_0 - a > 0$; (1) applied to $\tilde\rho$ gives $\frac{c}{24}\int y\,(\partial \log \tilde\rho')^2 dy = \frac{c}{24}\int (x-a) w^2 dx$. The same value is given by the exact identity for finite-amplitude local unitaries (not the linearised first law), $S_{N_a}(\omega_-\|\omega) = \langle U^*\Omega, -\log \Delta_{N_a} U^*\Omega\rangle = 2\pi \int (x-a)\langle T\rangle_{\omega_-} dx$, obtained from the transformation law of T and $\int (x-a)\mathrm{Sch}(\rho)\, dx = -\frac{1}{2}\int (x-a) w^2 dx$. Differentiating in a gives the remaining formulas. □

Corollary 6.2. *$G_a(f) = \partial_\theta^2 D_a(\mathrm{Exp}\,\theta f)|_{\theta=0} = \frac{c}{12}\int (x-a) f''(x)^2 dx$, which is > 0 for every non-zero test field f, and $G_0 - G_a = \frac{ca}{12}\int f''^2$.*

Proposition 6.3 (Noise does not reduce the response)**.** *Let $E : M \to B$ be a normal conditional expectation and $\iota : B \to M$ the inclusion. If two normal states φ, ψ on M factor through E ($\varphi = \varphi|_B \circ E$, $\psi = \psi|_B \circ E$), then $S_M(\varphi\|\psi) = S_B(\varphi|_B\|\psi|_B)$.*

Proof. Monotonicity under the restriction ι gives $S_B \le S_M$; monotonicity under the unital completely positive map E gives $S_M(\varphi\|\psi) \le S_B$. □

In $\mathcal{A}_{SU(2)_1}$, for $U \in \mathrm{Vir}_1$, the states ω and $\omega_\pm$ factor through $Q_{SO(3)}$: put $W_+ = U$, $W_- = U^*$, so that $\omega_\pm(x) = \omega(W_\pm^* x W_\pm)$ with $W_\pm \in M^{SO(3)}$; since $Q_{SO(3)}$ is $M^{SO(3)}$-bimodular and $\omega \circ Q_{SO(3)} = \omega$, $\omega_\pm(Q_{SO(3)}x) = \omega(Q_{SO(3)}(W_\pm^* x W_\pm)) = \omega_\pm(x)$; hence the information carried by an excitation of the selected algebra is the same in the full accessible algebra M as in Vir_1; the noise does not degrade it. This statement is about states generated inside the selected algebra, not about arbitrary information.

Remark 6.4 (What the identity is not). The identity is a statement about relative entropy on a family of algebras, and $-\partial_a D_a$ is a null-energy integral. It is not a Clausius relation for a thermodynamic entropy, and it does not produce an area or a curvature: moving a region in flat vacuum changes the algebra and its modular generator without creating curvature. The Hessian of the relative entropy corresponds to the Bogoliubov–Kubo–Mori metric, not to the symmetric-logarithmic-derivative quantum Fisher information. The structure is that of the half-sided modular inclusions used by Ceyhan and Faulkner [11] to relate shape derivatives of relative entropy to null energy, and null-plane algebras carry a copy of such a structure for each transverse position [12]; a transverse family with coupling is where area and focusing could enter, and is not treated here.

7 Limits of the results

Exact algebras and finite tasks. Exact fixed algebras and hulls are discontinuous in the dynamics. Two qubits with local Z-dephasing at rate κ and coupling $\epsilon X_1 X_2$ have the fixed algebra span$\{1, Z_1 Z_2\}$ for every $\epsilon \neq 0$, versus a four-dimensional diagonal algebra at $\epsilon = 0$. For $|\epsilon| < \kappa$, the slowest nonzero rate is $2(\kappa - \sqrt{\kappa^2 - \epsilon^2})$, so four slow modes persist on a window of order κ/ϵ^2. Their span is a rotated commutative algebra, but finite discrimination errors, not its exact fixedness, determine a reader's available tasks. HAFF-B treats such windows and the continuity of finite task risk [22].

Nor does the new generation theorem give a cheap continuum limit. For the normalized bump $g(u) = e^{4-1/(u(1-u))}$, the bracket mass at $\varepsilon = .05$ is about 1.1×10^{-31}. Dividing by that mass and taking spectral, Trotter and smooth limits are legitimate algebraic steps; they supply no bounded-control or finite-time certificate.

There is nevertheless a useful finite consequence of the constructive proof. Its scope is conditional on implementing the elementary flows, not on merely measuring the bounded seed.

Proposition 7.1 (A finite task consequence of flow generation)**.** *Take f, g, a, b as in Theorem 5.7, with $f_c(x) = xg(\log x - c)$. Suppose the elementary gates $e^{itT(f_c)}$, for finitely specified real t and $c \geq 0$, can be implemented to arbitrary specified error. For every compact stress field h supported in (b, ∞), every joint density operator ρ on the vacuum Hilbert space and a retained reference, and every $\delta > 0$, a finite word of such gates and their inverses approximates the channel of $e^{iT(h)}$ on ρ within trace distance δ. The statement holds uniformly over a specified finite list of joint states. If slot j is implemented within half-diamond error ϵ_j, the additional trace-distance error is at most $\sum_j \epsilon_j$.*

Proof. The proof of Theorem 5.7 uses conjugations, real linear combinations implemented by Trotter products, and fixed-compact smooth limits. For each constructed flow these give iterated strong limits of phase-adjusted finite words in the elementary flow unitaries. Scalar phases do not change the resulting channels. Their adjoints converge strongly too, since the limits are unitary and the inverse-time construction applies. On any finite set of test vectors, successive approximations can be chosen in the order in which these limits occur to produce one finite word with arbitrarily small vector error. On the separable vacuum Hilbert space a diagonal choice gives strong convergence of a sequence of words to $e^{iT(h)}$.

Strong convergence of uniformly bounded unitaries implies trace-norm convergence of conjugation on every trace-class joint state, first on finite-rank operators on the field and reference and then by trace-norm approximation. The same choice works for a finite list of states. For imperfect gates, telescoping through the same circuit and contractivity of channels gives the sum of slot errors, including the retained reference. □

This proposition is not uniform over all input states and gives no rate or optimal cost. Its targets are the stress flows constructed above, not every unitary or channel in the generated von Neumann algebra. Each finite approximation has a finite word length, gate precision, maximum modular shift and preparation budget that must be charged. $W^*(O_f)$ alone does not license its spectral functions as hardware. In a neutral current realization, that permission has a concrete implementation rather than an independent quadratic-control assumption.

Corollary 7.2 (Finite flows from local current control)**.** *Use the specified circle realization with neutral current Hamiltonian $H = L_0$, restricted at input to its $c = 1$ Virasoro vacuum subspace. Suppose signed local linear-current coupling, degenerate-helper rotations, record phase gates and calibrated clock preparation are available during coherent control intervals. Then the elementary-flow permission of Proposition 7.1 can be implemented by a finite positive autonomous program. For every fixed target there and $E < \infty$, its joint error can be made arbitrarily small, uniformly over inputs with $\mathrm{tr}(H\rho) \leq E$, retaining the old references, helper and original clock. The conclusion also holds on a specified finite list of joint states without a uniform energy bound.*

Proof. Q-RAIF C's current-stress compiler [1] selects finite current quadratures using full-period linear pulses, synthesizes their squares with a retained Pauli helper, and proves convergence of the current-factor cutoff to the Sugawara stress tensor. Its positive single-clock realization includes free propagation and both signs of each gate. Map the finitely many compact line smearings to the specified circle chart, accounting for scalar coordinate phases. On $P_M = 1_{[0,M]}(H)$, strong finite-word convergence becomes operator-norm convergence because P_M has finite rank. Approximate the finite set of prefix vectors when compiling the word and telescope the errors. For a joint mean-energy input the discarded proof tail costs at most $2\sqrt{E/(M+1)}$, independently of reference dimension. For a finite list of arbitrary joint states, their spectral tails instead tend to zero by trace-class convergence. The proof uses these projections only to bound errors, not as physical measurements or postselection. □

The physical choice has a bounded future test. Use the Fourier convention

$$f_n = (2\pi)^{-1} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta.$$

Take real compact f with $b = \mathrm{Im}\, f_2 \neq 0$, and $q_1 = (a_1 + a_1^*)/\sqrt{2}$. The vacuum flow satisfies

$$\langle e^{itT(f)}\Omega, q_1^2 e^{itT(f)}\Omega \rangle = \tfrac{1}{2} - bt + O(t^2).$$

Only the first-mode pair term contributes to this derivative. A later current Ramsey test of $e^{i\lambda q_1}$ therefore has $p'_+(0) = b\lambda^2 e^{-\lambda^2/4}/4 \neq 0$. A retained binary record selecting the two signs changes this later readout for sufficiently small nonzero t and implementation error. The test requires another paid linear pulse, helper and measurement. Intermediate current pulses can leave the Virasoro vacuum subspace; their endpoint leakage is included in the joint error. They are not protected by the always-on $SO(3)$ noise merely because the target stress flow is invariant. The carrier, coherent-control interval and growing precision budgets remain inputs; the finite program does not establish the exact Cut limit or its maintenance.

Foundation I gives a sharper, explicit bridge for finite pointer tasks: fresh partially polarized records approximate a sharp pointer instrument in half-diamond distance by their exact majority error, and errors add through finite adaptive circuits [20]. HAFF-A supplies a different finite achievement: actual calibrated fragment readouts certify distinct correlation-graph geometries

inside a stated GHZ protocol [21]. Neither result asserts that a finite apparatus has become an exact type-III Cut.

Sharing a carrier algebra does not fix its state: the vacuum and a local Virasoro excitation are both fixed by $D^{SO(3)}$ but have $D_0 > 0$. Conversely the two tensor-factor algebras intersect only in scalars while the correlated state $(1-\epsilon)(|00\rangle\langle 00| + |11\rangle\langle 11|)/2 + \epsilon I/4$ has Z disagreement $\epsilon/2$. Foundation II treats this distinction.

8 What the correspondence says about the independence assumption

A physical world can exist without a person looking at it. The harder question here concerns a complete description: does a perspective add an independently variable structure, or does it express relations already present in that world? The mathematical model isolates one structural part of this question. An accessible algebra and a state determine modular dynamics; a standard Cut and its vacuum determine a geometric net. Neither object is identified with experience.

Independence of variation, not ontological priority. Within the standard class, geometry and the modular triple determine each other. Fixing the net and vacuum fixes the modular flow of its half-lines; fixing a standard Cut returns the net. A geometric description and this structural proxy for a perspective cannot then be varied as two independent pieces of data. The equivalence is symmetric. A net with a vacuum remains a legitimate starting specification; it contains its half-line Cuts without waiting for an observer. Conversely, the standard triple is enough to reconstruct that description. Choosing either notation first settles no question about what exists first.

The choice of usable structure is a different question. Theorem 5.7 makes progress on it: once a local bump seed, carrier and modular dynamics are given, cyclicity, properness and standardness are consequences, not further choices. A reconstructed translation group is obtained without an independent translation permission. The result no longer leaves every Cut as an external input. It still inherits the state and carrier that define the modular flow. Noise selection also acts on a specified carrier: two internal rotations select the Virasoro subnet, but do not create its vacuum dynamics from nothing.

From a selected boundary to a maintained perspective. The seed-generated Cut supplies a boundary within which local operations can be reconstructed. A physical reader must additionally acquire records, retain them, choose operations and survive the disturbance its operations cause. The finite instruments of HAFF-E and Foundation I show how this extra layer can have explicit errors and resources. The geometry certification of HAFF-A then shows that changing actual record protocols can change what geometry is certified under the same dictionary. Those finite graphs and the chiral modular net are different geometric objects; no common continuum limit is asserted between them.

This distinction supports a stronger mechanism than treating perspective as a passive label. A record can change a subsequent coupling and thereby change the state or source, not merely the description. The subject line of the series asks when a retained history makes such changes useful and when a system must distinguish world drift from drift of its own interface. The present Cut theorem supplies an ideal structural boundary to that programme; it does not require a human reader, and it does not replace the programme with a net-theoretic definition of consciousness.

Conjecture 8.1 (Maintained modular boundaries)**.** *In a suitable local carrier and stationary reference state, a physical record-and-control process can maintain a hierarchy of finite task permissions whose modular covariance and nesting errors tend to zero as the charged control, memory and preparation resources increase, and whose ideal limit is a standard seed-generated Cut. Changing the retained control history can change the maintained boundary and hence subsequent response and access.*

The single-orbit construction identifies a nonempty ideal limit. Corollary 7.2 implements the finite words from specified local current controls, uniformly on bounded-energy inputs with retained references. The remaining task is to form those primitive permissions and sustain the control process, with explicit preparation, transport and maintenance costs on the same history. The claim would fail for a proposed process if its covariance error stayed bounded away from zero at increasing budgets, if disturbance destroyed standardness in every admissible stationary limit, or if different histories only relabelled an unchanged instrument. A family of task-level bounds, rather than algebra dimension alone, is the relevant evidence.

Experience and competing readings. The explanatory-gap argument [8] is not refuted by exchanging a geometric description for an algebraic one. A reductionist can read the Cut as a convenient organization of observables; a relational reading can treat the same dependencies as evidence that a perspective and its usable world are jointly constituted. Both readings must accept the same correspondence and finite interventions. Neither obtains experience from it. Thermal time [9] motivates the modular proxy, but state-relative modular evolution exists whether or not the system reports anything.

A stronger claim about first-person organization requires additional distinctions: private history versus public calibration, causal self-reading versus a model supplied from outside, and reporting behavior versus experience. Consciousness III develops that bridge as its own problem. Here the strongest supported conclusion is structural: the boundary, its state response and its geometric reconstruction are constrained together, and one such boundary can be generated from local seed history in a given carrier. That is a restriction on independent specification, not a proof that the universe needs an observer.

Null-plane copies have separate transverse source and coupling requirements [12, 11]; Consciousness II and HAFF-D address those physical interfaces.

9 Open problems

1. **Physical formation.** Derive the carrier, stationary reference and availability of the primitive local current controls from a common process, and test Conjecture 8.1. The seed-generated standard Cut is constructed in the specified model; primitive modular dynamics and a finite maintenance protocol are not.
2. **General seeds and inequivalent Cuts.** Conjecture 5.10 outside the positive log-bump cyclicity theorem; whether a seed-generated standard triple differs from all ordinary translated half-line triples. Modular-only generation is proved for that bump class, not for every nonzero stress seed.
3. **Finite resources.** Determine explicit gate costs and convergence rates for Corollary 7.2. It already supplies uniform approximability on bounded-energy inputs with arbitrary retained references; it does not give an optimal budget or a maintained Cut limit.

4. **Transverse families and gravity.** Coupling between null-plane copies, area and focusing; the selection of the algebra family and the physical source enter here. An invisible source is not thereby absent; changing modular data and changing actual stress are different tests. For ball families in higher dimensions Consciousness II identifies the variation of area minus $(d-1)/R$ times volume as minus the weighted linearized Einstein charge. On a conformal net in Minkowski space, its supplied MC_η consistency law with the CHM bulk modular energy, after the declared boundary-charge subtraction and under that paper's source and functional assumptions, is equivalent to the linearized Einstein equation [19]; null-plane couplings remain open.

5. **Selection.** Why this noise group; how unrecorded transformations are aligned or reduced and at what cost; the imprint of the discarded group in the sectors of the selected algebra (for SU(2) level one: the subcategory of Virasoro sectors of lowest weight j^2 contained in the vacuum representation).

6. **Unbounded generators.** Which non-multiplicative semigroups are compatible with a strict scaling covariance $E_t\sigma_s = \sigma_s E_{e^{2\pi s}t}$ is unclassified: it is known that such a semigroup preserves the vacuum, that it is trivial if it commutes with σ, and that it is zero if its generator is bounded.

A Proof of Theorem 5.1

Write $B = \mathrm{Vir}_c(0,\infty)$, $\mathrm{Hull} = \mathrm{Hull}_{\sigma,T}(O_f)$, and for a von Neumann algebra $N \subset M$ $\mathcal{V}_N = \{h \in C_c^\infty((0,\infty),\mathbb{R}) : e^{iuT(h)} \in N\ \forall u \in \mathbb{R}\}$.

Step 0: Hull $\subset B$. B is admissible: $\sigma_s = \mathrm{Ad}\,V(\delta_{e^{-2\pi s}})$ preserves the supports $(0,\infty)$ and the generators $T(h)$ go to $T(\delta_* h)$; T_a, $a \geq 0$, moves supports into $(a,\infty) \subset (0,\infty)$; and $O_f \in B$.

Step 1: spectral projections. $O_f = \psi(T(f))$ with the injective Borel function $\psi = \frac{2}{\pi}\arctan$. Hence ψ^{-1} is Borel, every bounded Borel function of $T(f)$ is a bounded Borel function of O_f, and so $e^{iuT(f)} \in W^*(O_f) \subset N$ for every admissible $N \ni O_f$. In particular $f \in \mathcal{V}_N$; $\mathcal{V}_N$ is closed under multiplication by real scalars since u ranges over $\mathbb{R}$.

Step 2: affine images. For $\varphi(x) = \lambda x + b$, $\lambda > 0$, $b \geq 0$, the Schwarzian vanishes, so $V(\varphi)T(h)V(\varphi)^* = T(\varphi_* h)$ with $(\varphi_* h)(y) = \lambda h((y-b)/\lambda)$. If $h \in \mathcal{V}_N$ then $e^{iuT(\varphi_* h)} = T_b\big(\sigma_s(e^{iuT(h)})\big)$ with $e^{-2\pi s} = \lambda$ belongs to N, because N is stable under σ and under T_b. So $\varphi_* h \in \mathcal{V}_N$.

Step 3: linear and limit closure. If $h_1, h_2 \in \mathcal{V}_N$ then $h_1 + h_2 \in \mathcal{V}_N$: $T(h_1) + T(h_2)$ equals $T(h_1 + h_2)$ on the finite-energy vectors, a common core on which $T(h_1 + h_2)$ is essentially self-adjoint [10, Thm. 4.4], so the Trotter product formula $e^{iu(A+B)} = \text{s-}\lim(e^{iuA/n}e^{iuB/n})^n$ gives $e^{iuT(h_1+h_2)} \in N$, N being strongly closed. If $h_n \in \mathcal{V}_N$ have supports in a fixed compact subset of $(0,\infty)$ and $h_n \to h$ in C^∞ (hence in $\|\cdot\|_{3/2}$), then $e^{iuT(h_n)} \to e^{iuT(h)}$ strongly [10, Prop. 4.5], and $h \in \mathcal{V}_N$.

Step 4: the case $m = \int f \neq 0$. Let $h \in C_c^\infty((0,\infty),\mathbb{R})$ and put $f_{\varepsilon,b}(y) = \varepsilon f((y-b)/\varepsilon)$, $k_{\varepsilon,b} = f_{\varepsilon,b}/(\varepsilon^2 m)$, so $\int k_{\varepsilon,b} = 1$ and $k_{\varepsilon,b} \in \mathcal{V}_N$ by Steps 1–3. Then

$$h_\varepsilon(y) := \int_{b\geq 0} h(b) k_{\varepsilon,b}(y)\, db = \frac{1}{m}\int f(w)\, h(y-\varepsilon w)\, dw \longrightarrow h(y)$$

in C^∞, with supports in a fixed compact subset of $(0,\infty)$ for small ε. Only $b \geq 0$ is used: h has compact positive support, and $h(y-\varepsilon w) = 0$ whenever $b = y - \varepsilon w < 0$. Riemann sums of the integral are finite real combinations of the k_{ε,b_i}, which lie in $\mathcal{V}_N$ and converge to h_ε in C^∞ with fixed compact support, so $h_\varepsilon \in \mathcal{V}_N$ by Step 3, and then $h \in \mathcal{V}_N$. Hence $\mathcal{V}_N = C_c^\infty((0,\infty),\mathbb{R})$.

Step 5: the case $\int f = 0$. Let $\varphi_s = \mathrm{Exp}(sf)$. By Step 1, $e^{isT(f)} \in N$, and $e^{isT(f)} = V(\varphi_s)$ up to a phase, so conjugation by it maps $\mathcal{V}_N$ to itself via $h \mapsto (\varphi_s)_* h$. (For a general diffeomorphism the conjugation sends $T(h)$ to $T(\varphi_* h)$ plus a real multiple of the identity, the Schwarzian term; the multiple exponentiates to a phase, which does not affect membership in N since $N \ni 1$.) With $f_\delta(y) = f(y-\delta)$, $\delta \geq 0$, we get $(\varphi_s)_* f_\delta \in \mathcal{V}_N$. Since $\frac{d}{ds}(\varphi_s)_* g = -[f,g]$ with $[X,Y] = XY' - X'Y$ and $\int [f, f_\delta] = 2\int f f_\delta'$,

$$\frac{d}{ds}\Big|_0 \int (\varphi_s)_* f_\delta\, dy = -2\int f(y) f'(y-\delta)\, dy = 2C'(\delta), \qquad C(\delta) = \int f(y) f(y-\delta)\, dy.$$

C is even, smooth, compactly supported and $C(0) = \|f\|^2 > 0$, so $C'(\delta_0) \neq 0$ for some $\delta_0 > 0$. For small $s \neq 0$ the field $g = (\varphi_s)_* f_{\delta_0}$ has $\int g \neq 0$, compact support in $(0,\infty)$ and $g \in \mathcal{V}_N$; apply Step 4 to g.

Step 6. All $e^{iuT(h)}$, $h \in C_c^\infty((0,\infty),\mathbb{R})$, lie in N, so $N \supset \{T(h)\}'' = B$. With Step 0, Hull $= B$. $\square$

B The Virasoro vacuum module in Fock space

In the Fock space of the $U(1)$ current with $[J_m, J_n] = m\delta_{m+n,0}$, $J_n\Omega = 0$ for $n \geq 0$, take $L_n = \frac{1}{2}\sum_k :J_{n-k}J_k:$. We built the descendants $L_{-n_1}\cdots L_{-n_k}\Omega$ ($n_i \geq 2$) at each level with exact rational arithmetic and computed their rank, and compared with the number of partitions into an even number of parts (the $\mathbb{Z}_2$-even Fock space).

level n	0	1	2	3	4	5	6	7	8	9
dim Virasoro vacuum module	1	0	1	1	2	2	4	4	7	8
$p(n) - p(n-1)$	1	0	1	1	2	2	4	4	7	8
dim even Fock space	1	0	1	1	3	3	6	7	12	14

The first level where the even space is strictly larger is 4. At level 4, $\langle L_{-4}\Omega, L_{-4}\Omega\rangle = 5 = \frac{c}{12}(4^3-4)$ with $c=1$, the Gram matrix on $\{L_{-4}\Omega, L_{-2}^2\Omega\}$ is $\left(\begin{smallmatrix} 5 & 3 \\ 3 & 9/2 \end{smallmatrix}\right)$ with determinant $27/2$, and $u = \frac{1}{2}J_{-1}^4\Omega - J_{-3}J_{-1}\Omega + \frac{3}{4}J_{-2}^2\Omega$ satisfies $L_1 u = L_2 u = L_3 u = L_4 u = 0$, $\langle u, L_{-4}\Omega\rangle = \langle u, L_{-2}^2\Omega\rangle = 0$, $\|u\|^2 = \frac{1}{4}\cdot 24 + 3 + \frac{9}{16}\cdot 8 = \frac{27}{2}$; the three vectors have rank 3. The generating function of the even dimensions agrees through order 23 with $\sum_{p\geq 0}(t^{4p^2} - t^{(2p+1)^2})\prod_n (1-t^n)^{-1}$, and the $SU(2)_1$ vacuum character identity $\sum_m z^m t^{m^2} P(t) = \sum_j \chi_j(z)(t^{j^2} - t^{(j+1)^2})P(t)$, $P(t) = \prod(1-t^n)^{-1}$, holds coefficient by coefficient for $|m| \leq 5$ through order 23. These finite checks are consistent with the decomposition of the even Fock space as $\bigoplus_p L(1, 4p^2)$ and with the character identity behind Theorem 4.6, but a check through finitely many levels does not prove an infinite decomposition; only the level-4 computation is used in Proposition 4.8, and the identity of characters used for Theorem 4.6 is Rehren's.

C Numerical checks

Cut displacement. For $f(x) = e^{4-1/(z(1-z))}(1 + 0.7(z - \frac{1}{2}))$, $z = x - 2$, supported in $(2, 3)$, $c = 1$, $\rho = \mathrm{Exp}(\theta f)$, the integrals of Theorem 6.1 were evaluated by quadrature after integrating the flow and its first three variational derivatives. For $\theta = 0.05$:

a	0	1	1.5	1.9
D_a/D_0	1.000	0.604	0.406	0.2476
$D_0 - D_a$	0	4.5641e-2	6.8461e-2	8.6718e-2
$2\pi aE$	0	4.5641e-2	6.8461e-2	8.6718e-2

and both $-\partial_a D_a$ and $2\pi \int \langle T \rangle dx$ are 4.5640958×10^{-2}. At 2001 and 4001 spatial points the displayed values agree to the shown precision. This is a quadrature diagnostic, not an interval-certified error bound. For $\theta = 0.2$, $D_a/D_0 = 1, 0.593, 0.390, 0.227$ at the same a.

Single-orbit closure. The analytic proof uses the allowed kernel integral (3), without negative recentering. For $g(u) = e^{4-1/(u(1-u))}$ on $(0, 1)$, numerical quadrature checks the kernel's non-negative sign and positive mass, and convergence in the first three derivative norms for targets supported strictly to the right of 1. A separate geometric-flow check evaluates the pushforward difference quotient of Lemma 5.6. These diagnostics check signs and a finite selection of derivatives; the proof of C^∞ and operator closure is the analytic argument, not the grid. Small kernel mass is also recorded, since normalization can be extremely costly and cannot certify finite-resource generation.

Finite-dimensional checks. The identities of Theorems 4.4 and 4.5 were checked in the spin representation $[1, 1, 2]$ of $SO(3)$ on M_{11}: the fixed algebra is 5-dimensional for the two-axis noise, the axis noises do not commute ($\|[Q_z, Q_x]\|_F = 1.90$), the dissipative identity of Theorem 4.5 holds for a fixed complex random probe (both sides 363.2646789268), and for $L = Q_x + Q_z - 2\,\mathrm{id}$, $\|e^{tL} - Q_{SO(3)}\|$ falls from 0.60653 at $t = 1$ to 9.4×10^{-14} at $t = 60$. The two specified irrational rotations also have a five-dimensional fixed algebra in this representation. These are finite-dimensional analogues, not statements about the net.

D Boundaries of shell and scale rules

A shell containing a full norm ball generates the ambient algebra and cannot select a proper Cut; a unit-sphere version has the same defect. Modular covariance must respect the antilinearity identity $J\Delta^{is}J = \Delta^{is}$. The vacuum modular generator has zero variance, while a truncated one-sided variance can diverge and have a regularization-dependent finite part. Approximate innerness does not imply innerness of an automorphism. Faithfulness gives separation in the stated representation, not cyclicity. These boundaries are separate from the local stress-seed proof. The complete earlier-route analysis is retained in the supplementary source record.

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Higher-Dimensional Consistency of Geometry and Modular Flow: Physical Source Identification and a Second-Order Extension

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Abstract

An observer reads ball algebras at different positions, sizes and rest frames. We ask when those readings agree with one geometry. In a conformal theory on Minkowski space, the first variation of the geodesic ball's boundary area minus $(d-1)/R$ times its volume equals minus the CHM-weighted linearized Einstein tensor at every radius. The conditional law $\eta\,\delta F_B + \delta\langle K_B\rangle = 0$ is therefore equivalent to $G^{\rm lin} = (2\pi/\eta)\delta\langle T\rangle$.

A nonzero-frequency source identifies this intrinsic linear response up to normalization even when internal length scales are allowed. Local coherent and neutral scalar preparations provide smooth finite-energy Radon witnesses; a bounded pulse inside a permitted shared hull supplies a more general positive-energy test, with local source regularity still required. A scalar's physical modular energy includes a boundary charge. Its subtraction, or the matched Wald density, is essential: raw energy cannot satisfy the pure-area bulk law on all small balls.

One frame fixes only the normal constraint. Boost comparisons determine the sourced tensor, with small-rapidity stress noise amplified as β^{-2}. Free Weyl curvature is invisible at linear order. Stress-containing selected algebras give the same sourced tensor but different relative entropies. Under the supplied radial readout model, conditional multipole information falls as $e^{-4k\log k}$.

We extend the local conditional law to second order. A quadratic extrinsic-curvature correction gives the full second-order Einstein residual from all small boosted balls. A bent slice of flat spacetime shows why the correction or a geodesic-slice prescription is necessary. For the prepared conformal scalar, changed propagation, metric variation and $\xi G\Phi^2$ supply explicit nonzero next-order source terms. The common cone, area density and consistency mechanism remain inputs. Curved stress readings are not assumed to be curved modular Hamiltonians; the common quantum source and nonlinear modular response form a strong, testable extension rather than an established consequence of the linear law.

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1 Introduction

Consciousness I asks whether a physical description can be fixed before any perspective is specified [1]. In its chiral conformal net, the geometry and modular data of a standard Cut determine each other. Transverse directions change the question. A higher-dimensional observer has a family of ball algebras, with different centres, radii and rest frames. Can one geometry account for all their readings?

For conformal matter in Minkowski spacetime these regions carry the bulk Casini–Huerta–Myers (CHM) stress charge [7]. We first construct its geometric partner rather than postulating a two-derivative conserved tensor. For every smooth perturbation the area–volume variation of a geodesic ball equals the weighted integral of the linearized Einstein tensor. The Gauss equation and intrinsic gauge invariance explain why that tensor appears. Comparing it with the CHM charge gives a conditional same-spacetime Einstein law, without a holographic dictionary. Its coefficient is supplied, and its physical consistency mechanism remains to be explained.

The next question is whether an experiment identifies the response law. A homogeneous source does not. A source with nonzero-frequency weight and an open set of boost directions identifies the entire compact intrinsic response at one tested radius, up to normalization. We retain the proof for tempered Radon Fourier measures, and provide actual scalar preparations with finite positive energy. A neutral pulse belongs to a proper stable hull; a shared non-scalar hull with translation permission can also supply a bounded positive-energy pulse. Control permission, smooth local source readings and source-apparatus costs are distinct inputs.

The physical one-sided scalar energy includes a surface charge. Keeping it in the ledger yields a raw-energy obstruction and a matched Wald coefficient; dropping it would falsely identify a geometric response. The resulting tensor is common to stress-containing selected algebras. Their relative entropies differ, not the emitted stress. A single rest frame leaves momentum and spatial stresses underdetermined, and no linear ball reading fixes free gravitational radiation.

Second order is not just a finer version of the same construction. Slice extrinsic curvature, changed source propagation, quantum reference stress and curved modular flow become relevant. We derive an extrinsic-corrected leading small-ball law and prove its two-jet equivalence to the second-order Einstein equation. A flat bent-slice counterexample and an explicit next-order scalar source test it. Section 7 states the remaining common-geometry extension with its source, readout and intervention conditions.

The linear results require a conformal net, the bulk CHM datum, and a common Minkowski cone. Q-RAIF A studies conditions for common propagation cones [5]; this paper does not derive the cone of its boosted family. Its exact conformal net is a supplied fixed-point theory. Approximate cones in a lattice realization do not by themselves establish the exact CHM modular flow used here; nonconformal corrections require a separate limit estimate [24]. The area density η and modular consistency are also inputs. Additional edge energy must be unchanged, separately read and subtracted, or directly bypassed by bulk stress readings. At second order we supply a corrected local *stress* consistency condition, not a curved CHM formula. Neither the Gauss identity nor stress conservation makes that condition physically true by itself.

Sections 2–4 establish the linear law. Sections 5–6 supply preparations, identification and partial-observer limits; Section 7 extends the geometry and source. Section 8 prices radial resolution, and the conclusion returns to the series' question of access and a common world. Indexed appendices retain the coefficient, spectral-measure, source and entropy calculations.

2 The modular family of balls

2.1 Balls, diamonds and their modular Hamiltonians

Work in Minkowski space $\mathbb{R}^{1,d}$, $d \geq 2$, with $\eta = \mathrm{diag}(-1, 1, \ldots, 1)$. A ball $B = B(x_0, R, u)$ is fixed by a centre x_0, a radius R and a future unit timelike vector u: it is the set of points x with $u \cdot (x - x_0) = 0$ and $|x - x_0| < R$, an open ball in the hyperplane $\Sigma_u(x_0)$ orthogonal to u. Its causal diamond is $D(B)$. We write

$$w_R(y) = \frac{R^2 - |y - x_0|^2}{2R}, \qquad T_{uu} = T_{\mu\nu} u^\mu u^\nu,$$

and Δ_u for the Laplacian of Σ_u.

Let $\mathcal{A}$ be a conformal net on $\mathbb{R}^{1,d}$ with vacuum Ω. The modular group of $(\mathcal{A}(D(B)), \Omega)$ acts geometrically, by the flow of the conformal Killing vector that preserves $D(B)$ [13, 14]. In coordinates with $x_0 = 0$ and $u = \partial_t$,

$$\zeta_B = \frac{1}{2R}\Big[(R^2 - t^2 - r^2)\,\partial_t - 2t\, x^i \partial_i\Big], \tag{1}$$

normalized so that its surface gravity at the edge ∂B is one; the modular period is then 2π, as for the boost of a wedge. The one-sided CHM modular energy is the stress charge

$$K_B = 2\pi \int_\Sigma T_{\mu\nu} \zeta_B^\mu n^\nu \, dV = 2\pi \int_B w_R T_{uu} \, d^d x. \tag{2}$$

Here Σ is a Cauchy surface of the diamond and the second expression uses its central ball. In a theory without an additional edge charge, a faithful type I regulator identifies it with $H_B = -\log \rho_B$, up to the additive normalization constant [7]. For a free conformal scalar the physical one-sided energy includes a surface term [8]; Section 5.4 keeps that term separate. In the continuum K_B denotes the bulk charge and its vacuum-subtracted expectations, with the required stress-smearing limits assumed finite.

The full modular generator is a different object: $\widehat{K}_B = -\log \Delta_B$. It annihilates the vacuum and implements modular covariance. At a regulated bipartition,

$$\widehat{K}_B = H_B \otimes 1 - 1 \otimes H_{B^c}^{\mathsf{T}}.$$

Thus it agrees with the physical one-sided energy in expectation differences for a unitary local to $D(B)$, whose complement state is unchanged. It agrees with the bulk charge as well when any boundary contribution is unchanged. Arbitrary state variations need not have this property. Throughout, $\delta\langle K_B \rangle$ means the vacuum-subtracted CHM charge, and modular covariance uses $\widehat{K}_B$.

2.2 Two kinds of variation

HAFF-D separates two ways in which a modular Hamiltonian can change [3], and both occur in the ball family.

A *state variation* keeps the region and changes the state: $\delta\langle K_B \rangle = 2\pi \int_B w_R \, \delta\langle T \rangle_{uu}$. For a unitary $W \in \mathcal{A}(D(B))$ with unchanged boundary charge, Araki's relative entropy of $\omega_W = \langle W\Omega, \cdot\, W\Omega \rangle$ is

$$S_B(\omega_W \| \omega) = \langle W\Omega, \widehat{K}_B W\Omega \rangle = \delta\langle K_B \rangle_{\omega_W}, \tag{3}$$

which Consciousness I uses in $1+1$ dimensions [1, 15]. For a first-order perturbation $\rho = \sigma + \epsilon\,\delta\rho$ of a regulated (type I) version, the first law is $\delta S = \delta\langle H_B\rangle$; it equals the bulk $\delta\langle K_B\rangle$ when the boundary variation is zero.

A *region variation* keeps the state and moves the ball. The family of all diamonds is one orbit of the conformal group, of dimension $2(d+1)$: $d+1$ for the centre, one for the radius and d for the rest frame. A one-parameter conformal group e^{-iaP_ξ} moves $D(B)$ to $D(B_a)$ and $\widehat{K}_{B_a} = e^{-iaP_\xi}\widehat{K}_B e^{iaP_\xi}$, so

$$\delta\widehat{K}_B = -i\,\delta a\,[P_\xi, \widehat{K}_B], \qquad \langle\Omega, \delta\widehat{K}_B\,\Omega\rangle = 0, \tag{4}$$

because $P_\xi\Omega = 0$ and $\widehat{K}_B\Omega = 0$. This is the variation formula of HAFF-D for a half-sided modular family. Null translations of a wedge give half-sided modular inclusions [16, 17], and balls are conformal images of wedges, so the supplied ball family has the region interface required by HAFF-D's conditional holographic matching. Vacuum covariance concerns the full generator.

This transitivity concerns a single diamond, not an arbitrary pair. A nested pair of balls carries the inversive-distance modulus; it is not determined by a dimension count for the orbit of one ball.

2.3 Conservation from the diamond

The algebra of the diamond is the same for every Cauchy surface of $D(B)$. For its bulk stress charge, independence of the first form of (2) from Σ is exactly conservation and tracelessness.

Proposition 2.1 (Conservation from the diamond)**.** *Let $T_{\mu\nu}$ be a smooth symmetric tensor field on an open set $U \subset \mathbb{R}^{1,d}$. The charge $Q_\Sigma(B) = \int_\Sigma T_{\mu\nu}\zeta_B^\mu n^\nu dV$ is the same for all Cauchy surfaces Σ of $D(B)$, for every diamond with $\overline{D(B)} \subset U$, if and only if $\partial^\mu T_{\mu\nu} = 0$ and $T^\mu{}_\mu = 0$ in U.*

Proof. ζ_B is a conformal Killing vector, $\partial_\mu\zeta_\nu + \partial_\nu\zeta_\mu = \frac{2}{d+1}(\partial\cdot\zeta_B)\eta_{\mu\nu}$, with $\partial\cdot\zeta_B = -(d+1)t/R$ in the coordinates of (1). Hence

$$\partial^\mu\big(T_{\mu\nu}\zeta_B^\nu\big) = \zeta_B^\nu\,\partial^\mu T_{\mu\nu} + \frac{\partial\cdot\zeta_B}{d+1}\,T^\mu{}_\mu.$$

Two Cauchy surfaces of $D(B)$ share the edge ∂B and bound a region of $D(B)$; if the right side vanishes, Stokes' theorem gives equal charges. Conversely, deforming Σ near an interior point x shows that the left side vanishes at every $x \in D(B)$, for every diamond. As B varies over diamonds containing a fixed x, the pairs $(\zeta_B(x), \partial\cdot\zeta_B(x))$ span $\mathbb{R}^{d+1}\times\mathbb{R}$, so both terms vanish separately. □

For the anomaly-free flat conformal stress both properties hold. The proposition identifies what slice-independent stress-charge representation requires: a modular datum is attached to a region's algebra, not to a slice; provided this same bulk stress charge represents it on every Cauchy surface, the stated equality forces conservation and a vanishing trace. Conservation is therefore not an observation to be placed beside the Bianchi identity; it is a property of the modular family.

The finite-radius charge has a controlled expansion with leading coefficient $c_0 = V_d/(d+2)$; its exact Fourier transfer and all radial coefficients are given in Appendix A.1. In particular, an open set of normals recovers a smooth source tensor from its leading small-ball charges (Corollary A.2).

3 The geometric functional of a ball

3.1 What a ball functional may read

Let $h_{\mu\nu}$ be a smooth metric perturbation of Minkowski space, and h_{ij} the perturbation of the metric induced on $\Sigma_u(x_0)$.

Definition 3.1 (Ball functional)**.** A *ball functional* assigns to every ball $B = B(x_0, R, u)$ and every h a number $\delta F_B[h]$ such that:

(F1) it is linear in h;

(F2) it depends only on the induced perturbation h_{ij} on $\overline{B}$;

(F3) it is unchanged by $h_{ij} \mapsto h_{ij} + \partial_i\xi_j + \partial_j\xi_i$ for every smooth vector field ξ on a neighbourhood of $\overline{B}$ in Σ_u;

(F4) it is invariant under the orthogonal group of Σ_u fixing x_0;

(F5) it has the same form for every centre and every rest frame, its dependence on R and on any constants being arbitrary;

(F6) it is a distribution of finite order supported in $\overline{B}$.

No scale covariance is assumed, so the functional may contain any number of length scales besides R. The restrictive condition is (F2): the functional reads the intrinsic geometry of the ball. Three considerations support it. The region to which the modular datum belongs is the ball on its slice, whose geometric data are the induced metric and the extrinsic curvature. The extrinsic curvature reverses sign under reflection of time about Σ_u, while $\delta\langle K_B\rangle$ does not change when the state is reflected in a theory with that symmetry; a linear functional matched to the modular energy for all states cannot depend on it. And second time derivatives of h are not data of the region at all: reading them presupposes an evolution law for the metric, which is what one wants to obtain. Proposition 4.7 shows what happens when (F2) is dropped.

Lemma 3.2 (Ball functionals see only the Einstein tensor)**.** *Let δF satisfy (F1)–(F4) and (F6), and let $\tau_u = \partial_i\partial_j h_{ij} - \Delta_u h_{kk}$ be the linearized scalar curvature of Σ_u.*

(a) For a plane wave $h_{ij} = \mathrm{Re}(\varepsilon_{ij}e^{ip\cdot x})$ on Σ_u, $\delta F_B[h] = \mathrm{Re}\big(m_B(|p|)\,\hat{\tau}\,e^{ip\cdot x_0}\big)$ with $\hat{\tau} = p^2\varepsilon_{kk} - p_ip_j\varepsilon_{ij}$ and m_B an even entire function, the transfer function *of the functional.*

(b) For polynomial h, $\delta F_B[h] = \sum_k m_k\,\Delta_u^k\tau_u(x_0)$, where $m_B(q) = \sum_k m_k(-q^2)^k$.

(c) $\tau_u = 2\,G^{\mathrm{lin}}_{uu}$.

(d) There is a rotation-invariant scalar distribution μ_B, supported in $\overline{B}$, such that for every smooth h on a neighbourhood of that ball,

$$\delta F_B[h] = \langle \mu_B, \tau_u[h]\rangle, \qquad \widehat{\mu}_B(p) = m_B(|p|).$$

No growth condition on h outside the ball is required.

The proof is in Appendix A.3.

At linear order about a flat slice, the Gauss equation equates half the intrinsic scalar curvature with the normal–normal Einstein component. The quadratic extrinsic term is essential beyond that order (Section 7). The Bianchi identity, which makes G^{lin} divergence-free, therefore comes with the functional; it is not a separate premise.

Corollary 3.3 (The Weyl curvature is invisible)**.** *If h is smooth and $G^{\mathrm{lin}} = 0$, every ball functional vanishes on h for every ball. In particular, for $d \geq 3$ a free gravitational wave, whose Riemann tensor is pure Weyl curvature, gives zero for every ball, every rest frame and every functional satisfying (F1)–(F4) and (F6).*

3.2 The functional

For a ball $B = B(x_0, R, u)$ and a metric $g = \eta + \epsilon h$, let B_ϵ be the geodesic ball of radius R about x_0 in the metric induced on $\Sigma_u(x_0)$, A_ϵ the area of its boundary and V_ϵ its volume. Define

$$F_B[g] = A(\partial B_\epsilon) - \frac{d-1}{R}\, V(B_\epsilon), \qquad \delta F_B[h] = \frac{d}{d\epsilon} F_B[\eta + \epsilon h]\Big|_{\epsilon=0}. \tag{5}$$

For $d = 2$ the area is the length of the boundary circle.

Theorem 3.4 (Exact identity)**.** *For every smooth h and every ball,*

$$\delta F_B[h] = -\frac{1}{2}\int_B w_R\, \tau_u\, d^d x = -\int_B w_R\, G^{\mathrm{lin}}_{uu}\, d^d x. \tag{6}$$

Proof. Put $x_0 = 0$ and write points of Σ_u as $s\hat{n}$. To first order the geodesic distance from 0 to $s\hat{n}$ is $s + \frac{1}{2}\int_0^s h_{nn}(\sigma\hat{n})\, d\sigma$, because the length functional is stationary on the straight line. The boundary of B_ϵ is therefore $s = R - \epsilon\Phi(\hat{n})$ with $\Phi(\hat{n}) = \frac{1}{2}\int_0^R h_{nn}(\sigma\hat{n})\, d\sigma$. The area element of a surface changes by half the trace of h over its tangent space, and the volume element by half the full trace, so

$$\delta A = \frac{1}{2}\oint_{S_R} \big(h_{kk} - h_{nn}\big)\, dA - (d-1)R^{d-2}\oint \Phi\, d\Omega, \qquad \delta V = \frac{1}{2}\int_B h_{kk}\, d^d x - R^{d-1}\oint \Phi\, d\Omega.$$

In $\delta A - \frac{d-1}{R}\delta V$ the terms in Φ cancel:

$$\delta F_B = \frac{1}{2}\oint_{S_R} \big(h_{kk} - h_{nn}\big)\, dA - \frac{d-1}{2R}\int_B h_{kk}\, d^d x.$$

On the other side use $w_R = 0$ and $\partial_n w_R = -1$ on S_R, $\partial_i w_R = -x_i/R$ and $\partial_i\partial_j w_R = -\delta_{ij}/R$. Integrating by parts twice, $\int_B w_R\, \partial_i\partial_j h_{ij} = \oint h_{nn}\, dA - \frac{1}{R}\int_B h_{kk}$ and $\int_B w_R\, \Delta h_{kk} = \oint h_{kk}\, dA - \frac{d}{R}\int_B h_{kk}$. Subtracting gives $\int_B w_R \tau_u = \oint (h_{nn} - h_{kk})\, dA + \frac{d-1}{R}\int_B h_{kk}$, which is $-2\,\delta F_B$. The second equality in (6) is Lemma 3.2(c). □

The geometric identity can also be evaluated without the linearized formulas, by shooting geodesics with their Jacobi fields in the metric $\delta + \epsilon h$ and differencing in ϵ. For a non-conformally-flat perturbation, this evaluation agrees with the right side of (6) to relative accuracy 2×10^{-11} in $d = 2$ and 10^{-10} in $d = 3$.

The content of Theorem 3.4 is known: it is the flat-space, linear geometric part of the first law of causal diamonds of Jacobson and Visser [21], there obtained from the covariant phase space of general relativity with the Einstein constraint imposed. The proof above uses no field equation. That is what the next section needs: (6) holds for every perturbation, whatever its dynamics.

3.3 Where the coefficient comes from

The volume term in (5) is the fixed-volume prescription of Jacobson's entanglement-equilibrium argument [19], where it is motivated physically. In the modular family it has a structural origin.

Proposition 3.5 (The coefficient)**.** *Let $F_\lambda = A - \lambda V$ for geodesic balls.*

(a) Fix a reference radius R_0 and hold λ_0 fixed when varying R. Then

$$\frac{d}{dR}\left[\Omega_{d-1}\left(R^{d-1} - \frac{\lambda_0 R^d}{d}\right)\right]\Bigg|_{R=R_0} = 0 \quad \Longleftrightarrow \quad \lambda_0 = \frac{d-1}{R_0}.$$

Centre and frame moves leave the flat value unchanged for every λ_0. This fixes the Lagrange multiplier for the variation at R_0.

(b) Write $\delta F_\lambda = \sum_k \gamma_k(\lambda)\, R^{d+1+2k} \Delta_u^k \tau_u(x_0)$ for polynomial h. Then

$$\frac{\gamma_k(\lambda)}{-c_k/2} = \frac{d+2k+2-\lambda R}{2k+3},$$

which is independent of k if and only if $\lambda = (d-1)/R$. Thus δF_λ has the modular weight w_R at every order in R exactly for this coefficient.

The proof is in Appendix A.4.

Agreement of a geometric functional with the modular data at leading order in R does not fix λ; only the coefficient of the Einstein equation would change. Matching the full kernel fixes it to the fixed-reference multiplier. The family $F_R = A_R - (d-1)V_R/R$ itself is not stationary: $F_R[\eta] = \Omega_{d-1} R^{d-1}/d$, so its radius derivative is $\Omega_{d-1}(d-1)R^{d-2}/d$. These two variations must be distinguished.

4 Modular consistency and the linearized Einstein equation

Definition 4.1 (Modular consistency)**.** Let a state have smooth stress-tensor expectation $\delta\langle T\rangle$, let h be a metric perturbation, δF a ball functional and $\eta > 0$ a constant. The pair satisfies (MC_η) on a family of balls if

$$\eta\, \delta F_B[h] + \delta\langle K_B\rangle = 0 \tag{7}$$

for every ball of the family.

Here K_B is the bulk datum defined by (2). If the physical one-sided modular energy is $H_B = K_B + Q_B$, the same condition uses $\delta\langle H_B\rangle - \delta\langle Q_B\rangle$. It uses the raw physical energy only when the boundary variation vanishes.

The constant η has the dimension of an inverse area. For first-order perturbations with unchanged boundary charge, where $\delta S_B = \delta\langle K_B\rangle$, and with $\delta F = \delta F_B$ of (5), condition (7) says that $\eta A + S$ is stationary at fixed volume, which is Jacobson's entanglement equilibrium [19]. We keep the modular energy, not the entropy, in (7); Remark 4.8 and Section 6 explain why. In words, (MC_η) says that the modular energy of every ball is balanced by one geometric response of the ball, with one constant. If a boundary charge varies, the first-order entropic version uses $\delta S_B - \delta\langle Q_B\rangle$, not the uncorrected entropy variation.

4.1 Equivalence

Theorem 4.2 (Equivalence)**.** *Let $U \subset \mathbb{R}^{1,d}$ be open, and let $\delta\langle T\rangle$ and h be smooth on U. With δF the functional* (5)*, the following are equivalent.*

(a) (MC_η) *holds for all balls $\overline{B} \subset U$, in all rest frames and at all radii.*

(b) (MC_η) *holds to leading order in R at every $x_0 \in U$, for all u in some nonempty open set of rest frames.*

(c) $G^{\mathrm{lin}}_{\mu\nu} = \dfrac{2\pi}{\eta}\,\delta\langle T\rangle_{\mu\nu}$ *in U.*

Proof. (c)⇒(a): by Theorem 3.4, $\eta\,\delta F_B = -\eta\int_B w_R G^{\mathrm{lin}}_{uu} = -2\pi\int_B w_R\delta\langle T\rangle_{uu} = -\delta\langle K_B\rangle$. (a) ⇒(b) is immediate. (b)⇒(c): the leading terms of both sides of (7) carry the same factor $V_d R^{d+1}/(d+2)$ (Proposition A.1 and Theorem 3.4), so $\eta G^{\mathrm{lin}}_{uu}(x_0) = 2\pi\delta\langle T\rangle_{uu}(x_0)$ for an open set of u, and Corollary A.2 gives the tensor equation. □

Normalization. The factor $V_d/(d+2)$ appears on both sides and cancels, so the coupling is $2\pi/\eta$ in every dimension. With $\eta = 1/4G$ it is $8\pi G$: the 2π is the modular period of a ball and the $1/4$ is the Bekenstein–Hawking area density.

All radii from leading order. Agreement at leading order already forces the field equation, and the field equation makes the two sides agree at every radius. The finite-size corrections of the modular side and of the geometric side are therefore identical, term by term in (24); for a Fourier mode both have the transfer function f of (25). Given the field equation, the full ball-dependent functional agrees with the modular data on every ball; this is the converse direction.

Conservation and trace. The Einstein tensor is divergence-free by the Bianchi identity and $\delta\langle T\rangle$ is conserved and traceless by Proposition 2.1; the equation is consistent with both and requires $R^{\mathrm{lin}} = 0$ in U. Neither property is an additional premise.

Background. The background is Minkowski space, so no cosmological term appears. For diamonds in de Sitter or anti-de Sitter space the first law of Jacobson and Visser contains the cosmological constant and its variation [21]; the corresponding version of Theorem 3.4 is not derived here.

4.2 Uniqueness

The canonical identity holds for every metric perturbation. Identifying that response from data requires a source that excites more than the zero-frequency channel.

Lemma 4.3 (Conserved amplitudes under frame variation)**.** *Let $k \neq 0$, and let t, g be symmetric complex tensors with $k^\mu t_{\mu\nu} = k^\mu g_{\mu\nu} = 0$, $t \neq 0$. On a nonempty open set of future unit timelike vectors suppose $g_{uu} = a(q_u)t_{uu}$, where $q_u^2 = k^2 + (u\cdot k)^2$ and a is analytic on an interval of attained momenta. Then $g = a_0 t$ for a constant a_0. Tracelessness is not needed.*

Proof. For timelike k, choose its rest frame. Conservation sets $g_{0a} = t_{0a} = 0$. Write $u = (\sqrt{1+r^2}, rn)$. At fixed $r > 0$, equality on an open patch of the sphere implies $g_{ij} = a(q_r)t_{ij}$ by equality of quadratic forms. The nonzero tensor t fixes the same scalar as r varies.

For spacelike k, put it along the first spatial coordinate. Conservation sets $g_{1a} = t_{1a} = 0$. At fixed $u^1 = b$, the remaining coordinates run over the timelike hyperboloid of squared norm $-1-b^2$. A homogeneous quadratic form vanishing on an open patch of this hyperboloid vanishes on the open timelike cone by analytic continuation and homogeneity, hence identically. Thus $g = a(q_b)t$, with one scalar independent of b.

For null k, take $k \propto (1, 1, 0, \ldots)$. Conservation gives $g_{0a} + g_{1a} = t_{0a} + t_{1a} = 0$. At fixed $r = u^0 - u^1 > 0$, the transverse coordinates v vary freely, q_u is fixed, and

$$g_{uu} = g_{00} r^2 + 2 r g_{0A} v^A + g_{AB} v^A v^B .$$

Equality on an open patch is equality of these polynomials. It gives $g = a(q_r)t$; again $t \neq 0$ fixes the scalar. An open frame set contains patches with an interval of r or b, away from any isolated critical value of q_u. □

Theorem 4.4 (Nonzero-frequency identification)**.** *Fix $R > 0$ and let δF obey (F1)–(F6), with transfer function m_R. Suppose*

$$\delta\langle T\rangle_{ab}(x) = \mathrm{Re}(t_{ab} e^{ik\cdot x}), \qquad G^{\mathrm{lin}}_{ab}[h](x) = \mathrm{Re}(g_{ab} e^{ik\cdot x}),$$

where $k \neq 0$, $t \neq 0$, and t, g are conserved symmetric amplitudes. If (MC_η) holds on a nonempty open set of centres and on a nonempty open set of rest frames at this radius, then

$$m_R(q) = -\frac{c}{2}\widehat{w}_R(q) \quad \textit{for every } q, \qquad g_{ab} = \frac{2\pi}{c\eta} t_{ab}, \qquad c \in \mathbb{R} \setminus \{0\}.$$

Consequently $\delta F = c\, \delta F_B$ at this radius on arbitrary smooth metric inputs. If the same witness satisfies consistency at further radii, the same c applies at each tested radius. A hypothesis restricted to $R < R_0$ identifies only that range.

More generally let the spacetime Fourier transforms of $\delta\langle T\rangle$ and $G^{\mathrm{lin}}[h]$ be tempered tensor-valued complex Radon measures, meaning that their total variations are locally finite and have at most polynomial growth, with the source measure nonzero away from $k = 0$. With consistency at all spacetime centres and an open frame set, the response is identified at this radius and $G^{\mathrm{lin}}[h] = (2\pi/(c\eta))\delta\langle T\rangle$ as distributions. This includes continuous Fourier densities and sources stationary in one frame with a continuous spatial spectrum. General tempered distributions need a separate argument.

The proof is in Appendix A.5.

One radius is enough because changing the rest frame turns a nonzero spacetime frequency into an interval of spatial frequencies, and the compact support of the response supplies analytic continuation. Exact identification does not imply stable identification from finitely many noisy frames. The theorem concerns response laws tested by such data, rather than a physical derivation of the modular-consistency hypothesis.

Corollary 4.5 (The spectral criterion is necessary and sufficient)**.** *Let $d \geq 2$, fix a radius, and suppose a nonzero smooth source and a smooth metric satisfy (MC_η) for some functional obeying (F1)–(F6), at every centre and on an open set of frames. Suppose both tensor Fourier transforms belong to the tempered Radon class of Theorem 4.4. The data identify the canonical functional up to normalization if and only if the source measure has nonzero weight away from zero. If it is supported at zero, both tensors are constant and the data test only $m_R(0)$.*

The proof is in Appendix A.6.

The nonzero-source hypothesis excludes a degenerate experiment: if both source and Einstein tensor vanish, every admissible functional fits. Polynomial sources fall outside the Radon class; their Fourier transforms contain derivatives of δ_0. A polynomial tensor of degree n probes only the finitely many coefficients m_k with $2k \leq n$, so it also leaves higher spatial derivatives undetermined.

Example 4.6 (A homogeneous source leaves the response open)**.** In $D = d + 1 \geq 3$ spacetime dimensions put $T_{ab} = \mathrm{diag}(d, 1, \ldots, 1)$, $\kappa = 2\pi/\eta$, $x^2 = \eta_{ab}x^a x^b$, and $Txx = T_{ab}x^a x^b$. The quadratic metric

$$h_{ab} = -\frac{\kappa}{2(D-2)}\big(T_{ab}x^2 + \eta_{ab}Txx\big)$$

has $R^{\mathrm{lin}} = 0$ and $G^{\mathrm{lin}}_{ab} = \kappa T_{ab}$. For every $\ell \neq 0$, the intrinsic functional

$$\delta F_B^{\mathrm{new}}[h] = \delta F_B[h] + \ell^2 \int_B w_R \Delta_u \tau_u[h]\, d^d x$$

obeys (F1)–(F6) and satisfies exactly the same consistency condition on this pair at every centre, radius and frame. Its extra term vanishes because G^{lin} is constant. It is not a multiple of the canonical functional: its transfer function is $(-\frac{1}{2} - \ell^2 q^2)\widehat{w}_R(q)$. A homogeneous source tests only the zero-momentum response.

The Einstein tensor appears through intrinsic slice curvature and the Gauss equation. Nonzero-frequency data then identify its two-derivative response: no independent assumption excluding internal length scales is needed. Actual source preparation is therefore part of what makes a geometric law identifiable.

4.3 What fails without each hypothesis

Proposition 4.7. *(a)* One order in R is not enough. *The area of the geodesic sphere at fixed radius, $\lambda = 0$ in Proposition 3.5, satisfies* (MC_η) *to leading order with $G^{\mathrm{lin}} = \frac{6\pi}{(d+2)\eta}\delta\langle T\rangle$, but its coefficient of order R^{d+3} differs from the modular one by the factor $1 - \frac{2(d-1)}{5(d+2)}$ (-16% for $d = 3$). Consistency at that order would require $\Delta_u \delta\langle T\rangle_{uu} = 0$ in every frame. For a Fourier mode the two transfer functions differ by 0.29%, 1.2%, 5.2% and 14% at $R|k_u| = 0.5, 1, 2, 3$ ($d = 3$).*

(b) Reading off the slice. *The functional $\delta F_B^{(\ell)} = -\int_B w_R\,(G^{\mathrm{lin}}_{uu} - \ell^2 \Box G^{\mathrm{lin}}_{uu})$ satisfies (F1) and (F3)–(F6) but not (F2).* (MC_η) *holds for it at all radii and in all frames if and only if $(1 - \ell^2\Box)G^{\mathrm{lin}}_{\mu\nu} = (2\pi/\eta)\delta\langle T\rangle_{\mu\nu}$, a fourth-order equation.*

(c) Reading only the trace-free part. *The functional $-\int_B w_R\,(R^{\mathrm{lin}}_{uu} + R^{\mathrm{lin}}/(d+1))$ gives $R^{\mathrm{lin}}_{\mu\nu} - \frac{1}{d+1}\eta_{\mu\nu}R^{\mathrm{lin}} = (2\pi/\eta)\delta\langle T\rangle_{\mu\nu}$; the Bianchi identity makes R^{lin} constant, and the result is $G^{\mathrm{lin}}_{\mu\nu} + \Lambda\eta_{\mu\nu} = (2\pi/\eta)\delta\langle T\rangle_{\mu\nu}$ with an integration constant $\Lambda = \frac{d-1}{2(d+1)}R^{\mathrm{lin}}$.*

Proof. (a) From Proposition 3.5(b) with $\lambda = 0$ the ratios to the modular coefficients are $(d + 2)/3$ and $(d+4)/5$ at the first two orders; the transfer functions are computed from the linearized geodesic ball. (b) and (c) follow as in Theorem 4.2, with the operator inside the integral in place of G^{lin}_{uu}; in (c), $\partial^\mu R^{\mathrm{lin}}_{\mu\nu} = \frac{1}{2}\partial_\nu R^{\mathrm{lin}}$ and $\partial^\mu \delta\langle T\rangle_{\mu\nu} = 0$ give $\big(\frac{1}{2} - \frac{1}{d+1}\big)\partial_\nu R^{\mathrm{lin}} = 0$. □

Part (b) is the general lesson. Higher-derivative laws arise when the geometric response of a region depends on how the geometry evolves away from the region's slice; the analysis of higher-curvature entanglement equilibrium by Bueno et al. fits this pattern [25]. Part (c) shows where a cosmological constant could enter as an integration constant: only through a functional blind to the trace, which the intrinsic one is not.

Remark 4.8 (Energy, not entropy). For a unitary $W \in \mathcal{A}(D(B))$ with unchanged boundary charge, the entropy of the ball does not change at a regulator preserving that local unitary, while $\delta\langle K_B\rangle = S_B(\omega_W\|\omega) \geq 0$, with strict inequality whenever the local state changes, by (3).

For a coherent excitation strictly local to $D(B)$, the modular energy and the relative entropy of that ball are equal and of second order in the amplitude. A condition written with entropy, $\eta\,\delta F_B + \delta S_B = 0$, would then require $\delta F_B = 0$ on that ball, in conflict with the semiclassical Einstein response to a nontrivial preparation. Condition (7) has no such problem. Entanglement equilibrium agrees with it for perturbations whose modular energy is of first order, and only for those.

5 What an observer's ball data fix

An observer does not have all balls. It has balls near its worldline, of sizes between a resolution and a scale of interest, in the rest frames it can compare. This section asks what such data fix through (MC_η), and with what stability.

5.1 One rest frame

Proposition 5.1 (One rest frame gives the Hamiltonian constraint)**.** *Fix u.*

(a) (MC_η) *for all balls in the frame u, at all centres in U and all radii, is equivalent to $G^{\mathrm{lin}}_{uu} = (2\pi/\eta)\delta\langle T\rangle_{uu}$ in U.*

(b) Let $\delta\langle T\rangle$ be stationary in the frame u, and let $\delta\langle T\rangle'$ be its time reverse, $T'_{00} = T_{00}$, $T'_{0i} = -T_{0i}$, $T'_{ij} = T_{ij}$. Then $\delta\langle T\rangle'$ is conserved, traceless and stationary, and the modular energies of all balls at rest in u are the same for both sources. For a ball boosted with rapidity β in the direction n they differ by $2\pi\int_B w_R\, 2\sinh(2\beta)\, n^i T_{0i}$.

(c) Consequently, if h satisfies $G^{\mathrm{lin}} = (2\pi/\eta)\delta\langle T\rangle$, the pair $(\delta\langle T\rangle', h)$ satisfies (MC_η) *for every ball at rest in u, although $G^{\mathrm{lin}}_{0i} = -(2\pi/\eta)T'_{0i}$.*

Proof. (a) is Theorem 4.2 with a single frame, where the leading term gives only the uu component. (b) Stationarity and conservation give $\partial_i T_{0i} = -\partial_t T_{00} = 0$ and $\partial_j T_{ij} = -\partial_t T_{0i} = 0$, which are the conservation equations of $\delta\langle T\rangle'$; the trace is unchanged. Balls at rest in u read only T_{00}. For $u = (\cosh\beta, \sinh\beta\, n)$, $T_{uu} - T'_{uu} = 4\cosh\beta\sinh\beta\, n^i T_{0i}$. (c) follows from (a) and (b). □

A body rotating one way and the same body rotating the other way have identical ball data in their common rest frame, and opposite frame dragging. The momentum constraint and the spatial components of the field equation, which carry the dynamics, are the part of the geometry that only comparison of rest frames reveals. In this precise sense the multi-directional character of the data is where the gravitational dynamics sits.

5.2 Several rest frames and their stability

By Corollary A.2 any open set of rest frames suffices in exact arithmetic. With noise and finite balls the question is quantitative.

Proposition 5.2 (Rapidity sets the precision)**.** *Let the observer use its rest frame and boosts of rapidity β in a fixed set of directions for which the leading-order readings determine a traceless symmetric tensor, and let each normalized reading $y_a = \delta\langle T\rangle_{u_a u_a}(x_0) + e_a$ carry an independent error of variance σ^2. As $\beta \to 0$ the least-squares estimate has root-mean-square error of order σ in T_{00}, σ/β in T_{0i} and σ/β^2 in T_{ij}.*

Proof. With $u_a = (1+\frac{\beta^2}{2}, \beta n_a)+O(\beta^3)$, $y_a = T_{00}+2\beta\, n_a^i T_{0i}+\beta^2(T_{00}+n_a^i n_a^j T_{ij})+O(\beta^3)+e_a$. The columns of the design matrix for (T_{00}, T_{0i}, T_{ij}) scale as $1, \beta, \beta^2$, and for a direction set whose leading blocks are nondegenerate the inverse scales as $1, \beta^{-1}, \beta^{-2}$. □

Finite balls add a bias. A reading at radius R multiplies each Fourier component by $f(R|k_u|)$ (Proposition A.1), and $|k_u|$ grows like e^β for large boosts. For the stresses the bias then grows like $R^2 e^{2\beta}|k|^2$ while the noise falls like $\sigma e^{-2\beta}$, and the best error obtainable with one radius scales as $\sqrt{\sigma}\, R\, |k|$. Table 1 shows both regimes for one mode in $d = 3$, with thirteen frames (the rest frame and boosts along the six coordinate half-axes and six face diagonals). A finer scan gives an optimal error proportional to $R^{1.03}$ when noise dominates and to $\sigma^{0.48}$ between $\sigma = 10^{-4}$ and 10^{-2} at $R = 0.02L$; at very small σ the rest-frame bias, proportional to R^2, dominates. Since the transfer function is known exactly, readings at several radii can remove the finite-size bias. What cannot be removed is the amplification of noise by a small range of rapidities.

	$\sigma = 10^{-4}$			$\sigma = 10^{-2}$		
β	T_{00}	T_{0i}	T_{ij}	T_{00}	T_{0i}	T_{ij}
0.05	3.2×10^{-5}	1.1×10^{-3}	8.8×10^{-2}	3.3×10^{-3}	1.0×10^{-1}	9.3
0.2	3.1×10^{-5}	2.7×10^{-4}	5.8×10^{-3}	3.1×10^{-3}	2.7×10^{-2}	5.8×10^{-1}
0.7	3.0×10^{-5}	7.6×10^{-5}	4.3×10^{-4}	1.9×10^{-3}	5.3×10^{-3}	3.8×10^{-2}
1.0	5.3×10^{-5}	9.5×10^{-5}	3.4×10^{-4}	1.1×10^{-3}	3.1×10^{-3}	1.7×10^{-2}
2.0	4.5×10^{-4}	7.3×10^{-4}	2.2×10^{-3}	4.6×10^{-4}	8.2×10^{-4}	2.7×10^{-3}

Table 1: Root-mean-square reconstruction error of a unit-norm traceless conserved source $\mathrm{Re}(\varepsilon_{\mu\nu} e^{ik\cdot x})$, $k = (1, 0.6, 0.3, -0.2)/L$, from leading-order readings of balls of radius $R = 0.05L$ in thirteen frames of rapidity β, 200 noise draws.

5.3 Free radiation

By Corollary 3.3, (MC_η) fixes h only up to solutions of $G^{\mathrm{lin}} = 0$. For $d \geq 3$ these include gauge modes and free gravitational waves. This is exactly the freedom that the Einstein equation itself leaves. No ball functional registers a free wave at linear order, in any frame and at any radius. At second order its quadratic geometry enters $G^{(2)}[h, h]$. Moving that term to an effective source is an additional convention, not an extra matter stress tensor. Shape-sensitive regions can also probe tidal data: a deformed region couples to the tidal part of the curvature. In $2+1$ dimensions the Riemann tensor is algebraically determined by the Ricci tensor, so $G^{\mathrm{lin}} = 0$ implies a locally pure-gauge perturbation on a contractible domain. Global identifications and boundary conditions can still distinguish metrics.

5.4 A locally prepared source and its boundary datum

The source witness of Theorem 4.4 can be prepared by a local operation. For a free scalar one must also specify which part of its modular energy is being read. We use the scalar boundary prescription of Casini, Mazzitelli and Testé [8, Eq. (58)], with the boundary-condition derivation of Herzog and Nishioka [11, Secs. 1–2]. The prescription agrees with thermal numerical calculations and Rényi-entropy checks [9, 10]. For the conformal improved stress tensor it gives

$$H_B = K_B + Q_B, \qquad Q_B = 2\pi\xi \int_{\partial B} \sum_a :\phi_a^2:\, dA, \qquad \xi = \frac{d-1}{4d}. \tag{8}$$

There is one real component for a real scalar and two for a complex scalar. The formula uses vacuum-subtracted expectations. A surface expectation is defined as the limit of smooth collar smearings of the normal-ordered field square, whose weights converge to surface measure. No sharp boundary operator on arbitrary states is assumed.

Proposition 5.3 (Local preparation supplies a nonzero spectral witness)**.** *In the free massless complex scalar vacuum, let f be a real smooth compactly supported spacetime function with nonzero one-particle class. Put $W_\theta = \exp[i\theta\phi_1(f)]$, $\theta \neq 0$, and $\varphi_f = \Delta_{\mathrm{PJ}} f$, with the sign convention fixed so that $\omega_{W_\theta}(\phi_1(x)) = \theta\varphi_f(x)$. Then*

$$\delta\langle T\rangle_{\mu\nu} = \theta^2 \left[\partial_\mu\varphi_f\partial_\nu\varphi_f - \frac{1}{2}\eta_{\mu\nu}(\partial\varphi_f)^2 + \xi(\eta_{\mu\nu}\Box - \partial_\mu\partial_\nu)\varphi_f^2\right]. \tag{9}$$

It is smooth, conserved and traceless, has finite strictly positive total spatial energy, and has a finite tensor-valued Radon Fourier measure with nonzero mass away from $k = 0$. All polynomial moments of that measure are finite. Its boundary-energy variation is

$$\delta\langle Q_B\rangle = 2\pi\xi\theta^2 \int_{\partial B} \varphi_f^2 \, dA. \tag{10}$$

If supp *f is compactly contained in $D(B)$, this term vanishes for that ball. It need not vanish for the same state on other balls.*

The proof is in Appendix B.1.

Example 5.4 (A common state, source and bulk response)**.** Take the prepared state above and $\kappa = 2\pi/\eta$. A smooth linearized response $G^{\mathrm{lin}}[h] = \kappa\delta\langle T\rangle$ exists with constraint-compatible initial data. For example, write $\bar{h}_{\mu\nu} = h_{\mu\nu} - \eta_{\mu\nu}h^\alpha{}_\alpha/2$ and solve $\Box\bar{h}_{\mu\nu} = -2\kappa\delta\langle T\rangle_{\mu\nu}$. At a fixed time t_0, choose smooth Poisson solutions

$$\Delta a = -2\kappa\delta\langle T\rangle_{00}(t_0,\cdot), \qquad \Delta b_j = 2\kappa\delta\langle T\rangle_{0j}(t_0,\cdot).$$

Set $\bar{h}_{00} = a$, all other initial components zero, $\partial_t\bar{h}_{0\nu} = 0$, and $\partial_t\bar{h}_{ij} = \partial_i b_j + \partial_j b_i - \delta_{ij}\partial_\ell b_\ell$. The Lorenz constraint $C_\nu = \partial^\mu\bar{h}_{\mu\nu}$ and its first time derivative vanish initially. Conservation gives $\Box C_\nu = 0$, so they vanish throughout the linear evolution and $G^{\mathrm{lin}}[h] = \kappa\delta\langle T\rangle$. Compact smooth sources on a slice admit these Poisson solutions; in $d = 2$ logarithmic behaviour at spatial infinity is allowed. The construction is smooth and tempered, and is perturbatively small on each fixed compact domain as $\theta \to 0$.

For this pair Theorem 4.2 gives, for every ball,

$$\eta\,\delta F_B[h] + \delta\langle H_B\rangle - \delta\langle Q_B\rangle = 0. \tag{11}$$

Both $\widehat{\delta\langle T\rangle}$ and $\widehat{G^{\mathrm{lin}}[h]} = \kappa\widehat{\delta\langle T\rangle}$ meet the Radon hypotheses. Hence one radius, all centres and an open frame family identify the canonical intrinsic response from these edge-subtracted readings. The matter state, its physical source and the response law coexist here; the Einstein coupling and the constraint-compatible gravitational data were supplied, not generated by the Weyl preparation.

The subtraction has operational content. It requires access to calibrated collar readings of $\sum_a :\phi_a^2:$, in addition to the modular energy or the bulk stress readings. Their limiting expectations are finite in this coherent family, but no finite sample, resolution or preparation cost for the collar limit has been derived. The total-energy formula counts the prepared radiation, not the switching, phase-reference or reset costs of the source apparatus. If the bulk stress charge is read directly, the subtraction is unnecessary; calling that charge the raw physical H_B would still be incorrect.

Corollary 5.5 (Raw scalar energy cannot obey the bulk law on all small balls)**.** *At every centre x_0 with $\varphi_f(x_0) \neq 0$, and in any one fixed rest frame, no smooth metric perturbation satisfies $\eta\,\delta F_B[h] + \delta\langle H_B\rangle = 0$ for all sufficiently small radii. Thus the obstruction is visible at each point of a nonempty open set for every nonzero coherent preparation.*

Proof. At a point with $\varphi_f(x_0) \neq 0$,

$$\delta\langle Q_B\rangle = 2\pi\xi\theta^2\Omega_{d-1}R^{d-1}\varphi_f(x_0)^2 + O(R^{d+1}).$$

The bulk charge and the canonical intrinsic geometric variation are both $O(R^{d+1})$ for smooth source and metric fields. They cannot cancel the nonzero R^{d-1} term at that centre, even without varying the rest frame. □

Remark 5.6 (The same curvature coupling fixes the surface subtraction). The boundary coefficient has a gravitational counterpart, not merely a readout calibration. For the nonminimally coupled scalar Lagrangian

$$\mathcal{L} = \left(\frac{1}{16\pi G} - \frac{\xi}{2}\Phi^2\right)\mathcal{R} - \frac{1}{2}\sum_a (\nabla\phi_a)^2, \qquad \Phi^2 = \sum_a \phi_a^2,$$

the binormal curvature derivative gives the Wald surface density [20, Eqs. (53), (93)]

$$S_{\mathrm{W}}[\Sigma] = \int_\Sigma \left(\frac{1}{4G} - 2\pi\xi\Phi^2\right) dA.$$

The nonminimal scalar contribution and its relation to the entropy contact term are established results; see, for example, Donnelly and Wall [12, Eq. (5), Sec. II.A]. Here the same coefficient is used in the ball-family consistency law. Indeed, for the curvature coefficient c, $\partial\mathcal{L}/\partial R_{abcd} = c(g^{ac}g^{bd} - g^{ad}g^{bc})/2$; contraction with a binormal of norm -2 and multiplication by -2π give $4\pi c$. In the coherent witness the vacuum-subtracted quadratic profile is $\delta\langle\Phi^2\rangle = \theta^2\varphi_f^2$. At the retained order, with $h = O(\theta^2)$, its surface-measure correction is $O(\theta^4)$. For $\eta = 1/(4G)$, the Wald-matched ball quantity, with the same supplied volume term as F_B, is consequently

$$\mathcal{B}_B := \delta S_{\mathrm{W}}[\partial B] - \eta\frac{d-1}{R}\delta V(B) = \eta\,\delta F_B - \delta\langle Q_B\rangle.$$

Thus $\mathcal{B}_B + \delta\langle H_B\rangle = \eta\,\delta F_B + \delta\langle K_B\rangle$: the bulk consistency law is exactly restored. This matches a local surface density and fixes its matter coefficient from the same curvature coupling. It is a coefficient identity for the ball consistency law, not a thermodynamic first law for arbitrary diamonds.[1] The pure-area/raw-energy obstruction above is unchanged.

6 Which algebra the observer has

Consciousness I and Foundation I select an observer's algebra by noise and by a readable seed, and the selected algebra is in general smaller than the full local algebra [1, 4]. HAFF-D conjectures that gravitational phenomena are tied to the selection of observable algebras [3]. In the ball family the question is precise: does the geometry fixed by (MC_η) depend on which algebra is selected?

[1] An arbitrary ball edge is not a Killing horizon; MC, a generalized volume and a nonlinear second law do not follow from this density match.

For a bounded collection of scalar preparations and tested stress or collar smearings, choose a containing diamond D_0. The $U(1)$ Haar expectation E on this diamond preserves the vacuum and fixes the invariant readings. The semigroup $\mathcal{E}_t = e^{-\gamma t}\mathrm{id} + (1 - e^{-\gamma t})E$ is vacuum-stationary. Foundation I's stationary-source and stable-seed theorems [4] place the modular hull of any such invariant seed inside the proper fixed algebra. The charged coherent preparation above requires a separate phase-referenced source. A neutral quadratic seed supplies another preparation, entirely inside the same fixed algebra.

Proposition 6.1 (A neutral preparation inside a stable hull)**.** *In the free complex scalar theory of Proposition 5.3, let $X_a = \phi_a(f)$, $V_f = \omega(X_a^2) > 0$, and*

$$Q_f = \tfrac{1}{2}(X_1^2 + X_2^2), \qquad U_\theta = e^{i\theta Q_f}.$$

The strongly commuting fields X_1, X_2 define a self-adjoint Q_f by their joint spectral calculus. Both Q_f and U_θ are $U(1)$-invariant and local to any diamond containing supp f*. Include* arctan Q_f*, together with the desired bounded stress and collar readings, in the stable seed above. Its hull contains U_θ for every θ, while remaining inside the proper fixed algebra.*

Put $v_f = \varphi_f$, and define $H_f(x) = \frac{1}{2}\omega(\{X_a, \phi_a(x)\})$ for either fixed component $a = 1, 2$, with no sum over a; internal symmetry makes the two functions equal. Let $t[v]$ denote the quadratic tensor in brackets in (9)*, and polarize it as $t[v, z] = (t[v + z] - t[v] - t[z])/2$. The exact source is*

$$\delta\langle T\rangle_{\mu\nu} = 4\theta\, t_{\mu\nu}[v_f, H_f] + 2\theta^2 V_f\, t_{\mu\nu}[v_f]. \tag{12}$$

For every $\theta \neq 0$, it is smooth, conserved and traceless, has compact support on each spatial slice and has a finite Radon Fourier measure with all polynomial moments finite and nonzero weight away from zero. Its total energy is

$$\int \delta\langle T\rangle_{00}\, d^d x = \theta^2 V_f \int \left[(\partial_t v_f)^2 + |\nabla v_f|^2\right] d^d x > 0.$$

Its boundary datum is $\delta\langle\Phi^2\rangle = 4\theta v_f H_f + 2\theta^2 V_f v_f^2$.

The proof is in Appendix B.2.

The constraint construction of Example 5.4 gives $G^{\mathrm{lin}}[h] = (2\pi/\eta)\delta\langle T\rangle$, so this preparation identifies the canonical response. Its inputs are permission to apply the neutral quadratic pulse and the complete centre and frame family. Containing the stress tensor need not supply Q_f. Nor does neutrality by itself guarantee a nonzero tangent: the local conformal Wick square $:\Phi^2:(g)$ can have zero first-order vacuum stress response. The spatially smeared square used here is a different operator, and its positive finite-amplitude energy is computed explicitly. The noise fixes the invariant readings, although the prepared state need not be stationary.

There is a more general finite-energy route that uses a bounded observable already in the selected algebra. It needs no particular quadratic seed. Local source regularity remains a separate condition.

Proposition 6.2 (A bounded accessible preparation with positive energy)**.** *Let $H \geq 0$ be the translation generator, with* ker $H = \mathbb{C}\Omega$*. Let $\mathcal{N}$ be a selected von Neumann algebra for which Ω is separating. Suppose it contains a bounded non-scalar self-adjoint C, together with $\alpha_t(C) = e^{itH} C e^{-itH}$ for $0 < t < \tau$, where $\tau > 0$. For a sufficiently narrow real nonnegative $f \in C_c^\infty$ in that interval, $\int f = 1$, put*

$$B = \int f(t)\alpha_t(C)dt, \qquad U_\theta = e^{i\theta B}.$$

Then B is bounded, self-adjoint and non-scalar, $U_\theta \in \mathcal{N}$, and its vacuum excitation has finite energy for every θ. For every θ for which U_θ is non-scalar, in particular $0 < |\theta|\|B\| < \pi$,

$$0 < E_\theta := \langle U_\theta\Omega, HU_\theta\Omega\rangle \le |\theta|\|[H, B]\|.$$

With $K = [H, B]$, $\psi = (C - \langle\Omega, C\Omega\rangle)\Omega$ and the spectral measure μ_ψ of H, the quadratic coefficient is

$$\begin{aligned} E_2 &= \langle B\Omega, HB\Omega\rangle = \tfrac{1}{2}\langle\Omega, [B, [H, B]]\Omega\rangle \\ &= \int_{(0,\infty)} \lambda|\widehat{f}(\lambda)|^2\, d\mu_\psi(\lambda) > 0, \qquad E_2 \le \|B\|\|K\|, \end{aligned} \tag{13}$$

where $\widehat{f}(\lambda) = \int f(t)e^{it\lambda}dt$. An explicit remainder is

$$|E_\theta - \theta^2 E_2| \le 2|\theta|^3\|B\|^2\|K\| + \tfrac{1}{2}\theta^4\|B\|^3\|K\|.$$

If the state's vacuum-subtracted energy density is integrable on a time slice and its integral is E_θ, this gives a nonzero integrated source. Implementation permission and its cost are additional inputs.

The proof is in Appendix B.3.

The smoothed operator has bounded iterated commutators $\mathrm{ad}_H^m(B) = i^m \int f^{(m)}(t)\alpha_t(C)dt$. Repeated use of the product rule in the exponential series therefore gives $U_\theta\Omega \in \bigcap_m \mathrm{Dom}\, H^m$; all energy moments are finite. If μ_ψ has no atoms, narrowness is unnecessary: the nonzero entire function $\widehat{f}$ has only isolated zeros, which cannot carry that spectral measure. With atoms, a broad smearing can cancel an excitation exactly, so this refinement requires the stated spectral condition.

Corollary 6.3 (A common pulse inside a shared hull)**.** *Let several selected algebras $\mathcal{N}_j$ act in one vacuum representation with one positive generator H,* $\ker H = \mathbb{C}\Omega$*, and Ω separating for an ambient algebra containing them. Suppose each is invariant under* $\mathrm{Ad}\, e^{itH}$ *for $t \ge 0$. If their intersection $\mathcal{N}_\cap$ is non-scalar, it contains one common bounded pulse U_θ with finite strictly positive H-energy.*

Proof. The intersection is a von Neumann algebra, inherits separation and forward translation invariance, and contains a bounded non-scalar self-adjoint C. Apply Proposition 6.2 in $\mathcal{N}_\cap$. □

Proposition 6.4 (Finite algebras cannot carry this vacuum permission)**.** *Let $\mathcal{N}$ be finite-dimensional, let $H \ge 0$ have* $\ker H = \mathbb{C}\Omega$*, and suppose Ω is separating for $\mathcal{N}$. If* $\mathrm{Ad}\, e^{itH}(\mathcal{N}) \subseteq \mathcal{N}$ *for every $t \ge 0$, then $\mathcal{N} = \mathbb{C}1$. The ambient Hilbert space need not be finite-dimensional.*

Proof. Injectivity and finite dimension turn each inclusion into equality, so the automorphisms extend to negative times. Continuity fixes every minimal central projection. On each matrix block choose a self-adjoint generator K and matrix units E_{ab} in its eigenbasis: $\alpha_t(E_{ab}) = e^{it(k_a - k_b)}E_{ab}$. Since $H\Omega = 0$ and separation makes every nonzero $E_{ab}\Omega$ nonzero, this vector is an H-eigenvector with eigenvalue $k_a - k_b$. Positivity applied to E_{ab} and E_{ba} makes all differences zero. Thus the automorphisms fix $\mathcal{N}$, and $C\Omega$ lies in the one-dimensional kernel of H for every $C \in \mathcal{N}$. Separation then gives $C = c1$. □

The common pulse of Corollary 6.3 consequently uses an infinite field algebra when its full forward permission is exact. A finite observer can implement a finite-duration pulse or an explicitly controlled approximation; its controller Hamiltonian cannot simply be identified with the vacuum translation generator. The bounded-pulse construction requires only its stated finite interval and is unaffected.

Foundation I's Γ_3-hulls have precisely this forward translation closure [4, Theorem on Cut inheritance]. Thus a non-scalar shared intersection of those hulls, with the common representation hypotheses above, supplies a common energy-orientation test without adding Q_f. The positive generator must be the same for the readers: in a null Cut model it is null momentum, not automatically the time-translation Hamiltonian of this higher-dimensional theory. A merely modular-stable Γ_2-hull does not inherit translation permission. Realizing the pulse and reading its energy still require controls and an energy ledger.

There is a common physical-H instance in $3+1$ dimensions. For two independent free conformal complex scalar nets, take the future-cone algebra $\mathcal{A}(V_+)$ and the two selected nets $\mathcal{A}_1^{U(1)}\bar{\otimes}\mathcal{A}_2$ and $\mathcal{A}_1\bar{\otimes}\mathcal{A}_2^{U(1)}$. Time translations move V_+ into itself, the vacuum is separating there, and the physical H_1+H_2 has unique zero-energy vacuum [13, 14]. Each fixed algebra is taken on its own vacuum-cyclic subspace. The same physical Hamiltonian restricts to it because the Haar projection commutes with translations. Their intersection contains the neutral preparation of Proposition 6.1, the total stress and the invariant collar readings at each tested diamond. Thus the reference cone algebra and the selected ball family coexist in one matter model; they are distinct region algebras, not all operators inside one cone. The generic bounded pulse keeps its separate source-regularity condition. Shared implementation, calibrated collar limits and MC_η remain supplied permissions and consistency data.

Corollary 6.5 (A positive pulse supplies the spectral witness)**.** *Suppose a pulse from Proposition 6.2 has a smooth source $\delta\langle T\rangle$ in the tempered Radon Fourier class, and its vacuum-subtracted energy density is integrable on a time slice with integral $E_\theta > 0$. Then its source Fourier measure has nonzero weight away from $k=0$. If the response also belongs to that class and the full centre and open-frame consistency hypotheses of Theorem 4.4 hold, this source identifies the canonical response up to normalization.*

Proof. A Radon measure supported only at zero is a multiple of δ_0; the source would be constant. Its energy density is then constant on the infinite spatial slice. Integrability forces that component to vanish, contradicting $E_\theta > 0$. The identification follows from Theorem 4.4. □

This witness uses finite amplitude, not necessarily a nonzero first-order stress tangent. In fact $\partial_\theta E_\theta|_0 = i\langle\Omega,[H,B]\Omega\rangle = 0$ for every such vacuum pulse. A local tangent depends separately on $-2\,\mathrm{Im}\langle\Omega,T(x)B\Omega\rangle$. Proposition 6.1 computes both the regularity and a source explicitly; the general bounded construction does not supply Radon regularity from finite energy moments alone. Neither route establishes finite-resource identification of the entire geometric response.

Proposition 6.6 (Selection is invisible to bulk modular energy)**.** *For each ball let $\mathcal{A}_{\mathrm{sel}}(B)\subset\mathcal{A}(D(B))$ be a von Neumann subalgebra with a normal conditional expectation E_B that preserves the vacuum state, and suppose $\mathcal{A}_{\mathrm{sel}}(B)$ contains the bounded functions of stress-tensor smearings in $D(B)$; fixed-point algebras of a compact internal symmetry are examples. Then the modular group of $(\mathcal{A}_{\mathrm{sel}}(B),\omega)$ is the restriction of that of $\mathcal{A}(D(B))$, the quantity $\delta\langle K_B\rangle = 2\pi\int_B w_R\,\psi(T_{uu})$ is determined by the restriction of the state ψ to $\mathcal{A}_{\mathrm{sel}}(B)$, and (MC_η) is the same condition for every such choice. Every such selected algebra therefore fixes the same matter-sourced Einstein tensor. For $d\geq 3$, the metric additionally requires common initial*

or boundary data and is defined modulo gauge; for $d = 2$, it is locally fixed up to gauge on contractible domains. If the physical energy has an extra boundary term, the bulk datum must be read directly or that term must also be accessible and subtracted.

Proof. The first statement is Takesaki's theorem on conditional expectations [26]. The second holds because the stress-tensor smearings are affiliated with $\mathcal{A}_{\text{sel}}(B)$. The last follows from Theorem 4.2. □

Relative entropy behaves differently, as Foundation I found for current excitations of the SU(2) level-one net [4]. The same argument works in any dimension.

Proposition 6.7 (Selection changes entropy)**.** *Let $\mathcal{A}$ be the net of a free massless conformally coupled complex scalar $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$ with its internal $U(1)$, let $\mathcal{A}_{\text{sel}}(B)$ be the $U(1)$-fixed points of $\mathcal{A}(D(B))$ and E_B the average over the group, and let $W = e^{i\theta\phi_1(f)}$, $\theta \neq 0$, where f is real, supported in $D(B)$, and has nonzero one-particle class. A nonzero test function alone would not suffice, since $f = \Box g$ can represent the zero field.*

(a) $S_{\mathcal{A}_{\text{sel}}(B)}(\omega_W \| \omega) < S_{\mathcal{A}(D(B))}(\omega_W \| \omega) = \delta\langle K_B \rangle_{\omega_W}$.

(b) As $\theta \to 0$, the modular energy is quadratic in θ, and the restricted state has vanishing first variation. At a faithful finite-dimensional regulator, write $\rho_\theta^{\text{inv}} = \rho_0 + \theta^2\chi + O(\theta^4)$. Then

$$S(\rho_\theta^{\text{inv}} \| \rho_0) = \frac{\theta^4}{2} g_{\text{BKM},\rho_0}(\chi, \chi) + O(\theta^6).$$

The continuum quartic law additionally requires a finite BKM tangent norm and uniform control of the entropy remainder; smooth field smearing and faithfulness alone do not establish these conditions.

The proof is in Appendix B.4.

A local coherent displacement preserves the regulated entropy of the full algebra. Its modular energy equals its relative entropy. The invariant algebra strictly loses relative entropy while retaining the stress charge; when the quartic expansion is controlled, its entropy increase exhausts that charge to order θ^2.

Appendix C computes the invariant mode coefficient and separates the exponential same-charge contribution from neutral-pair terms. A bounded BKM tangent norm is only one requirement: a continuum quartic entropy law still needs uniform control of the remainder as the regulators are removed. The strict entropy inequality above is independent of that expansion.

Conjecture 6.8 (Selection enters through second-order quantities or through the source)**.** *In the conformal ball setting, a change of the selected algebra at a fixed state changes the geometry reconstructed from ball modular data only through quantities of second order in the state perturbation, such as relative entropy, Fisher information and canonical energy, or through physical operations that change the stress tensor.*

Mechanism. The first-order datum is a function of the restriction of the state to stress-tensor observables, which every relevant algebra contains (Proposition 6.6). What sees more of the state is relative entropy, which depends on the algebra by monotonicity, and in holographic theories its second-order part for balls is the canonical energy that carries the second-order gravitational equations [28, 29].

Support. Propositions 6.6 and 6.7; Foundation I's strict inequality in $1 + 1$ dimensions [4]; the result of HAFF-D that a record alone does not move a gravitational source, while a preparation does [3].

Strongest objection. At second order the world still has one metric. The relative entropy of a smaller algebra measures what its observer cannot distinguish, so a second-order condition built from it may describe a coarser reconstruction rather than a different geometry. If so, the conjecture is a statement about the reach of an observer's reconstruction, which an observer-relative account of geometry has to quantify anyway, and not about a second spacetime.

Test. Compute $S_{\mathcal{A}(D(B))} - S_{\mathcal{A}_{\mathrm{sel}}(B)}$ for the charged coherent excitation of Proposition 6.7 through order θ^4, and compare the second-order consistency conditions built from the two relative entropies.

7 Second-order geometry and a source that follows it

The linear identity holds at every radius. At second order two of its inputs change: the intrinsic curvature need not equal twice the normal Einstein tensor, and the flat CHM charge need not be the modular Hamiltonian of the curved theory. We can resolve the first issue and identify the source terms needed for the second. This does not supply a nonlinear modular law.

7.1 A geometric two-jet consistency law

Put $c_0 = V_d/(d+2)$. Let g_ϵ be a smooth family of Lorentzian metrics with $g_0 = \eta$. At p, choose smooth spacelike hypersurfaces $\Sigma_{\epsilon,u}$ whose flat member is the hyperplane normal to u. Let n_ϵ be the future unit normal, q_ϵ its induced metric, and K_ϵ its second fundamental form. The family, including its centre and normal prescription, is part of the experiment. For the intrinsic geodesic ball of proper radius R, define

$$\begin{aligned} \mathfrak{F}_{R,u}(g_\epsilon) &= A_\epsilon - \frac{d-1}{R} V_\epsilon - \frac{\Omega_{d-1}}{d} R^{d-1} - \frac{c_0}{2} R^{d+1} \mathcal{Q}_\epsilon(p), \\ \mathcal{Q}_\epsilon &= (\mathrm{tr}_{q_\epsilon} K_\epsilon)^2 - K_{\epsilon,ij} K_\epsilon^{ij}. \end{aligned} \tag{14}$$

The last term is quadratic about a flat slice. It does not change any of the linear results, but it changes the second-order claim. This is a geometric correction dictated by the Gauss constraint, not an entropy or modular correction derived from a quantum state. The proper radius R and its multiplier $(d-1)/R$ are held fixed as ϵ varies. A coordinate sphere of unchanged coordinate radius is a different experiment: its geodesic radius and boundary measure change. The radial check below explicitly inverts that distance change.

For a smooth symmetric source T_ϵ, define a *stress reading*

$$\mathfrak{K}_{R,u}(T_\epsilon, g_\epsilon) = 2\pi \int_{B_\epsilon} \frac{R^2 - r^2}{2R} T_{\epsilon,\mu\nu} n_\epsilon^\mu n_\epsilon^\nu \, dV_\epsilon. \tag{15}$$

Here r is intrinsic geodesic distance, and the normal field on the slice is used inside the integral. In flat spacetime this is the bulk CHM charge. On a general curved metric it is only the specified local stress reading. No assertion of geometric modular flow on the curved diamond is being made.

Proposition 7.1 (Leading nonlinear ball identity)**.** *On a compact family with bounded smooth geometric and source derivatives, spacelike slices and radii below their normal injectivity radius,*

$$\mathfrak{F}_{R,u} = -c_0 R^{d+1} G_{\mu\nu}(g_\epsilon)(p) n_\epsilon^\mu n_\epsilon^\nu + O(R^{d+3}), \tag{16}$$

$$\mathfrak{K}_{R,u} = 2\pi c_0 R^{d+1} T_{\epsilon,\mu\nu}(p) n_\epsilon^\mu n_\epsilon^\nu + O(R^{d+3}). \tag{17}$$

The remainders and their first two ϵ derivatives are uniform when the corresponding mixed derivatives are bounded.

Proof. In intrinsic normal coordinates the volume density is $1-\mathrm{Ric}_{ij}\,x^ix^j/6+O(|x|^3)$. Angular averaging removes the odd terms and gives

$$V = V_d R^d \left(1 - \frac{{}^{(d)}\mathcal{R}}{6(d+2)} R^2\right) + O(R^{d+4}), \qquad A = \Omega_{d-1} R^{d-1} \left(1 - \frac{{}^{(d)}\mathcal{R}}{6d} R^2\right) + O(R^{d+3}).$$

Thus the area–volume difference minus its flat value is $-c_0 R^{d+1}\,{}^{(d)}\mathcal{R}/2 + O(R^{d+3})$. The exact Gauss equation ${}^{(d)}\mathcal{R} + \mathcal{Q} = 2G_{nn}$ proves the first statement. Taylor expansion of the smooth scalar T_{nn} and the volume density proves the second; its constant integral is $c_0 R^{d+1}$ and its odd terms average to zero. Differentiating these local expansions establishes the stated uniformity. Sufficient bounded mixed derivatives are assumed, rather than inferred from finite energy alone. □

Write, with no factorial in the second coefficient,

$$g_\epsilon = \eta + \epsilon h + \epsilon^2 j + O(\epsilon^3), \qquad T_\epsilon = \epsilon t_1 + \epsilon^2 t_2 + O(\epsilon^3), \tag{18}$$

$$G(g_\epsilon) = \epsilon G^{\mathrm{lin}}[h] + \epsilon^2 \big(G^{\mathrm{lin}}[j] + G^{(2)}[h,h]\big) + O(\epsilon^3). \tag{19}$$

$G^{(2)}$ is the quadratic coefficient of the covariant Einstein tensor in this fixed smooth identification of the metrics. Coefficients of the source use that same identification and the same renormalization convention. Appendix D gives its explicit coordinate formula and the source variation used below.

Theorem 7.2 (Second-order small-ball equivalence)**.** *Suppose $G^{\mathrm{lin}}[h] = (2\pi/\eta)t_1$ on an open domain and the two-jet families above have the regularity of Proposition 7.1. Then the following are equivalent:*

(a) At every p and for u in a nonempty open frame set,

$$\lim_{R\to 0} R^{-d-1}[\epsilon^2]\big(\eta\mathfrak{F}_{R,u} + \mathfrak{K}_{R,u}\big) = 0.$$

(b) The second-order tensor equation is

$$G^{\mathrm{lin}}[j] + G^{(2)}[h,h] = \frac{2\pi}{\eta} t_2.$$

Here $[\epsilon^2]$ denotes the quadratic Taylor coefficient. For one frame only the corresponding normal constraint follows.

Proof. Set

$$E(g_\epsilon) = G(g_\epsilon) - (2\pi/\eta)T_\epsilon.$$

Proposition 7.1 gives the leading residual $-\eta c_0 R^{d+1} E_{n_\epsilon n_\epsilon}$. Its zeroth and first coefficients vanish. Consequently changes of the normal at order ϵ multiply the vanishing first residual and do not alter the quadratic coefficient. That coefficient is $-\eta c_0 (G^{\mathrm{lin}}[j] + G^{(2)}[h,h] - (2\pi/\eta)t_2)_{uu}$. Tensor recovery from an open frame set proves both directions. □

This is a second-order conditional law in the same spacetime, not a holographic result. Unlike Theorem 4.2, it gives no exact finite-radius identity. Terms quadratic in curvature, its derivatives, the ball prescription and curved modular flow must be accounted for there. Nor does it classify all quadratic response functionals: adding terms whose small-ball leading coefficient vanishes leaves this test unchanged.

The extrinsic correction can instead be avoided by a particular physical region prescription. The local exponential image at p of $n_\epsilon^\perp$ is a spacelike geodesic hypersurface with $K_\epsilon(p) = 0$. On that family the uncorrected area–volume difference has the same leading law. It is not the same region as a ball on an arbitrary fixed laboratory slice. Regions, normals and preparations cannot be changed while interpreting their readings as the same experiment.

Example 7.3 (A bent slice in flat spacetime)**.** Keep the ambient metric exactly Minkowski and the source zero. Near the origin choose $t = \epsilon\alpha|x|^2/2$. Its normal there is ∂_t, and $K_{ij} = \pm\epsilon\alpha\delta_{ij} + O(\epsilon^2)$, so $\mathcal{Q} = \epsilon^2 d(d-1)\alpha^2 + O(\epsilon^3)$ and ${}^{(d)}\mathcal{R} = -\mathcal{Q}$. The raw ball difference is

$$[\epsilon^2]\left(F_R - \frac{\Omega_{d-1}}{d}R^{d-1}\right) = \frac{\Omega_{d-1}(d-1)}{2(d+2)}\alpha^2 R^{d+1},$$

although $G = 0$. To check the coefficient directly, its radial metric is $(1-\epsilon^2\alpha^2 s^2)ds^2 + s^2 d\Omega^2$; invert its proper-radius integral and expand its area and volume. The correction in Eq. (14) cancels this term exactly at quadratic order. Without a slice prescription or this correction, a nonlinear extension would assign a source to flat spacetime merely by bending a slice.

7.2 Why the flat prepared source cannot simply be reused

Even if a quantum stress expectation is smooth, a second-order response must use the total source in its perturbed metric. With a common covariant renormalization prescription and no gravitational anomaly, its Ward law is $\nabla^\mu T_{\mu\nu} = 0$. Expanding that identity gives

$$\partial^\mu t_{2,\mu\nu} = -\mathcal{D}_h[t_1]_\nu, \tag{20}$$

where

$$\begin{aligned}\mathcal{D}_h[t]_\nu &= -h^{\mu\alpha}\partial_\alpha t_{\mu\nu} - \eta^{\mu\alpha}\Gamma^{(1)\lambda}_{\alpha\mu}t_{\lambda\nu} - \eta^{\mu\alpha}\Gamma^{(1)\lambda}_{\alpha\nu}t_{\mu\lambda},\\ \Gamma^{(1)\lambda}_{\alpha\beta} &= \tfrac{1}{2}\eta^{\lambda\rho}(\partial_\alpha h_{\beta\rho} + \partial_\beta h_{\alpha\rho} - \partial_\rho h_{\alpha\beta}).\end{aligned}$$

Thus keeping only a conserved flat t_1 is generally not a second-order source: its curved divergence is $\epsilon^2\mathcal{D}_h[t_1]$. The expanded Bianchi identity demands precisely this correction.

The coherent preparation permits an explicit classical calculation of the missing terms. Take the same nonminimal scalar coupling as in Remark 5.6, now varied in the curved metric. For one real mean field Φ, its stress is

$$T_{\mu\nu}[g,\Phi] = \partial_\mu\Phi\partial_\nu\Phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\Phi)^2 + \xi(g_{\mu\nu}\Box_g - \nabla_\mu\nabla_\nu)\Phi^2 + \xi G_{\mu\nu}(g)\Phi^2. \tag{21}$$

Its divergence is $(\Box_g\Phi - \xi\mathcal{R}_g\Phi)\partial_\nu\Phi$; the derivative of the last term cancels the curvature derivative from the improvement. Its trace vanishes on shell for the conformal value $\xi = (d-1)/(4d)$ at the classical level.

Set $\epsilon = \theta^2$ for the Weyl witness, and take $g = \eta + \theta^2 h_2 + \theta^4 h_4 + \cdots$ and $\Phi = \theta v + \theta^3 v_3 + \cdots$. Then

$$\Box v_3 = -\delta\Box_{h_2}v + \xi\mathcal{R}^{\text{lin}}[h_2]v, \tag{22}$$

$$t_2 = t[v], \qquad t_4 = 2t[v,v_3] + \delta_g t_{h_2}[v] + \xi G^{\text{lin}}[h_2]v^2. \tag{23}$$

Here $t_g[v]$ comprises the first three terms of Eq. (21); $\delta_g t_{h_2}[v]$ varies them with v fixed, and $t[v, z]$ is their flat polarization used in Proposition 6.1. In coordinates,

$$\delta\Box_h v = -h^{\mu\nu}\partial_\mu\partial_\nu v - (\partial_\mu h^{\mu\nu} - \tfrac{1}{2}\partial^\nu h^\alpha{}_\alpha)\partial_\nu v.$$

On any fixed finite causal domain with smooth h_2, the driven flat wave equation for v_3 has smooth solutions after its Cauchy data are specified. Taking zero correction data is one preparation prescription, not the same as keeping the field unchanged through the perturbed evolution. Equations (22)–(23) obey Eq. (20) at order θ^4. The curvature response, changed propagation and metric variation are calculable requirements, not optional control overhead. This is an actual two-jet source calculation within the supplied classical scalar action; it neither derives that action nor proves a quantum source apparatus realizes it.

This calculation has nonempty second-order content. Choose smooth compact Cauchy data with $v(p) = B \neq 0, \partial_t v(p) = A \neq 0$ and spatially constant data near p. They can be represented by a local Weyl preparation using a smooth time-slice cutoff. Then $t_{2,00}(p) = A^2/2 > 0$. Choose a constraint-compatible h_2 as in Example 5.4 and use a linearized normal-coordinate gauge at p, so $h_2(p) = 0$ and $\partial h_2(p) = 0$. Prescribe zero v_3 and first time derivative on that Cauchy slice. Tracefreeness gives $\mathcal{R}^{\text{lin}}[h_2](p) = 0$; Eq. (22) then gives zero second time derivative of v_3 at p as well. All second derivatives entering the polarized stress consequently vanish there, and $\delta_g t_{h_2}[v](p) = 0$. Therefore

$$t_{4,\mu\nu}(p) = \xi\kappa B^2 t_{2,\mu\nu}(p), \qquad t_{4,00}(p) = \xi\kappa B^2 A^2/2 > 0, \quad \kappa = 2\pi/\eta.$$

This is one supplied local preparation and coordinate prescription, not a gauge-invariant attribution to one separate term on every slice. The sum in Eq. (23) is the covariant source.

For a coherent displacement of a Hadamard reference on the *same* metric, with unchanged connected two-point function, point splitting gives exactly this state-dependent stress difference: the added two-point function is $\Phi(x)\Phi(y)$ and the state-independent counterterms cancel in the difference. The total gravitational source additionally contains the reference's renormalized stress, metric-dependent vacuum response and the preparation/control sources. Those terms cannot be declared zero by changing the reference independently on every ball. The scalar surface expectation and its measure also acquire order-θ^4 corrections. A Wald coefficient match remains useful, but does not turn a generic curved diamond into a Killing horizon or supply its modular Hamiltonian.

7.3 A strong common-geometry extension

Conjecture 7.4 (A causal source and a common second-order geometry)**.** *There is a physically preparable family in a common positive-energy covariant matter theory, with a fixed source reference and supplied gravitational data, for which the corrected small-ball stress consistency of Theorem 7.2 is the local limit of the actual modular/Wald response of its selected region family. It has the following additional properties:*

1. *The renormalized total source and its two-jet Ward identity include the controller, old reference, support and changed propagation, rather than only the outgoing mean pulse.*
2. *The geometric slice prescription and its quadratic extrinsic term are common to the observers. Their source readings and boundary subtractions identify one $G^{\text{lin}}[j] + G^{(2)}[h, h]$ with one normalization; changing a stress-containing algebra alone does not create a second sourced spacetime.*

3. *Selection changes relative-entropy reconstruction or changes the source through an actual intervention. A returned geometric record can then change a permitted preparation; deleting the record after emission does not undo its stress.*

4. *Finite-radius corrections have one causal source/geometry account, with separately controlled readout and collar errors. The small-ball limit is uniform in the prepared family, rather than obtained by spending unbounded resources on each ball independently.*

The flat Weyl and neutral pulses provide nonzero-frequency source witnesses for the first order; the scalar two-jet above provides an explicit propagation and Ward test for the next order. HAFF-D provides finite source–response–record returns in a supplied JT sector. Q-RAIF C provides an interacting quantum source with retained references and a common classical covariant action [6]. These are different models and dimensions. None yet proves this conjecture's common higher-dimensional quantum source or its actual curved modular response. The missing bridge is substantive: flat CHM covariance does not persist for arbitrary g, quantum reference stress can gravitate, and a finite positive regulator does not establish a renormalized covariant Ward law.

The bent-slice example is a geometric falsification test. A second is to insert the flat witness unchanged at order θ^4: unless $\mathcal{D}_{h_2}[t_2] = 0$ exceptionally, it fails the source Ward identity. A third compares charged and invariant-algebra relative entropies on one physical preparation while keeping the complete source fixed. Any resulting difference is a reconstruction difference unless a new source operation is demonstrated. These tests preserve the proposed connection between physical access and geometry without assigning gravitational stress to information that has merely been forgotten.

8 Higher multipoles and finite reading

Under a supplied radial Gaussian readout model, the ball coefficients give conditional information $\log I_k = -4k \log k + O(k)$. The resolved order consequently grows as $\log N/(4 \log\log N)$, as in Consciousness III's information-spectrum bound [2]. Appendix B.6 retains the normalized directions, noise assumptions, exact Schur complement and proof. This is a world-source reading model, not a self-description theorem.

9 Relation to other work, and limits

Entanglement equilibrium. Jacobson [19] compares, in a small geodesic ball, the variation of the area at fixed volume with the CHM modular energy, and obtains the Einstein equation from all rest frames at leading order; the coefficient $\Omega_{d-1}R^{d+1}/(d(d+2))$ is the c_0 of (24). The present paper adds the following: the fixed-volume coefficient is the fixed-reference variational multiplier and follows from full-kernel matching (Proposition 3.5); the comparison is exact at every radius (Theorem 4.2); among functionals of the intrinsic geometry, internal scales allowed, nonzero-frequency witnesses identify it up to normalization (Theorem 4.4); the condition has to be written with modular energy rather than entropy (Remark 4.8, Proposition 6.7); and the results say what an observer's partial data fix (Section 5).

The first law of causal diamonds. Jacobson and Visser [21] derive, from the covariant phase space of general relativity, a first law relating the variations of the edge area, the volume of the maximal slice, the cosmological constant and the matter Hamiltonian of a diamond in a

maximally symmetric space. Theorem 3.4 is its flat-space, linear geometric content, proved without a field equation.

Holography. Faulkner et al. [22] show that the first law for all boundary balls of a holographic CFT is equivalent to the linearized Einstein equation of the bulk; HAFF-D uses this for a supplied half-sided modular family [3]. There the geometry is the bulk geometry and the Ryu–Takayanagi formula supplies the geometric side [23]. Here the geometry is that of the spacetime in which the balls live, no dictionary is used, and (MC_η) takes the place of the Ryu–Takayanagi formula. The present result is a same-spacetime conditional route, with (MC_η) as its explicit hypothesis.

Non-conformal matter and higher curvature. For non-conformal fields the modular Hamiltonian of a ball is not the CHM charge, and Casini, Galante and Myers showed that for relevant deformations of low dimension the corrections dominate the small-ball limit [24]. The modular results and scalar preparations here use conformal matter; the leading two-jet geometric test only requires its stated smooth source. Higher-curvature versions of entanglement equilibrium use the Wald entropy and a generalized volume [25], consistent with Proposition 4.7(b).

Limits. The exact finite-radius results are linear in h about Minkowski space, with conformal matter and the specified bulk modular datum. Section 7 extends the local small-ball law to a geometric two-jet, with a separate stress consistency input. A physical boundary charge cannot be discarded when it varies; the scalar example makes both its compensating readout and the failure of the raw-energy law explicit. The constant η is supplied. (MC_η) is a hypothesis: the equivalence theorem identifies its field equation, and nonzero-frequency witnesses identify its intrinsic response up to normalization. They do not say why the modular energy of a ball should be balanced by a geometric response at all. Only balls are used.

10 Access, common geometry and the remaining mechanism

The ball family gives a precise sense in which different perspectives must agree with one geometry. Given the linear consistency condition, its intrinsic response is the Einstein tensor. A prepared nonzero-frequency source identifies that response; homogeneous or zero sources do not. The metric additionally needs gauge and common gravitational data. Comparison of rest frames recovers components inaccessible to a single frame, and finite rapidity or radius sets the precision of that recovery.

Stress-containing selected algebras share this sourced tensor. A coherent preparation loses relative entropy when restricted to an invariant algebra, while its stress remains. This distinguishes an observer's reach from a second physical geometry. The difference matters to Consciousness III's finite self-models without implying that a modular energy or a ball algebra is itself experienced [2].

The two-jet extension makes the next step concrete. An intrinsic ball alone can confuse a bent slice with a gravitational source; the Gauss correction removes that error. All small boosted balls then test one second-order residual, including the quadratic first-order geometry. The scalar source follows its perturbed propagation and curvature coupling. It cannot be held flat while its geometry is evolved. These are necessary parts of a causal selection–source–response mechanism, not its full derivation.

The central question is why the actual modular data obey the required geometric consistency at all. HAFF-D addresses physical selection and returned source response [3]; the present paper gives their higher-dimensional consistency and identification conditions. A common renormalized quantum source, curved modular response and controlled finite-radius law remain the high-value bridge in Conjecture 7.4. Curved backgrounds, non-conformal matter, the origin of η and shape-sensitive tidal readings remain further directions.

A Linear ball kernels and response identification

A.1 The small-ball series and exact Fourier transfer

Proposition A.1 (Small-ball series)**.** *Let $\delta\langle T\rangle$ be smooth near x_0. For every u and every N,*

$$\begin{aligned}\delta\langle K_B\rangle &= 2\pi \sum_{k=0}^{N} c_k\, R^{d+1+2k}\, \Delta_u^k\, \delta\langle T\rangle_{uu}(x_0) + O\big(R^{d+3+2N}\big),\\ c_k &= \frac{\Omega_{d-1}}{4^k\, k!\, (d/2)_k\, (d+2k)(d+2k+2)},\end{aligned} \tag{24}$$

with $\Omega_{d-1} = 2\pi^{d/2}/\Gamma(d/2)$ and $(a)_k$ the Pochhammer symbol. For a single Fourier component $\delta\langle T\rangle_{\mu\nu} = \mathrm{Re}(\varepsilon_{\mu\nu} e^{ik\cdot x})$, exactly,

$$\delta\langle K_B\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, f\big(R\,|k_u|\big)\, \mathrm{Re}\big(\varepsilon_{uu} e^{ik\cdot x_0}\big), \qquad f(z) = \Gamma\big(\tfrac{d}{2}+2\big)\Big(\frac{2}{z}\Big)^{d/2+1} J_{d/2+1}(z), \tag{25}$$

where $|k_u|^2 = k\cdot k + (u\cdot k)^2$ is the squared spatial momentum in the frame u, $V_d = \Omega_{d-1}/d$ and $f(z) = 1 - z^2/(2(d+4)) + O(z^4)$.

The proof is in Appendix A.2.

The leading coefficient is $c_0 = V_d/(d+2)$. For smooth inputs this is a finite-order asymptotic expansion; an infinite convergent series requires additional analyticity. The Fourier formula is exact. Section 8 uses these radial coefficients.

Corollary A.2 (Tensor recovery)**.** *The leading terms of* (24) *for all u in any nonempty open subset of the unit hyperboloid determine $\delta\langle T\rangle_{\mu\nu}(x_0)$.*

Proof. If $A_{\mu\nu}u^\mu u^\nu = 0$ on an open subset of the hyperboloid, it vanishes on all of it by analyticity, hence on the open cone of timelike vectors by homogeneity, so $A = 0$. □

A single frame gives only $\delta\langle T\rangle_{uu}$ at each point; Section 5.1 shows that this is a real loss, not an artefact of the leading order.

A.2 The small-ball coefficients

Proof. The spherical mean of a smooth function over the sphere of radius r about x_0 in Σ_u is $\sum_k r^{2k}\Delta_u^k g(x_0)/(4^k k!(d/2)_k)$ up to $O(r^{2N+2})$ (Pizzetti's formula [18]), and $\int_0^R \frac{R^2-r^2}{2R} r^{d-1+2k} dr = R^{d+1+2k}/((d+2k)(d+2k+2))$. For the Fourier component, $\int_{|y|<1}(1-|y|^2)e^{ip\cdot y}d^dy = \pi^{d/2}(2/p)^{d/2+1}J_{d/2+1}(p)$, and $e^{ik\cdot x}$ restricted to Σ_u is a plane wave of momentum $|k_u|$. □

A.3 Intrinsic distributional classification

Proof. Translate the centre to zero. By (F6) write $\delta F_B[h] = \sum_{ij}\langle L_{ij}, h_{ij}\rangle$, with a symmetric tensor distribution L supported in the closed ball. Gauge invariance gives $\partial_i L_{ij} = 0$, and rotation invariance makes $\sigma = \sum_i L_{ii}$ a radial distribution. A constant metric perturbation is pure gauge, so $\langle \sigma, 1\rangle = 0$.

Let Γ be the radial fundamental solution of the spatial Laplacian, $\Delta\Gamma = \delta_0$, and set $\mu_B = -\Gamma * \sigma/(d-1)$. This distribution is supported in the closed ball. Indeed, for $|x| > R$ the function $y \mapsto \Gamma(x-y)$ is harmonic on a neighbourhood of the ball. Its spherical average in y is the constant $\Gamma(x)$, by the mean-value property. Pairing with a radial distribution therefore gives $(\Gamma * \sigma)(x) = \Gamma(x)\langle\sigma, 1\rangle = 0$. The argument includes the logarithmic fundamental solution in $d = 2$.

For real $p \neq 0$, orthogonal covariance implies $\widehat{L}_{ij} = a(|p|)\delta_{ij} + b(|p|)p_i p_j$. Transversality and the trace then give

$$\widehat{L}_{ij}(p) = (|p|^2\delta_{ij} - p_i p_j)\frac{\widehat{\sigma}(p)}{(d-1)|p|^2} = (|p|^2\delta_{ij} - p_i p_j)\widehat{\mu}_B(p).$$

Both sides are Fourier transforms of compactly supported distributions, so equality extends through zero and identifies the distributions. Thus $L_{ij} = (\partial_i\partial_j - \delta_{ij}\Delta)\mu_B$. Distributional integration by parts proves (d) for arbitrary smooth h. The compact radial distribution μ_B has an even entire Fourier transform along any coordinate axis, which proves (a). Differentiation at zero proves (b). For (c), the Gauss equation ${}^{(d)}R + K^2 - K_{ij}K^{ij} = 2G_{\mu\nu}n^\mu n^\nu$ has $K_{ij} = O(h)$ about the flat slice, so at linear order $\tau_u = 2G^{\mathrm{lin}}_{uu}$. □

A.4 Fixed-reference multiplier and full kernel

Proof. (a) is a one-line differentiation. For (b), the functional depends on h only through τ_u (Lemma 3.2), so it can be evaluated on the conformally flat perturbation $h_{ij} = 2\phi\,\delta_{ij}$, for which $\tau_u = -2(d-1)\Delta\phi$. With the spherical means $\sum_k m_k s^{2k}$ of ϕ, the linearized geodesic ball gives $\delta A = (d-1)\Omega_{d-1}\sum_k m_k R^{d-1+2k}\frac{2k}{2k+1}$ and $\delta V = \Omega_{d-1}\sum_k m_k R^{d+2k}\frac{2k(d-1)}{(2k+1)(d+2k)}$; comparison with Proposition A.1 gives the ratio. □

A.5 The nonzero-frequency and Radon identification proof

Proof. On any open set of centres, the real and imaginary parts of a nonzero plane wave are linearly independent. Lemma 3.2(d) therefore gives

$$2\eta m_R(q_u)g_{uu} + 2\pi\widehat{w}_R(q_u)t_{uu} = 0, \qquad \widehat{w}_R(q) = \frac{V_d R^{d+1}}{d+2}f(Rq).$$

The entire function $\widehat{w}_R$ is nonzero. The transfer function m_R cannot vanish identically: otherwise t_{uu} would vanish on an open frame set and tensor recovery would give $t = 0$. Exclude the isolated zeros of m_R and $\widehat{w}_R$, and restrict to a frame patch whose q_u varies over an interval. There

$$g_{uu} = a(q_u)t_{uu}, \qquad a(q) = -\frac{\pi\widehat{w}_R(q)}{\eta m_R(q)}.$$

Lemma 4.3 gives $g = a_0 t$; the scalar is nonzero on this patch. The entire identity $2\eta a_0 m_R + 2\pi\widehat{w}_R = 0$ follows by analytic continuation. Set $c = 2\pi/(\eta a_0)$ to obtain the normalization. The real, reflection-invariant functional has a real transfer function on real momenta, so a_0 and

c are real. A compactly supported tensor distribution is determined by its Fourier transform, so equality on plane waves identifies the functional on all smooth inputs. The fixed nonzero amplitudes fix a_0, hence the same c, at every other tested radius.

For the measure extension, dominate all tensor components by the locally finite measure $\mu = \sum_{a,b}(|\widehat{\delta\langle T\rangle}_{ab}| + |\widehat{G}_{ab}|)$, and write their Radon–Nikodym densities as $t_{ab}(k), g_{ab}(k)$. Lemma 3.2(d) represents the centre dependence as convolution with a compact distribution on the slice. Its Fourier multiplier is $m_R(q_u(k))$, a smooth function of k whose real-axis growth is polynomial. Multiplication therefore preserves the tempered Radon class. Conservation holds for these densities almost everywhere. Fourier-transform consistency for a countable dense set of frames in the open frame set. Intersect their full-measure sets: at each remaining k, the displayed coefficient equation holds on the dense frame set, hence on the entire open set by continuity in u. The source hypothesis supplies one such $k \neq 0$ with $t(k) \neq 0$. The amplitude proof at that k identifies $m_R = -c\widehat{w}_R/2$.

Substitute this identity back into the measure equation. For almost every k, $\widehat{w}_R(q_u)[g(k) - (2\pi/(c\eta))t(k)]_{uu} = 0$ on the open frame set. If $k \neq 0$, q_u varies over an interval and the entire function $\widehat{w}_R$ has only isolated zeros. The tensor contraction therefore vanishes on an open frame patch and tensor recovery sets the bracket to zero. At $k = 0$, $\widehat{w}_R(0) > 0$ gives the same conclusion directly. Thus the tensor measures, and hence the distributions, obey the field equation. This uses densities with respect to a common measure, without dividing distributions. At any additional tested radius repeat the argument with that radius's own exceptional null set. The already established field equation fixes the same constant c, since the source is nonzero. No common full-measure set for uncountably many radii is required. □

A.6 The zero-frequency converse

Proof. Sufficiency is Theorem 4.4. A Radon measure supported at zero is a multiple of δ_0, so the remaining nonzero source is constant. In the common-measure proof above, its atom forces $m_R(0) \neq 0$. At almost every $k \neq 0$ the source density is zero and consistency gives $m_R(q_u)g_{uu} = 0$. Since m_R is nonzero and entire, varying the frame forces $g(k) = 0$. The response tensor is therefore constant as well. Adding $\ell^2 \int_B w_R \Delta_u \tau_u$, with $\ell \neq 0$, leaves these data unchanged but changes the transfer function by $-\ell^2 q^2 \widehat{w}_R(q)$, which is independent of the original transfer function because it vanishes at zero. □

B Prepared sources, bounded pulses and finite readings

B.1 The coherent source and its Radon measure

Proof. The Weyl commutator shifts the field by the classical solution $\theta\varphi_f$; normal ordering therefore gives (9) and $\delta\langle\sum_a :\phi_a^2:\rangle = \theta^2\varphi_f^2$. Since $\Box\varphi_f = 0$, differentiation gives conservation and zero trace. Finite propagation makes its Cauchy data compactly supported on every fixed time slice. The improvement in T_{00} is a spatial divergence, so

$$\int_{\mathbb{R}^d} \delta\langle T\rangle_{00}\, d^dx = \frac{\theta^2}{2}\int_{\mathbb{R}^d} \left[(\partial_t\varphi_f)^2 + |\nabla\varphi_f|^2\right] d^dx > 0.$$

Zero energy would make φ_f constant and, by compact spatial support, zero, contrary to the nonzero one-particle class.

Up to Fourier convention, the spacetime transform of φ_f is $\mathrm{sgn}(k^0)\delta(k^2)\widehat{f}(k)\, d^{d+1}k$. Its two mass-shell sheets carry densities proportional to $\widehat{f}(\pm|p|, p)\, d^dp/(2|p|)$. Near $p = 0$ the radial

total-variation integral is $O(r^{d-2}dr)$, integrable for $d \geq 2$; at infinity the restricted Schwartz function decreases faster than any power. Thus this is a finite complex measure with every polynomial moment finite. Derivatives multiply it by momenta. Products of these smooth fields have Fourier measures given by convolution; the improvement differentiates the convolution twice. These operations retain finite variation and every polynomial moment, proving the assertion for $\widehat{\delta\langle T\rangle}$. If that measure were supported only at zero, $\delta\langle T\rangle$ would be a constant tensor. Compact spatial support would force it to vanish, contradicting its positive total energy. Finally, a point on the edge of B is spacelike to every point in a compact subset of $D(B)$, so φ_f vanishes there and (10) is zero. □

B.2 The invariant quadratic source

Proof. The CCR and $[X_a, Q_f] = 0$ give the exact transformation $U_\theta^* \phi_a(x) U_\theta = \phi_a(x) + \theta v_f(x) X_a$. Applying it to the vacuum two-point function and the point-split stress tensor gives (12) and the boundary formula. Internal rotations preserve $X_1^2 + X_2^2$; injectivity of arctan on the spectrum implies that its bounded functional calculus contains U_θ. The stable-seed theorem then proves the hull assertions.

Both v_f and H_f are smooth real massless solutions. Their Fourier measures have the same rapidly decreasing mass-shell densities as in Proposition 5.3, with different phases on the two sheets. For $d \geq 2$ they have finite variation and every polynomial moment. The products and derivatives in (12) preserve these properties. Every term contains v_f or a derivative of it, so finite propagation gives compact spatial support. Polarization of the free-field identities proves conservation and zero trace.

The complex solution $\omega(X_a \phi_a(x))$ has only one frequency sign. Its real and imaginary parts have zero cross term in the integrated classical energy: in spatial Fourier variables the two amplitudes differ by multiplication by i. Since v_f is a real multiple of the imaginary part and H_f is its real part, $\int t_{00}[v_f, H_f]\, d^d x = 0$. The remaining energy is the displayed positive quadratic term. A Fourier measure supported only at zero would make the source constant, which contradicts compact spatial support and positive total energy. □

B.3 The bounded accessible pulse

Proof. The weak integral belongs to $\mathcal{N}$. Choose normalized nonnegative smearings whose supports shrink to a point $t_0 \in (0, \tau)$. Then $B\Omega \to \alpha_{t_0}(C)\Omega = e^{it_0 H} C\Omega$. Separation keeps $C\Omega$, hence this limit, off $\mathbb{C}\Omega$, so sufficiently narrow smearings remain non-scalar. Translation smoothing gives a bounded commutator $[H, B] = i \int f'(t) \alpha_t(C) dt$. Consequently U_θ preserves Dom H, with $\|[H, U_\theta]\| \leq |\theta| \|[H, B]\|$, and its vacuum excitation has finite energy. For $0 < |\theta| \|B\| < \pi$, the exponential is injective on the bounded spectrum of B, hence U_θ is non-scalar. If its energy were zero, the unique ground vector would give $U_\theta \Omega = z\Omega$, and separation would force $U_\theta = zI$, a contradiction. The upper bound follows from $E_\theta = \langle U_\theta \Omega, [H, U_\theta]\Omega \rangle$. The bounded commutator makes the exponential expansion valid in the graph norm of H. Since $H\Omega = 0$, the double commutator gives the quadratic term; the spectral theorem gives its integral and strict positivity. Put $R_\theta = (U_\theta - I - i\theta B)\Omega$. The exponential integral formula gives $\|R_\theta\| \leq \theta^2 \|B\|^2/2$ and $\|HR_\theta\| \leq \theta^2 \|B\| \|K\|$. Expanding the energy of $i\theta B\Omega + R_\theta$ gives the stated remainder. □

B.4 Strict entropy loss and the regulated coefficient

Proof. (a) The equality is (3); it is finite because a smooth coherent excitation has finite modular energy. Here the compactly supported preparation is strictly local to $D(B)$, so its surface variation (10) vanishes for this ball. Take a real test function g in $D(B)$ with nonzero commutator function $\Delta(f,g)$, and $x = \arctan \phi_1(g) \in \mathcal{A}(D(B))$. The rotation by π maps ϕ_1 to $-\phi_1$ and commutes with E_B, and arctan is odd, so $E_B(x) = 0$. In ω_W the field $\phi_1(g)$ has the vacuum Gaussian law shifted by $\theta\Delta(f,g) \neq 0$, so $\omega_W(x) \neq 0 = \omega_W(E_B(x))$. The state does not factor through E_B, and by Petz's criterion for sufficient subalgebras [27] the inequality is strict. (b) The coherent stress expectation is quadratic in its classical displacement. Rotation by π makes the restricted state even in θ. For a finite faithful density matrix, differentiating the logarithm gives

$$g_{\mathrm{BKM},\rho_0}(\chi,\chi) = \sum_{a,b} |\chi_{ab}|^2 \frac{\log p_a - \log p_b}{p_a - p_b},$$

with the diagonal value $1/p_a$; the first entropy derivative vanishes because $\operatorname{Tr}\chi = 0$. This proves the regulated assertion. □

B.5 Radial multipoles and finite reading

The higher terms of (24) are the radial multipoles of the local source, read through the modular weight. Consciousness III states a resolution order for any family of directions whose information falls as $e^{-\alpha j \log j}$, and notes that a physical instance needs normalized directions, a fluctuation kernel and a permitted measurement [2]. The ball family supplies the directions and the signal; the noise remains to be supplied.

Proposition B.1 (Fisher spectrum of the multipoles)**.** *Write the normalized reading of a ball as*

$$y(R) = \frac{\delta\langle K_B\rangle}{2\pi V_d R^{d+1}/(d+2)} = \sum_{k\geq 0} a_k \Big(\frac{R}{L}\Big)^{2k} \theta_k,$$

$$\theta_k = L^{2k}\Delta_u^k \delta\langle T\rangle_{uu}(x_0), \qquad a_k = \frac{d(d+2)}{4^k k!\,(d/2)_k\,(d+2k)(d+2k+2)},$$

exact when the analytic ball-average series converges on the radius range used. Suppose a record consists of n readings at radii drawn independently and uniformly from $(0,\rho L]$, each with an independent Gaussian error of variance σ^2. The Fisher information per record about θ_k, with $\theta_0,\dots,\theta_{k-1}$ unknown and all higher coefficients fixed and known (or with the series truncated at order k), is

$$I_k = \frac{n}{\sigma^2}\, a_k^2 \rho^{4k}\, \frac{1}{4k+1} \prod_{j<k} \frac{[2(k-j)]^2}{[2(k+j)+1]^2}, \qquad \log I_k = -4k\log k + O(k),$$

and ρ enters only at order k.

B.6 The radial Fisher Schur complement

Proof. The Fisher matrix is $F_{jk} = (n/\sigma^2)a_j a_k \rho^{2j+2k}/(2j+2k+1)$, a diagonally scaled Cauchy matrix with entries $1/(x_j+x_k)$, $x_j = 2j+\frac{1}{2}$. The information about θ_k with the lower parameters unknown is the last Schur complement of the leading $(k+1)\times(k+1)$ block, for which the Cauchy determinant gives the product. Stirling's formula gives $\log a_k = -2k\log k + O(k)$, and the product is $e^{O(k)}$. □

B.7 Numerical checks and applications

The closed form agrees with a Cholesky factorization at eighty-digit precision for $k < 14$ and $d = 2, 3, 4$. A readout weight comparable to a power of R, in the radius distribution or in the noise variance, changes the Schur complements by factors $e^{O(k)}$ and leaves the leading term. With Consciousness III's tolerance c, the number of multipoles resolved from N records is $K(N) \sim \log N/(4 \log\log N)$; for $d = 3$, $\rho = 1$ and $c = 1$ one finds $K = 8$ at $\log N = 100$ and $K = 55$ at $\log N = 1000$. The factorial suppression is a property of the modular weight, so the resolution exponent $\alpha = 4$ has a physical signal behind it. The readout model, including its noise, is supplied, and the directions belong to the world as read by the observer, not to the observer's model of itself.

C Regularity of the quartic entropy law

For a displaced charged thermal mode, put $z = e^{-b}$, $p_n = (1 - z)z^n$. Phase averaging gives the tangent $\chi_n = |\alpha|^2 p_n[(1 - z)^2 n/z - (1 - z)]$. The geometric variance $z/(1 - z)^2$ yields

$$g_{\mathrm{BKM}}(\chi, \chi) = \sum_n \chi_n^2/p_n = |\alpha|^4(1 - z)^2/z = 4|\alpha|^4 \sinh^2(b/2). \tag{26}$$

Thus finite modular energy $\sum_j b_j|\alpha_j|^2$ does not alone control the diagonal BKM coefficient.

For finitely many independent modes of one charge, the invariant coherences also contribute. With $m = n + e_j - e_l$, expansion of the displacement gives

$$\chi_{mn} = \alpha_j \overline{\alpha}_l \sqrt{(n_j + 1)n_l}\, p_n(1 - z_j)(1 - z_l)/z_l.$$

Using $p_m/p_n = z_j/z_l$ in the spectral BKM kernel and averaging $(n_j + 1)n_l$ gives

$$g_{\mathrm{BKM}}(\chi, \chi) = \sum_{j,l} |\alpha_j|^2|\alpha_l|^2(1 - z_j)(1 - z_l)\frac{\log z_j - \log z_l}{z_j - z_l}, \tag{27}$$

with coincident value $(1 - z_j)^2/z_j$. Opposite charges add the ordered neutral-pair sum with kernel $(1 - z_j)(1 - z_l)(b_j + b_l)/(1 - z_j z_l)$. The mean-value theorem bounds the same-charge kernel by $e^{b_j} + e^{b_l}$; the neutral-pair kernel is at most $b_j + b_l$. Consequently, writing $A = \sum_j |\alpha_j|^2$, $E = \sum_j e^{b_j}|\alpha_j|^2$ and $L = \sum_j b_j|\alpha_j|^2$, the full regulated tangent obeys $g_{\mathrm{BKM}} \le 2A(E + L)$. The full mode expansion is retained as `CII_BKM_mode_calculation.tex` in the supplementary source record.

These estimates control the tangent, not the entropy remainder. For the continuum ball net, a quartic law still requires its modular BKM norm and uniform remainder control as occupation, mode and volume regulators are removed. Compact smooth spacetime smearing has not been shown here to supply the exponential moment E. The exact strict relative-entropy inequality is independent of this open regularity question.

D Explicit two-jet geometry and source bookkeeping

All index moves in the following coefficients use η, and antisymmetrization is written out rather than hidden in a factor. For $g = \eta + \epsilon h + \epsilon^2 j$, the connection coefficients are

$$\Gamma^{(1)\lambda}_{\mu\nu}[h] = \tfrac{1}{2}\eta^{\lambda\rho}(\partial_\mu h_{\nu\rho} + \partial_\nu h_{\mu\rho} - \partial_\rho h_{\mu\nu}),$$
$$\Gamma^{(2)\lambda}_{\mu\nu}[j, h] = \Gamma^{(1)\lambda}_{\mu\nu}[j] - h^\lambda{}_\rho \Gamma^{(1)\rho}_{\mu\nu}[h].$$

Define the quadratic part with $j = 0$ by $\gamma_2 = \Gamma^{(2)}[0, h]$. Then

$$r^{(2)}_{\mu\nu}[h,h] = \partial_\lambda \gamma^\lambda_{2\,\mu\nu} - \partial_\nu \gamma^\lambda_{2\,\mu\lambda} + \Gamma^{(1)\lambda}_{\lambda\rho}\Gamma^{(1)\rho}_{\mu\nu} - \Gamma^{(1)\lambda}_{\nu\rho}\Gamma^{(1)\rho}_{\mu\lambda},$$
$$G^{(2)}_{\mu\nu}[h,h] = r^{(2)}_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}(\eta^{\alpha\beta} r^{(2)}_{\alpha\beta} - h^{\alpha\beta} R^{\text{lin}}_{\alpha\beta}[h]) - \tfrac{1}{2} h_{\mu\nu} R^{\text{lin}}[h].$$

These coefficients include the inverse-metric correction. The full quadratic Einstein coefficient is $G^{\text{lin}}[j] + G^{(2)}[h,h]$, not $G^{\text{lin}}[j]$ alone. For a first-order vacuum wave, the second-order vacuum equation is $G^{\text{lin}}[j] = -G^{(2)}[h,h]$; interpreting its right side as a gravitational energy tensor requires a gauge/averaging convention.

For the scalar define $s = v^2$ and

$$C^\rho_h = \eta^{\alpha\beta}\Gamma^{(1)\rho}_{\alpha\beta}, \qquad \delta\Box_h s = -h^{\alpha\beta}\partial_\alpha\partial_\beta s - C^\rho_h \partial_\rho s.$$

Varying the first three terms of Eq. (21) at fixed v gives the fully explicit quantity

$$\begin{aligned}\delta_g t_{h,\mu\nu}[v] = &-\tfrac{1}{2} h_{\mu\nu}(\partial v)^2 + \tfrac{1}{2}\eta_{\mu\nu} h^{\alpha\beta}\partial_\alpha v \partial_\beta v \\ &+ \xi\big(h_{\mu\nu}\Box s + \eta_{\mu\nu}\delta\Box_h s + \Gamma^{(1)\rho}_{\mu\nu}\partial_\rho s\big).\end{aligned}$$

At a normal-coordinate point this vanishes when h and its first derivatives vanish. The remaining $\xi G^{\text{lin}}[h]v^2$ does not, which is the nonzero-source calculation in Section 7. The identity $\nabla^\mu[(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)s] = -R_{\nu\rho}\nabla^\rho s$, together with the Bianchi identity, reduces the full scalar stress divergence to its scalar equation times $\partial_\nu\Phi$. Its two-jet expansion gives Eq. (20), including the inverse metric and both connection terms. Counterterms cancel between two states only with one common metric and renormalization convention; the reference source remains in a total gravitational equation.

A finite algebraic check uses $g = e^{2\epsilon f}\eta$, $\Phi = \sqrt{\epsilon}e^{-(d-1)\epsilon f/2}v$, $f = t^2 + x^2/3 + tx/5$ and $v = 1 + 2t$. Direct rational-polynomial expansion through ϵ^2 in spacetime dimensions $3, 4, 5$ gives the conformal wave equation, vanishing trace and covariant divergence. Dropping $\xi G\Phi^2$ fails the divergence test. This tests the source identity on a prescribed conformal metric, not a solution of the coupled Einstein equations, and does not supply quantum renormalization or an actual curved modular Hamiltonian.

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Finite Self-Models and the Explanatory Gap: What a Continuing Observer Can Know About Itself

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Abstract

A continuing observer learns about itself through permitted operations, retained records and a description it can decode. With relative entropy as the loss, its risk splits exactly into unobservable, finite-record, compression and decoding terms. Continued neutral acquisition with sufficient growing memory reaches the observability limit; permanent failure and fixed-bit storage impose separate floors. Joint information prices actual working opportunities, including adaptive access. An unknown qubit state supplied once has optimal Bloch-ball risk $h(3/5) - 1/3 = 0.340$ nat. A drifting Gaussian self-parameter has stationary risk $\frac{1}{2}\ln(1 + P/v)$. We then distinguish matching routine reports from matching all permitted interventions. A finite pointer observer cannot passively distinguish a changed world, sensor or record frame, although they require different physical repairs. With two paid diagnostic permissions, its noisy restoration-failure risk plus read price is $29/120$, below $13/50$ for every fixed query list in the same menu; bounded finite gates preserve the advantage. Keeping the same body through repeated faults gives a stronger boundary: routine-only restoration success tends to at most $1/4$, for every history-dependent rule. Two paid internal pilots and a continuing two-step controller give failure approximately $.091$ in a specified noisy experiment, against $.195$ for the best enumerated memoryless map. Compressing causal records to confidence alone incurs a separate decision penalty. Confidence becomes causally useful when acquired distinctions return to the operation that needs correction. We propose that reflective first-person continuity depends on this maintained causal self-access, and specify selective-disruption, replay and retention contrasts. The proposal addresses the organization of reflection; the proved record boundaries do not establish phenomenal experience.

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1 Introduction

1.1 The question

An observer that is part of the world it describes must also describe itself: its own instruments, its own reliability and its own state. It can do so only from inside, with the operations available to it, from records it has kept, in a memory that is itself finite. This paper asks four questions about such self-description. What can a finite but continuing observer know about itself? Which limits are specific to self-description, as opposed to ordinary estimation? How does monitoring its own reliability extend what it can know? And what do the answers imply for the explanatory gap between a functional account of a subject and its experience?

The self-description model specifies an accessible algebra, a family of self-states, permitted instruments, acquired records, finite-bit memory and a decoder. Its loss is relative entropy between the true self-state and the description. The results distinguish finite evidence, restricted operations and properties of the target that is being read. A second, finite task in Section 7 prices an actual causal diagnosis and restoration. Its indicator loss and read prices are not measured in relative-entropy nats.

1.2 Position within the series

Consciousness I studies a noise-selected accessible algebra and standard Cuts inside its own vacuum representation, relating their modular data to geometry [2]. The selected algebra is not itself a Cut of the ambient algebra. Consciousness II carries modular data to balls in higher dimensions and studies their consistency with linearized gravity under its geometric inputs [3]. This paper asks what an observer can know about its own states and conditions of access. HAFF G motivates structural limits of such descriptions [4]. T-DOME supplies self-frames selected by history [6] and calibration and maintenance loops [7]; *The World as Imprint* gives a reader that estimates its own instrument [8]. The self-description bounds and causal diagnosis here connect those functional mechanisms to the Consciousness series' experiential question. The bridge has its own argument and competing explanations: none of the functional theorems supplies phenomenal presence by definition.

1.3 Results

- **Finiteness is not a floor** (Proposition 2.2). A two-outcome code reconstructs a short orbit of self-states exactly if the exact outcome probability is available. Loss enters through records and memory, not through dimension.
- **Exact anatomy of the loss** (Theorem 3.1). Unobservable, finite-record, compression and decoding terms add exactly. For the uniform Bloch ball read along one axis the unobservable term is $1/4$ nat and the finite-record term is an explicit sum.
- **What can be known** (Proposition 4.1, Theorem 4.2). With sufficient growing memory, fresh expressions identify exactly what their outcome law distinguishes. Every decoder obeys a non-asymptotic floor $2m^2/(\mathbb{E}J_Y + J_\pi)$ on any von Neumann algebra.
- **Access must continue** (Propositions 4.3–4.6). Actual working opportunities determine the available information. Repair restores the attainable fibre limit; in the continuous example, permanent failure and fixed-bit storage impose separate positive floors.

- **Limits that belong to a self.** The single-copy limit for the present state (Theorem 5.1); self-shaped unobservable fibres (Proposition 5.2); an exact and strictly positive stationary floor for a Gaussian self-parameter that drifts while it is read at a finite rate, and the resulting price of keeping up (Theorem 5.3, Corollary 5.4).
- **Monitoring** (Propositions 6.1 and 6.2). Record-conditioned risk is calibrated under a correct model; an effective implementation also needs a specified posterior and processing resources. Model errors confined to an unobservable fibre remain invisible.
- **Causal self-access** (Section 7). An interventional distance bounds every discriminating test. A pointer realization gives exact paid diagnosis and actual repair, separating identical routine reports from the internal causes they hide.
- **Resolution order** (Proposition A.1). Under a factorially decaying information spectrum, the number of self-directions resolvable to a fixed tolerance grows as $\log N/(\alpha \log\log N)$, with a two-sided Lambert-W asymptotic.
- **A bridge to experience** (Section 8). A hypothesis relating reflective access to dispositional self-knowledge, with its reasons, the strongest objection and distinguishing predictions.

1.4 Related work

Breuer proved that an observer cannot distinguish all present states of a system in which it is contained, classical or quantum [15]. Our single-copy limit and drift floor are quantitative relatives of that result: they say how much can be known, at what cost, and which parts cannot be known at all under given permissions. The Holevo bound limits the information a measurement extracts from one copy of a quantum state [14]. That Bayesian predictive states are optimal for averaged relative entropy is known for quantum statistical models [16], and the relative-entropy radius of a set of states equals its maximal Holevo information [17]. Posterior Cramér–Rao bounds for filtering are due to Tichavský, Muravchik and Nehorai [18]. Bayesian metacognition models distinguish coupled decision and confidence processes [19]. Report-based and no-report paradigms separately examine experience and the processes used to express it [20]. Our elementary testing and decision rules use standard tools; their role is to connect continuing records to a physically specified correction, and to state which experiential comparison those tools leave open.

2 Finite observers and self-description

2.1 Observer, self-states and permissions

Definition 2.1 (Finite self-describing observer)**.** A finite self-describing observer consists of

1. an accessible von Neumann algebra $\mathcal{A}_{\text{obs}}$, for example the stable accessible algebra of Consciousness I;
2. a family of normal self-states $\{\omega_\theta\}_{\theta\in\Theta}$ on $\mathcal{A}_{\text{obs}}$ with a prior π on Θ;
3. a permitted observation protocol Λ, a sequence of instruments whose classical outcomes form the record Y_N after N uses;
4. a memory that stores a compression $Z = f(Y_N)$ taking at most 2^B values;

5. a decoder that outputs a normal state σ_Z on $\mathcal{A}_{\text{obs}}$.

The self-description risk is $\mathbb{E}\, S(\omega_\Theta \| \sigma_Z)$, with S the Araki relative entropy [10]; the minimax risk replaces the prior average by a supremum over θ.

The protocol may be adaptive; then Y_N carries its full joint likelihood. Records, memory and decoding are physical operations of the observer, so their limits are its limits; no external reader is assumed. Relative entropy is the natural loss: it is defined for normal states on any von Neumann algebra, it is monotone under the observer's operations [11], and its local quadratic form is the Bogoliubov–Kubo–Mori metric [12], which Consciousness I computes exactly for half-line algebras of chiral nets along local diffeomorphisms [2].

2.2 Finiteness alone gives no floor

Proposition 2.2 (No floor from dimension)**.** *Let $s \mapsto \omega_s$ be a nonconstant smooth family of normal states on $\mathcal{A}_{\text{obs}}$, and choose an effect $0 \le M \le \mathbf{1}$ with $\frac{d}{ds}\omega_s(M)|_{s=0} \neq 0$. On a short enough segment, $p(s) = \omega_s(M)$ is injective, and the two-outcome code $\omega_s \mapsto (p(s), 1-p(s))$, which factors through the commutative algebra $\mathbb{C}^2$, followed by the recovery $(p(s), 1-p(s)) \mapsto \omega_s$, reconstructs every state on the segment with zero relative entropy.*

Proof. The map $\Phi(a,b) = aM + b(\mathbf{1} - M)$ is normal, unital and completely positive from $\mathbb{C}^2$ into $\mathcal{A}_{\text{obs}}$, and its dual sends ω_s to $(p(s), 1-p(s))$. Injectivity on a short segment follows from the inverse function theorem, and the recovery is defined on the image. □

The code is exact because it is given the probability $p(s)$, a real number. A record is not a probability. Finitely many outcomes of the effect determine $p(s)$ only up to sampling error, and finitely many bits only up to quantization. When those are charged, the record and compression terms of Theorem 3.1 below are positive for a continuous prior: for an affine family the decoded state is $\omega_{\mathbb{E}[s|Z]}$, which differs from ω_s with probability one. The quantity that a finite self-model fails to capture is therefore fixed by its records, its memory and its permissions, and those are what the rest of the paper prices.

3 The exact anatomy of self-description loss

In this section the algebra is a full matrix algebra and all states have full rank.

Theorem 3.1 (Four-term decomposition)**.** *Let $\Theta \to P \to Y \to Z$ be a Markov chain, where $P = p^{\Lambda}_{\Theta}$ is a sufficient specification of the protocol's record law, Y the record and Z its compression, and put $\rho_P = \mathbb{E}[\omega_\Theta \mid P]$, $\rho_Y = \mathbb{E}[\omega_\Theta \mid Y]$ and $\rho_Z = \mathbb{E}[\omega_\Theta \mid Z]$. For fresh independent uses, P is the one-use outcome law; for an adaptive protocol it must determine the full conditional experiment. For every decoder σ_Z,*

$$\mathbb{E}\, S(\omega_\Theta \| \sigma_Z) = \underbrace{\mathbb{E}\, S(\omega_\Theta \| \rho_P)}_{\textit{unobservable}} + \underbrace{\mathbb{E}\, S(\rho_P \| \rho_Y)}_{\textit{finite records}} + \underbrace{\mathbb{E}\, S(\rho_Y \| \rho_Z)}_{\textit{compression}} + \underbrace{\mathbb{E}\, S(\rho_Z \| \sigma_Z)}_{\textit{decoding}}. \tag{1}$$

The optimal decoder is $\sigma_Z = \rho_Z$, and the optimal risk is $\mathbb{E}\, S(\rho_Z) - \mathbb{E}\, S(\omega_\Theta)$, the average Holevo information of the posterior ensemble given Z.

Proof. For an ensemble with average $\bar\rho$ and any σ, $\mathbb{E}\,S(\rho\|\sigma) = \mathbb{E}\,S(\rho\|\bar\rho) + S(\bar\rho\|\sigma)$, because $\mathbb{E}\,\mathrm{tr}\,\rho(\log\bar\rho - \log\sigma) = \mathrm{tr}\,\bar\rho(\log\bar\rho - \log\sigma)$. Applied conditionally on Z with $\sigma = \sigma_Z$, it splits off the decoding term. For the remaining term, $S(\omega_\Theta\|\rho_Z) - S(\omega_\Theta\|\rho_Y) = \mathrm{tr}\,\omega_\Theta(\log\rho_Y - \log\rho_Z)$; conditioning on (Y,Z) and using $\mathbb{E}[\omega_\Theta \mid Y,Z] = \rho_Y$, which holds because Θ and Z are independent given Y, gives $\mathbb{E}\,S(\rho_Y\|\rho_Z)$. The same step at the level of (P,Y), with $\mathbb{E}[\omega_\Theta \mid P,Y] = \rho_P$, gives the record term. The optimal risk follows from $\mathbb{E}\,\mathrm{tr}\,\omega_\Theta\log\rho_Z = \mathbb{E}\,\mathrm{tr}\,\rho_Z\log\rho_Z$. □

The four terms come from one joint model, which is why they add. Bounds derived for different models, such as a $d/(2N)$ rate, a fibre radius and a quantization bound, cannot simply be summed. Each term has its own remedy: new permissions for the first, more records for the second, more memory for the third and a better decoder for the fourth.

Example 3.2 (The uniform Bloch ball read along one axis)**.** Let the self-state be a qubit $\omega_r = \frac{1}{2}(\mathbf{1}+r\cdot\sigma)$ with r uniform in the Bloch ball, and let the protocol measure σ_z on fresh preparations of ω_r. Then $p = (1+z)/2$ has prior $\mathrm{Beta}(2,2)$, and after N outcomes with K positive the posterior is $\mathrm{Beta}(K+2, N-K+2)$ and the optimal description is $\frac{1}{2}(\mathbf{1} + \hat z_K\sigma_z)$ with $\hat z_K = (2K-N)/(N+4)$. With h the binary entropy in nats, $\mathbb{E}\,S(\omega_r) = 1/3$ and $\mathbb{E}\,S(\omega_{(0,0,z)}) = 7/12$, so the unobservable term is exactly $1/4$ nat, and

$$R_N = \sum_{k=0}^{N} \frac{6(k+1)(N-k+1)}{(N+1)(N+2)(N+3)}\, h\!\left(\frac{k+2}{N+4}\right) - \frac{1}{3}. \tag{2}$$

Numerically $R_0 = 0.35981$, $R_{10} = 0.28408$, $R_{100} = 0.25477$, $R_{1000} = 0.25050$, and $N(R_N - \frac{1}{4}) = 0.4998$ at $N = 10^4$: the finite-record term behaves as $1/(2N)$ for the one identifiable direction.

Proof. The z-marginal of the uniform ball has density $\frac{3}{4}(1-z^2)$, which gives the Beta law. The posterior mean of z is $2\mathbb{E}[p \mid K] - 1$. The entropies follow by integrating $h((1+|r|)/2)$ against $3|r|^2\,d|r|$ and $h((1+|z|)/2)$ against $\frac{3}{4}(1-z^2)\,dz$. The risk is the average posterior Holevo information of Theorem 3.1, and the transverse components of r have posterior mean zero for every record. □

4 What a finite observer can know

4.1 Dispositions are learnable

Proposition 4.1 (Learnable dispositions)**.** *In finite dimension, suppose the protocol has finitely many outcomes and its uses are independent given Θ, each producing a fresh expression of the same self-parameter. The optimal risk R_N for uncompressed records is non-increasing and converges to $\mathbb{E}\,S(\omega_\Theta\|\mathbb{E}[\omega_\Theta \mid P])$. It converges to zero if and only if ω_Θ is, π-almost surely, a function of the outcome law P; in particular, if the protocol is informationally complete on the family.*

Proof. The conditional means $\mathbb{E}[\omega_\Theta \mid Y_N]$ form a bounded matrix martingale and converge; given Θ the empirical frequencies converge to P, so the limit is $\mathbb{E}[\omega_\Theta \mid P]$. The risk is $\mathbb{E}\,S(\rho_{Y_N}) - \mathbb{E}\,S(\omega_\Theta)$, monotone because ρ_{Y_N} is a martingale and entropy is concave, and continuous in the limit because entropy is continuous and bounded in finite dimension. The limit vanishes exactly when $\omega_\Theta = \mathbb{E}[\omega_\Theta \mid P]$ almost surely. □

The condition "fresh expression" matters. If all records copy a single outcome, they are nested and Markov but never improve on one use; repetition helps only when the self-parameter is expressed again. For the uniform Bloch ball with $N/3$ reads along each of $\sigma_x, \sigma_y, \sigma_z$, a Monte Carlo evaluation of the Bayes risk gives 0.290, 0.118 and 0.020 nat for $N = 3$, 30 and 300 (standard errors below 0.005), against the single-axis limit of $1/4$.

4.2 A floor for every decoder

Theorem 4.2 (Binary-effect floor)**.** *Let $\Theta = [a, b]$ carry a C^1 prior π vanishing at the endpoints with finite Fisher information J_π, and let $\{\omega_\theta\}$ be normal states on any von Neumann algebra. Suppose there is an effect M with $p(\theta) = \omega_\theta(M)$ monotone and $|p'(\theta)| \geq m > 0$, and that the record Y has a joint likelihood satisfying the regularity conditions of the scalar van Trees inequality, with Fisher information $J_Y(\theta)$. Then for every decoder σ_Y whose randomization is conditionally independent of Θ given Y (any correlated retained tape is included in Y),*

$$\mathbb{E}\, S(\omega_\Theta \| \sigma_Y) \;\geq\; \frac{2m^2}{\mathbb{E}_\pi J_Y(\Theta) + J_\pi}. \tag{3}$$

Proof. Let $q = \sigma_Y(M)$, let $\tilde{q}$ be q clipped to the interval between $p(a)$ and $p(b)$, and put $\hat{\theta} = p^{-1}(\tilde{q})$. Monotonicity of relative entropy under the measurement $\{M, \mathbf{1} - M\}$, Pinsker's inequality in nats for two Bernoulli laws, the fact that clipping to an interval containing $p(\theta)$ does not increase the distance to $p(\theta)$, and the mean value theorem give

$$S(\omega_\theta \| \sigma_Y) \;\geq\; D\big(\mathrm{Ber}(p(\theta)) \| \mathrm{Ber}(q)\big) \;\geq\; 2|p(\theta) - q|^2 \;\geq\; 2|p(\theta) - \tilde{q}|^2 \;\geq\; 2m^2 |\theta - \hat{\theta}|^2.$$

The van Trees inequality [13] bounds $\mathbb{E}(\hat{\theta} - \Theta)^2$ below by $1/(\mathbb{E}_\pi J_Y + J_\pi)$. □

The theorem needs no quadratic expansion of the Araki relative entropy and no restriction of the decoder to the family, so it holds on Type III algebras. For independent records $J_Y = NF$; for correlated records the joint information must be used, and it is not in general the independent value divided by a correlation time. As an example, take $\Theta = [-1, 1]$ with prior $\frac{15}{16}(1 - \theta^2)^2$, a commutative self-state $\mathrm{Ber}(0.5 + 0.2\theta)$ and N independent records of it, so that $m = 0.2$, $\mathbb{E}_\pi F = 0.16387$ and $J_\pi = 10$. The exact Bayes risks for $N = 0$, 10 and 100 are 0.01153, 0.00936 and 0.00345 nat; the floor gives 0.00800, 0.00687 and 0.00303.

4.3 Working opportunities and retained knowledge

A reading rate presupposes a working reader. Let A_t indicate whether a reading is actually acquired in wall-clock slot t, let $N(H) = \sum_{t=1}^{H} A_t$, and include access indicators, selected operations and acquired outcomes in the complete experimental record $\mathcal{R}_H$. This record may contain more than the observer retains. A floor derived for it also bounds any internal compression. The time labels specify the experiment; they do not grant a disabled processor an internal clock or free processing.

Proposition 4.3 (Information from actual access)**.** *Suppose the access and operation conditional probability kernels, given the pre-event complete-record history, are independent of a static parameter θ. Write $F_t(\theta)$ for the predictable conditional Fisher information of an acquired outcome, allowing dependence on that history. Suppose the joint likelihood is regular and the conditional scores have mean zero. Then*

$$J_{\mathcal{R}_H}(\theta) = \mathbb{E}_\theta \sum_{t=1}^{H} A_t F_t(\theta). \tag{4}$$

In particular, if $F_t \le F_{\max}$, every decoder in Theorem 4.2 obeys

$$\mathbb{E}\, S(\omega_\Theta \| \sigma) \ge \frac{2m^2}{J_\pi + \mathbb{E}_\pi J_{\mathcal{R}_H}} \ge \frac{2m^2}{J_\pi + F_{\max}\mathbb{E}_\pi N(H)}. \tag{5}$$

The identity allows access rules chosen from earlier outcomes and decisions to stop reading.

Proof. Factor the joint likelihood into conditional access, operation and outcome laws. Only acquired-outcome factors contribute to its derivative under the stated parameter-neutral access law. Their scores form martingale differences, so distinct scores are orthogonal in expectation. Their conditional second moments give (4). Apply Theorem 4.2 to the complete record and then to any retained record by data processing. □

If the access law itself depends on θ, its score must also be counted: an erasure pattern can carry information. In the access-gated loop of T-DOME III, the working body is not driven by its readings, and access is parameter-neutral given its stored latch. Equation (4) therefore applies to a static instrument in that loop. In its read-and-flip loop, the controlled body can instead make the access pattern informative about the instrument's sign; that pattern's score must then be counted [7].

For a concrete gate, start working, fail with probability $h \in (0, 1)$ after a working slot and recover with probability $r \in [0, 1]$ after a failed slot. Suppose this chain is independent of the self-parameter and the entire stream of fresh outcomes. Its mean acquired count is

$$\mathbb{E}N(H) = qH + (1-q)\frac{1-(1-h-r)^H}{h+r}, \qquad q = \frac{r}{h+r}. \tag{6}$$

Without repair, $\mathbb{E}N(H) = [1-(1-h)^H]/h \le 1/h$. In the affine Bernoulli example above, $F(\theta) \le 4/21$. At $h = 0.05$, (5) therefore gives a limiting floor of $21/3625 = 0.0057931$ nat. Independence of the gate sharpens it by using $\mathbb{E}_\pi F = 0.1638677$ instead of its maximum, giving 0.0060253 nat. These are decoder-independent lower bounds, not the exact Bayes risk.

Proposition 4.4 (Repair and the attainable observability limit)**.** *For the independent gate above and a finite-outcome protocol on a finite matrix algebra producing fresh expressions that are iid conditional on Θ, let R_n^* denote its optimal n-read risk with uncompressed records. The optimal wall-clock risk is*

$$R_H^* = \mathbb{E}R_{N(H)}^*.$$

If $r > 0$, then $N(H)/H \to q$ almost surely and

$$R_H^* \longrightarrow R_{\text{fibre}} := \mathbb{E}\, S(\omega_\Theta \| \mathbb{E}[\omega_\Theta \mid P]). \tag{7}$$

It tends to zero exactly when the self-state is almost surely determined by its permitted outcome law. If $r = 0$, then instead

$$R_H^* \longrightarrow \sum_{n \ge 1} h(1-h)^{n-1} R_n^*, \tag{8}$$

strictly above R_{fibre} whenever some finite R_n^* is above it.

Proof. The independent gate changes the count but not the conditional experiment. Its access history supplies no parameter information beyond that count. Positive repair makes the two-state chain recurrent with working fraction q, so infinitely many fresh expressions are acquired. Proposition 4.1, together with the finite-dimensional entropy bound, gives (7) by dominated convergence. With no repair the total count has geometric law $h(1-h)^{n-1}$, giving (8) and its strictness criterion. □

Proposition 4.5 (The fibre limit under predictable neutral access)**.** *Let a fixed finite-outcome instrument on a finite matrix algebra produce fresh outcomes with law $P = p_\Theta$ independently conditional on the static parameter Θ. Acquisition is selected before the current outcome, using the preceding complete history and parameter-independent random coins. Its conditional access kernels are parameter-neutral; no extra parameter-dependent status record is supplied. If $N(H) \to \infty$ almost surely under the prior, the optimal uncompressed wall-clock risk satisfies*

$$R_H^* \longrightarrow R_{\text{fibre}}.$$

Access may depend on previously acquired outcomes. The finite-horizon identity $R_H^* = \mathbb{E}R_{N(H)}^*$ additionally requires independence of access from the outcome stream, as in Proposition 4.4.

Proof. Let T_n be the slot of the n-th acquisition. The event $T_n = t$ is selected without the current fresh outcome, so conditional on Θ and the preceding acquired history its next outcome has law p_Θ. Successive conditioning proves that the acquired outcomes, in acquisition order, are iid conditional on Θ. Their empirical frequencies therefore converge to P. Factoring the full experimental likelihood, the access factors are parameter-neutral and only the acquired outcomes contribute to the posterior. Conditional on P, the acquired sequence and the access choices are independent of Θ. Since P is measurable from the infinite acquired sequence,

$$\mathbb{E}[\omega_\Theta \mid \mathcal{R}_\infty] = \mathbb{E}[\omega_\Theta \mid P].$$

The bounded matrix martingale $\rho_{\mathcal{R}_H} = \mathbb{E}[\omega_\Theta \mid \mathcal{R}_H]$ converges to this state. Continuity and boundedness of finite-dimensional entropy, together with the posterior predictive risk identity, give the claimed limit. At finite H, access selected from earlier outcomes can change their distribution conditional on $N(H)$; neutral likelihood factors do not justify averaging the fixed-count risks over that selected count. □

Repair restores opportunities while a nontrivial fibre remains. A recovering gate whose recovery probability is bounded below by a positive constant under every available latch acquires infinitely many fresh expressions; history-dependent latches are therefore covered by Proposition 4.5. Both limiting results require uncompressed or asymptotically sufficient growing memory. For k outcomes, their counts use $O(k \log(H+1))$ bits and are exactly sufficient under these neutral kernels. More generally, if retained descriptions satisfy $\mathbb{E}\|\rho_{Z_H} - \rho_{\mathcal{R}_H}\|_1 \to 0$, uniform continuity of matrix entropy gives the same limiting optimal risk. A fixed register has the separate floor below.

Proposition 4.6 (A fixed-memory floor)**.** *For $\pi(\theta) = \frac{15}{16}(1-\theta^2)^2$ on $[-1,1]$ and $\omega_\theta = \text{Ber}(1/2+\theta/5)$, every description whose complete retained information takes at most $M \le 2^B$ values obeys*

$$\mathbb{E}\, S(\omega_\Theta \| \sigma_Z) \ge \frac{9}{2560} M^{-2} \ge \frac{9}{2560} 2^{-2B} \quad \textit{nat}, \tag{9}$$

regardless of its acquisition history.

Proof. Clip the decoded Bernoulli parameters to $[0.3, 0.7]$ and invert them to centres $a_j \in [-1,1]$. Pinsker's inequality gives loss at least $(2/25)(\theta - a_j)^2$. On $[-1/2, 1/2]$ the prior is at least $135/256$. The best M-centre integrated squared error on this unit interval is at least $1/(12M^2)$: nearest-centre cells of lengths ℓ_j contribute at least $\ell_j^3/12$, and $\sum_j \ell_j^3 \ge 1/M^2$. Multiplication gives (9). Random encoding cannot improve the nearest-centre bound. Decoder randomness independent of Θ conditional on Z can be averaged out by convexity of relative entropy in its second argument; a retained correlated tape is additional information and belongs in Z. □

The three boundaries now separate. Lost working opportunities limit evidence; restricted operations leave fibres; fixed storage quantizes what the evidence could identify. Maintaining a reader, enlarging its permissions and enlarging its memory solve different problems. The memory floor is compatible with Proposition 2.2, whose exact real-valued probability is not a B-bit record. T-DOME III couples working opportunities to body maintenance [7]; the relational working graph of Brain Anchor determines which surviving channels can reconstruct an unread target [9]. Neither repair nor routing grants an internal monitor a way to detect a model error invisible to all its permitted experiments.

5 Limits that belong to a self

The estimation results also apply to external readers. For self-description, three boundaries are especially consequential: one preparation of the unknown occurrent state, permissions shaped by the self-model, and drift while the reader continues.

5.1 One copy of the present

A self-parameter that is re-expressed in each interaction can be sampled repeatedly. For the unknown occurrent state in this model there is only one preparation to interrogate: the observer is that copy. The reconstruction target is the initially prepared unknown state. Resetting it to a known state changes the target rather than reconstructing the old one. Repeated expressions of a stable parameter are a different resource.

Theorem 5.1 (Single-copy limit)**.** *In finite dimension, let the target be a state ω_Θ prepared once. Every record obtained by any sequence of instruments on that copy, adaptive or not, followed by any processing, has the outcome law of a single POVM $\{E_y\}$ on ω_Θ. Its optimal risk is*

$$R_1 = \chi - \sup_{\{E_y\}} \mathbb{E}_Y \, S\big(\rho_Y \| \bar{\rho}\big) \;\geq\; \chi - I_{\text{acc}}, \tag{10}$$

where $\bar{\rho} = \mathbb{E}\,\omega_\Theta$, $\chi = \mathbb{E}\,S(\omega_\Theta\|\bar{\rho})$ is the risk with no record, and I_{acc} is the accessible information of the ensemble. For the uniform Bloch ball, $R_1 = h(3/5) - 1/3 = 0.33968$ nat, against $\chi = \ln 2 - 1/3 = 0.35981$.

Proof. A sequence of instruments composes to one instrument, whose outcome statistics form a POVM. Theorem 3.1 with $\bar{\rho}$ as the decoder of the empty record gives $\chi = \mathbb{E}\,S(\omega_\Theta\|\rho_Y) + \mathbb{E}\,S(\rho_Y\|\bar{\rho})$, which is (10); and $S(\rho_Y\|\bar{\rho}) \leq D(\pi(\cdot \mid Y)\|\pi)$ by monotonicity under $\theta \mapsto \omega_\theta$, so the subtracted term is at most $I(\Theta;Y) \leq I_{\text{acc}}$. For the Bloch ball write $E_y = \alpha_y(\mathbf{1} + n_y \cdot \sigma)$ with $|n_y| \leq 1$, $\sum \alpha_y = 1$ and $\sum \alpha_y n_y = 0$. Then $\Pr(y) = \alpha_y$, and since $\mathbb{E}[r_i r_j] = \delta_{ij}/5$ the posterior Bloch vector is $n_y/5$. Hence $\mathbb{E}_Y S(\rho_Y\|\mathbf{1}/2) = \sum_y \alpha_y[\ln 2 - h(\frac{1}{2} + \frac{|n_y|}{10})] \leq \ln 2 - h(3/5)$, with equality when every $|n_y| = 1$, which proves the value. □

One copy of a qubit-sized part, under the uniform Bloch-ball prior, can be described only 0.020 nat better than with no record, a reduction of 5.6%. The same part, if it were an identifiable disposition re-expressed in fresh interactions, could be described to any precision with sufficient growing memory (Proposition 4.1). The contrast is a structural fact about these experiments: recurring couplings, gains and habits of reading supply repeated evidence, whereas the initially unknown state on its sole preparation supplies one POVM record. This gives a quantitative boundary relevant to Breuer's self-measurement problem [15].

5.2 Self-shaped blind spots

The unobservable term of Theorem 3.1 depends on the permitted protocol. When the protocol is chosen by the observer's own self-model, so does the blind spot.

Proposition 5.2 (Self-shaped fibres)**.** *Fix the observer's current self-model and its available history, and take expectations under the corresponding conditional prior. If the protocol at that stage is $\Lambda(\hat{\theta})$, its unobservable term is $\mathbb{E}\, S(\omega_\Theta \| \mathbb{E}[\omega_\Theta \mid P_{\Lambda(\hat{\theta})}])$, and its fibres $\{\theta : p_\theta^{\Lambda(\hat{\theta})} = p\}$ move when the self-model moves. In the budgeted reader of T-DOME II, with a rank-k foreground $\mathcal{P}$ selected by the self-frame, the fibres are the spheres $\{u : \mathcal{P}u = w\}$ in the complement of the foreground, so no number of records gathered through the frame locates the part of the task the frame misses [6].*

Proof. The first statement is Theorem 3.1 applied to the protocol in use. The second is the in-frame identifiability theorem of T-DOME II: the law of in-frame records depends on the coupling only through its in-frame component and the total output variance. □

The smallest case is a two-point fibre. When an even record law cannot tell θ from $-\theta$, as in the residual model of T-DOME III, which T-DOME II derives from a two-channel self-frame, the best estimate under true weights $w, 1-w$ is $\theta(2w-1)$ with mean squared error $4\theta^2 w(1-w)$ for every N, and the best description has risk equal to the Holevo information of the two states, $2g\theta^2 w(1-w)$ to leading order, with g the BKM metric along the fibre. For a qubit with base Bloch vector $(0, 0, \frac{1}{2})$ and $\theta = 0.05$, $w = 0.8$, the Holevo information is 8.795×10^{-4} against the leading term 8.789×10^{-4}. A single read in the direction that the self-frame excludes breaks the symmetry [6]. Blind spots of this kind are repaired by changing what the observer reads, which requires a new permitted operation rather than another decoding of the same records.

5.3 A floor for continuing self-knowledge

A continuing observer's self-parameters change while it reads them. The following theorem gives the exact floor in the Gaussian case.

Theorem 5.3 (Drift floor)**.** *Let the self-state be the law $\mathcal{N}(\theta_t, v)$ of a commutative register, let θ_0 have a Gaussian prior and $\theta_{t+1} = \theta_t + \eta_t$ with independent $\eta_t \sim \mathcal{N}(0, q)$, $q > 0$, and let the observer obtain at each step N independent readings $\theta_t + \epsilon$, $\epsilon \sim \mathcal{N}(0, r)$. Over all decoders that use the records up to time t, the optimal expected relative entropy converges to*

$$R_{\text{drift}} = \tfrac{1}{2} \ln\Big(1 + \frac{P}{v}\Big), \qquad P = \frac{-q + \sqrt{q^2 + 4qr/N}}{2}, \tag{11}$$

which is strictly positive and satisfies $R_{\text{drift}} = \sqrt{qr/N}/(2v)\,(1 + o(1))$ as $qr/N \to 0$ with $q/(r/N) \to 0$.

Proof. The relative-entropy identity of Theorem 3.1 holds for densities, so the optimal decoder is the posterior predictive law. With Gaussian prior and Gaussian records the posterior of θ_t is $\mathcal{N}(m_t, P_t)$, with P_t given by the Kalman recursion $P_t^{-1} = (P_{t-1} + q)^{-1} + N/r$, so the posterior predictive law is $\mathcal{N}(m_t, v + P_t)$, and

$$\mathbb{E}\, D\big(\mathcal{N}(\theta_t, v) \| \mathcal{N}(m_t, v + P_t)\big) = \tfrac{1}{2}\Big[\ln \frac{v + P_t}{v} + \frac{v}{v + P_t} + \frac{P_t}{v + P_t} - 1\Big] = \tfrac{1}{2} \ln\Big(1 + \frac{P_t}{v}\Big).$$

The recursion converges to the positive root of $NP^2 + NqP - qr = 0$, which is the stated P. The expansion follows from $P = \sqrt{qr/N}\,(1 + O(\sqrt{qN/r}))$. □

A Kalman simulation reproduces (11) to within 0.5% in the four cases computed, and the decoder $\mathcal{N}(m_t, v)$, which ignores its own uncertainty, does worse in each. For general families and records the posterior Cramér–Rao recursion $J_{t+1} = F + (q + J_t^{-1})^{-1}$ bounds the mean squared error of any parameter estimate by the inverse of its fixed point [18]; converting it into a relative-entropy floor for arbitrary decoders requires a bridge such as Theorem 4.2 on the range the drift explores, which is not attempted here.

Corollary 5.4 (The price of keeping up)**.** *In the setting of Theorem 5.3, allow controllable observation precision N/r with arbitrary real $N \geq 0$ and cost wN, measured in the same units as the loss. To leading order as $P/v \to 0$ and $qN/r \to 0$, the stationary loss plus reading cost per step is minimized at $N^* = \left(\sqrt{qr}/(4vw)\right)^{2/3}$, where it equals $3wN^*$. The minimal cost thus scales as $w^{1/3}q^{1/3}r^{1/3}v^{-2/3}$. For independent whole readings, this is the continuous-precision relaxation.*

Proof. Minimize $\sqrt{qr}/(2v\sqrt{N}) + wN$ over $N > 0$. □

With $\nu = \sqrt{q}$, $J = 1/r$ and $G = 1/v$, the minimal cost has the form of T-DOME I's scalar drift law [5]. T-DOME III proves the exact predictive-relative-entropy optimum for every predictable precision policy under the Gaussian prior used here [7]. Conditional on the full realized records and selected precisions, the posterior remains Gaussian, its variance obeys $P_t^{-1} = (P_{t-1} + q)^{-1} + N_t/r$, and its optimal risk is exactly $\frac{1}{2}\ln(1 + P_t/v)$. Telescoping this realized recursion gives the bound; a constant allowed precision attains it. This avoids converting an averaged mean-square bound through a concave logarithm. Integer reading counts retain the relaxation bound and require a separate attainable schedule. Numerical minimization of the exact continuous-precision cost agrees with the leading-order optimum to within 0.7% for $q \leq 10^{-4}$, $r = v = 1$, $w = 10^{-3}$.

A floor for self-knowledge under drift needs a typed observation model, and Theorem 5.3 supplies one: an observer with finite observation precision and nonzero drift has a positive tracking shortfall, with the stated square-root dependence in the slow-drift regime. The floor is not a consequence of finite dimension, which Proposition 2.2 excludes, but of a finite reading rate set against a nonzero drift.

6 Monitoring one's own reliability

Proposition 6.1 (Record-conditioned assessment)**.** *Suppose a regular conditional posterior exists, the decoder σ_R is measurable, and $L = S(\omega_\Theta \| \sigma_R)$ is measurable and integrable under the observer's model. Then $Q(R) = \mathbb{E}[L \mid R]$ satisfies $\mathbb{E}Q = \mathbb{E}L$ and*

$$\mathbb{E}[(L - Q)\mathbf{1}_C] = 0$$

for every event C determined by the record. Under a correct model this is a calibrated risk assessment, including for Araki relative entropy on a general von Neumann algebra. On a finite matrix algebra the optimal decoder $\rho_R = \mathbb{E}[\omega_\Theta \mid R]$ has

$$Q(R) = S(\rho_R) - \mathbb{E}[S(\omega_\Theta) \mid R],$$

where the individual-state entropies are von Neumann entropies.

Proof. The first identities follow from conditional expectation and its tower property. In finite dimension, $\mathbb{E}[\omega_\Theta \mid R] = \rho_R$ gives $-\mathbb{E}[\operatorname{tr} \omega_\Theta \log \rho_R \mid R] = S(\rho_R)$, proving the entropy difference. The difference formula is not used on Type III algebras. □

Being a function of a record is not an algorithm for evaluating it. An implemented monitor needs an effectively specified posterior, decoder and loss, an approximation rule and the storage and processing to execute it. The finite-register filters in T-DOME III supply a controller specification and quantization certificate for their transition family; their workspace, update throughput and cyclic erasure still require a physical implementation [7]. A finite calibrated monitor can exist without being a universal solver for arbitrary state families.

Monitoring extends what an observer can do with its self-knowledge in at least four ways that the series has made concrete. It can weight self-knowledge by its reliability: in the budgeted reader of T-DOME II, scaling the use of the self-frame by its computed reliability keeps it from ever doing worse than no memory and extends its useful range of drift by factors between 2.1 and 14 [6]. It can decide when to read more: the sequential reader of *The World as Imprint* stops acquiring references once its estimated reliability reaches a target [8]. It can allocate authority: in T-DOME III, reliability-weighted authority over a shared world reduces the damage from a failed sensor [7]. And Theorem 3.1 tells it which term dominates, hence whether more records, more memory, a better decoder or new permissions would help most.

Monitoring has a limit that mirrors the blind spot it serves.

Proposition 6.2 (What monitoring cannot see)**.** *In the two-point fibre $\{\theta, -\theta\}$ with true weights $w, 1-w$, let the observer assume weights $q, 1-q$. Its estimate is $\theta(2q-1)$; its true mean squared error exceeds the Bayes value $4\theta^2 w(1-w)$ by $4\theta^2(q-w)^2$, and its self-assessed error differs from its true error by $4\theta^2(w-q)(1-2q)$. No number of records within the protocol changes either quantity.*

Proof. The records do not depend on the sign, so the posterior on the fibre equals the assumed prior. Direct computation gives the true error $4\theta^2[w(1-q)^2 + (1-w)q^2]$ and the assessed error $4\theta^2 q(1-q)$, whose differences are as stated. □

An observer's risk assessment is calibrated under a correct model, and it can be confidently and undetectably wrong on an unobservable fibre. A monitoring rule can recommend new permissions, but the new operations must actually be available. When acting respects a symmetry of the joint initial, sensing and transition laws, it cannot distinguish points on that symmetry orbit. Invariance is an obstruction, not a sufficient condition for identification. T-DOME III supplies both this obstruction and a stationary body-moment construction; with independent sign inversions and bath flips, every fresh-record read-and-flip policy without a symmetry-breaking operation tends to chance [7]. Retained earlier physical records and labelled operations can provide relational standards. Their formation is compatible with the obstruction to absolute jointly relabelled orientation.

The model being monitored need not be reassigned at each calibration. T-DOME III also gives a continuing joint filter of a bounded instrument's gain, sign and controlled body, carried forward from its own records. A finite log-ratio register has a uniform quantization-risk bound, and a reference rule driven by that same history improves body risk under the same physical gates and reference cap [7]. This makes history-based model revision operational at finite memory without making the transition family, reference labels or processor resources self-originating. Monitoring its estimated risk remains conditional on that family being correct.

7 Causal self-access: world, sensing and retained records

A confidence score concerns whether a judgment is right. A model of one's access also concerns why it might be wrong, which part can be changed and what evidence would justify that change.

The same routine report can result from a change in the world, the sensor or the way retained records are read. Comparing reports alone does not distinguish these causes. Comparing the consequences of permitted interventions can. This distinction also limits what an experience hypothesis can test.

7.1 Which equivalence an experiment establishes

Definition 7.1 (Interventional record distance)**.** For two specified physical conditions ϑ, ϑ' and a common resource budget, let $\mathfrak{P}$ be their common set of permitted adaptive protocols. Each chooses operations from its acquired records and parameter-independent private coins. Include initial accessible information, operation labels, acquired outcomes and retained correlated tapes in the complete experimental record. Let P^{π}_{ϑ} be its law and put

$$\delta_{\mathfrak{P}}(\vartheta, \vartheta') = \sup_{\pi \in \mathfrak{P}} \|P^{\pi}_{\vartheta} - P^{\pi}_{\vartheta'}\|_{\mathrm{TV}}.$$

The conditions are record-equivalent for this experiment class when $\delta_{\mathfrak{P}} = 0$.

Proposition 7.2 (The boundary of a discriminating bridge)**.** *For equal prior odds, every protocol in $\mathfrak{P}$ and every decision based on its complete record has discrimination error at least $(1 - \delta_{\mathfrak{P}})/2$. For any record statistic $0 \le T \le 1$, the difference of its expected values in the two conditions is at most $\delta_{\mathfrak{P}}$. Thus no test in this class distinguishes two record-equivalent conditions, including through confidence reports.*

Proof. For a fixed protocol, the minimum equal-prior testing error is $(1 - \|P^{\pi}_{\vartheta} - P^{\pi}_{\vartheta'}\|_{\mathrm{TV}})/2$: choose the condition with the larger likelihood. The difference of expectations of a $[0, 1]$-valued statistic is bounded by the same total variation. Taking the supremum over permitted protocols proves both bounds. Record processing and independent randomization cannot enlarge the distance. □

Equality of routine reports is weaker than this equivalence. A selective perturbation, a new reference or an external physiological readout can enlarge the experiment class and make a difference measurable. It must be included as an additional physical permission, not as a consequence of identical laws. Conversely, the proposition does not decide whether two record-equivalent systems have identical experiences. It states when a proposed difference has no test in the declared class.

7.2 An internal diagnosis that changes actual repair

Consider three commuting pointer registers with values $(w, g, m) \in \{-1, 1\}^3$. They represent the controlled world's value, a sensor's orientation and a retained-record reading frame. The nominal state is $(1, 1, 1)$. Exactly one register has reversed, with equal prior probabilities. A routine record reads the product $y = mgw$, with any common independent report noise. The three causes therefore have identical routine laws for any number of such reads. Two additional internal permissions send a known pilot through the sensor while bypassing the record frame, or through the record frame while bypassing the sensor. With perfect pilots their outcomes are:

Reversed register	Routine y	Sensor pilot	Record-frame pilot
World	-1	$+1$	$+1$
Sensor	-1	-1	$+1$
Record frame	-1	$+1$	-1

After at most two pilots, one permitted repair flips a chosen register. The loss is failure to restore the actual triple to $(1, 1, 1)$, plus nonnegative price c for each pilot. This is a declared task with labelled standards. A reversed record frame is an invertible change of reading, not erasure of an unknown original record.

Proposition 7.3 (Paid causal diagnosis)**.** *In this one-fault model:*

1. *Every repair selected only from routine records has failure probability at least $2/3$, even with arbitrarily many such records. Nevertheless, any chosen register flip makes the noise-free routine report $+1$. Normalizing the report is not the restoration criterion.*
2. *With perfect pilots, the optimal risk over protocols using at most two pilots is*

$$\min\{2/3, 5c/3\}.$$

   For $c < 2/5$, query one internal stage; repair immediately if it is reversed, otherwise query the other. Restoration is exact and the mean pilot count is $5/3$.
3. *With independent pilot flip probability $\epsilon = 1/10$ and $c = 1/20$, the optimal risk in that same menu is $29/120$. Its restoration failure is $4/25$ and its mean pilot count is $49/30$. Every record-independent query list of length at most two, including random mixtures and repeated queries, has risk at least $13/50$, even though its repair may use all outcomes.*

Proof. Routine likelihoods are identical, so the three-way posterior stays uniform. An incorrect repair leaves two reversed registers; their product is still positive. For perfect pilots, a first negative sensor pilot identifies the sensor. A positive result leaves world and record frame equally likely. A second pilot then removes failure $1/2$ at cost c. Backward induction compares this rule, risk $5c/3$, to no query, risk $2/3$; stopping after one query cannot lower their minimum.

For noisy pilots write p for the posterior over the three causes. With n queries remaining its exact value is

$$V_n(p) = \min\left\{1 - \max_i p_i,\ \min_{a\in\{g,m\}}\left[c + \sum_{z=\pm1} p(z \mid p, a) V_{n-1}(p^{a,z})\right]\right\}. \tag{12}$$

All likelihoods are ϵ or $1 - \epsilon$. After the first sensor pilot, a negative result gives posterior $(1, 9, 1)/11$ with probability $11/30$. Neither remaining query improves the optimal repair probability, so repair the sensor. A positive result gives $(9, 1, 9)/19$ with probability $19/30$. A record-frame pilot then gives optimal conditional failure $14/95$; repeating the sensor query gives no improvement. Including its price,

$$V_2(1/3, 1/3, 1/3) = \frac{1}{20} + \frac{11}{30}\frac{2}{11} + \frac{19}{30}\left(\frac{1}{20} + \frac{14}{95}\right) = \frac{29}{120}.$$

This beats stopping or any other first query; symmetry covers the record-frame-first rule. The failure and mean query count follow by the same branch sums. The fixed lists have risks $2/3$ (none), $9/20$ (one), $1/2$ (twice the same port), and $13/50$ (both ports). Their mixtures cannot beat the minimum. □

The single-fault premise affects both value and stopping. If the triple reversal is also possible, two perfect pilots still identify all four conditions compatible with the routine product -1, but a uniform four-condition prior requires both pilots. Repairing the triple reversal also needs three flips, outside the stated single-repair menu. Nor do arbitrarily many fully correlated pilot copies have the independent likelihood used above. These changes require a different experiment, not just a new confidence threshold.

7.3 What a causal self-model must retain

A confidence number can be calibrated yet omit what an operation needs. A finite self-model for this task is not a second name for the whole raw history. It is a retained description of the observer's world–sensor–record relations, sufficient to select an allowed next question and the corresponding correction at the given prices. Which distinctions are sufficient depends on the task and the future operation menu.

Proposition 7.4 (The decision price of compressing self-access)**.** *Let R be the available acquired record in a specified finite causal task, $Z = f(R)$ its retained statistic, and $\mathcal{U}$ a finite common menu of operations. For bounded loss $\ell(\theta, u)$, include each operation's acquisition and actuation prices in ℓ. Put $L_u(R) = \mathbb{E}[\ell(\Theta, u) \mid R]$. The extra optimal decision risk imposed by retaining only Z is exactly*

$$\Delta_{\text{dec}} = \mathbb{E} \min_{u \in \mathcal{U}} \mathbb{E}[L_u(R) \mid Z] - \mathbb{E} \min_{u \in \mathcal{U}} L_u(R) \geq 0. \tag{13}$$

It is zero exactly when, for almost every $Z = z$, one action minimizes $L_u(R)$ almost surely in that retained-record cell. The action can be a whole permitted finite continuation policy; then its future query budget, available clock, records and costs must be part of the specified menu.

In the pointer task, randomly choose the first sensor or record-frame pilot with equal probabilities, and condition on its negative outcome. Their two posteriors are $(1, 9, 1)/11$ and $(1, 1, 9)/11$. Both have maximum confidence $9/11$. With no further pilot allowed, retaining the queried port gives repair failure $2/11$; when the complete remaining information is confidence alone (no port label or correlated tape), failure is $6/11$. The conditional decision penalty is $4/11$, separately from the physical price of erasing those distinctions.

Proof. Conditioning on Z commutes with each fixed action value. The minimum of finitely many linear functions is concave, giving (13). Select a minimizing action for each cell. Its loss above the record-optimal minimum is nonnegative and has zero expectation exactly when it vanishes almost surely. This proves the equality condition, including ties. For the example the two port labels have equal likelihood $11/30$ of a negative outcome. Erasing their distinction gives posterior $(1, 5, 5)/11$. Its best single repair succeeds with probability $5/11$, instead of $9/11$ when the port is retained. The read price already paid is common and cancels in the comparison; new diagnostic permissions would change the menu. □

This decision penalty is not another relative-entropy term to add to Theorem 3.1. It has the restoration task's units and a different loss. A state description accurate for routine prediction can be inadequate for repair; conversely, a sufficient causal statistic can allow redundant logs to be discarded without losing repair value. Keeping a complete calibrated controller state and its update rule can preserve every permitted future law, whether that state was learned here or supplied by copying a learned controller. The task distinguishes the information and its causal use, not an unobservable metaphysical privilege of its origin.

7.4 Finite physical operations and the role of history

The model has a finite pointer realization. Encode each sign in a qubit's Z basis, and prepare a fresh meter in $|0\rangle$ for a pilot. For the selected sensor or record-frame register j and duration $\tau > 0$, the positive Hamiltonian, in units $\hbar = 1$,

$$H_j = \frac{\pi}{2\tau} P_j^- \otimes (I - X_{\text{meter}}), \qquad P_j^- = (I - Z_j)/2,$$

acts for time τ as a controlled NOT. Measuring the meter with the supplied stochastic read noise gives the required likelihood. The repair uses $H_{\text{rep}} = \pi(I - X_j)/(2\tau)$ on the chosen register. The held triple is not re-prepared between pilots and repair. Only commuting pointer content is copied to the meter; this is not a second copy of an arbitrary unknown quantum state.

Each gate has $\int_0^\tau \|H\|^2 dt = \pi^2/\tau$. The noisy optimal rule therefore uses mean query-plus-repair control charge $(79/30)\pi^2/\tau$. This is a specified amplitude-duration charge, not net work or heat. Pilot preparation, noise, measurement, retained records, the reference labels and the clock remain supplied. The two pilot signs use at most two data bits; port, repair and stage labels and the finite decision table require additional storage. Rational branch probabilities do not require an exact continuous posterior register.

The comparison survives finite gates. If each of at most three gates has pulse-area error at most η, its unitary operator error is at most 2η and its channel's half-diamond distance is at most 2η. Telescoping the adaptive instrument and repair gives total variation at most 6η for the terminal task and record. Since their total loss lies in $[0, 1 + 2c]$, risk increases by at most $6\eta(1 + 2c)$. At $\eta = 10^{-4}$ the bound is .242327, still below the ideal fixed-list minimum .26.

Own-record return is experimentally relevant even when every repair looks successful through the routine port. Repair labels replayed from an independent trial are independent of the current cause; their restoration failure is $2/3$. Blocking the acquired pilots from the repair selector has the same boundary. In contrast, deleting redundant raw logs after retaining their sufficient diagnosis need not change the action. These are different memory interventions. Brain Anchor gives a noncommutative companion: equal held logical marginals and equal current gain confidence can require opposite future protection commands because the access-dependent damage histories differ [9]. Its world and environment are not this pointer example, but both locate the useful history in an actual return to permitted action.

7.5 The same body through repeated damage and correction

The single-fault experiment ends after repair. A continuing observer keeps the body left by that repair, including mistakes. Encode (w, g, m) as $x \in \mathbb{F}_2^3$, with zero the nominal body. At each interval, each coordinate flips independently with probability $0 < h < 1/2$. The routine meter reads parity $p = x_1 + x_2 + x_3$ exactly. The permitted repair is no flip or one labelled coordinate flip. There is no preparation of a new body between intervals.

Theorem 7.5 (Routine repair cannot maintain the hidden body)**.** *Start from the known nominal body, with independent fresh faults. Every randomized repair policy based only on the complete parity and action history has post-repair restoration success at interval t at most*

$$\frac{1}{4} + \frac{3}{4}(1 - 2h)^t. \tag{14}$$

Thus its limiting success is at most $1/4$. The policy that does nothing on even parity and flips a fixed coordinate on odd parity attains the limit, while displaying nominal parity after every repair.

Proof. Condition on the preceding complete history. The old body's parity is known, so the new parity reveals only the parity of the fresh fault. The even subgroup $E \subset \mathbb{F}_2^3$ has four elements. Translate each parity coset to E, and write u for its uniform law. The probabilities of even and odd faults are

$$a = (1 - h)^3 + 3h^2(1 - h), \qquad b = 3h(1 - h)^2 + h^3.$$

Even-conditioned convolution sends $q - u$ to $\alpha_0(q - u)$, with $\alpha_0 = (1 - h)(1 - 2h)/a$. Odd-conditioned convolution sends it to $-\alpha_1 P(q - u)$, where P is a translation permutation and $\alpha_1 = h(1 - 2h)/b$. These formulas follow from the one zero and three weight-two even faults, and the three weight-one and one weight-three odd faults. Their total-variation contractions are α_0, α_1. Every chosen repair is another known translation, preserving distance from the uniform coset law. Averaging over the next parity gives $a\alpha_0 + b\alpha_1 = 1 - 2h$. The initial distance from uniform is $3/4$, hence its expected value is at most $3(1 - 2h)^t/4$. Given a uniform coset, any allowed single repair can send at most one of its four states to zero. This proves the bound. The indicated policy always chooses a parity-compatible repair, and its success converges to $1/4$ as the conditional coset becomes uniform. □

The bound concerns the actual triple, not its displayed report. At $h = .05$, even at interval 100 no routine-only history can have success above .250020. Initial parameter labels, fault-correlated tapes or another internal diagnostic port would change its experiment class.

Now acquire the two internal pilots every interval, independently flipped with probability $\epsilon = .1$. Together with the parity meter they cost three readings per interval. A posterior q over the eight pre-repair body states is retained across damage and action. Let T be the known fresh-fault matrix, let $L_o(x)$ be the three-meter likelihood and let $A = \{0, e_1, e_2, e_3\}$ be the four repairs. For an action a, set $q^a_x = q_{x+a}$. Choose an action maximizing

$$q_a + \sum_o \max_{b \in A} (q^a T)_b L_o(b). \tag{15}$$

The first term is present restoration success. The second is the expected next success after fresh damage and the next paid pilot pair, using its optimum terminal repair. This is an exact two-step decision at the current posterior, used repeatedly. It is not an infinite-horizon optimality theorem. Bayes prediction, actual acquired outcomes and the known chosen flip update q; the actual damaged body is never reset to that estimate.

Controller	Restoration failure	Repair fraction
Routine parity only	.7506	.1354
Current pilots, specified decoded repair	.2730	.1255
Best enumerated memoryless pilot map	.19465	—
Persistent posterior, present-success rule	.2614	.1354
Persistent posterior, two-step rule	.09068	.19180

The two-step row averages three independent seed blocks, 91101–91103, each with 5000 held bodies, 700 intervals and the first 250 discarded for this comparison. Block means are .09088, .09086, .09030, with within-block standard errors about .00036. The first, second and fourth rows use a separate 16,000-body block, 600 intervals and 200 discarded intervals; the second row agrees with its exact rational stationary failure .2726062. These rows are finite empirical comparisons, not certificates of the two-step policy's stationary optimum. Every pilot controller uses the same three readings and the same four repairs. The memoryless comparison exhausts $4^8 = 65,536$ deterministic maps from current parity and the two pilot bits; it does not optimize all randomized or history-dependent rules. A direct rational enumeration of all retained branches gives the two-step policy's first three failure probabilities .02891, .056064106375, .076489639905.

History alone does not guarantee a better controller. Retaining a posterior but maximizing only present restoration performs worse than the best current-record map here. Considering

the effect of today's repair on tomorrow's body changes that result. The two-step rule also executes more repairs, so its lower failure is not a claim of lower total energy or a matched fixed pulse count. With three read prices c_Q and repair price c_R, its empirical per-interval charge is approximately $3c_Q + .19180c_R$, before computation, preparations, retention and thermal export.

This finite horizon need not require a real-valued posterior register. At $h = 1/20$, $\epsilon = 1/10$, every fresh-fault-plus-record kernel has integer numerator over $D = 800000$. Store eight unnormalized integer weights, update them with that kernel and permute after the actual repair. Starting from one, each weight is at most D^t, so $1 + \lceil t \log_2 D \rceil$ bits per weight suffice. Multiplying Eq. (15) by its common denominator makes its comparisons integer operations as well. Three acquired bits and two action bits per interval, arithmetic workspace and reversible garbage are additional resources. Saving every preceding weight vector for reversible execution costs quadratic, rather than constant, storage in the horizon. This gives finite exact computation in principle; the table's simulations use binary64, and no efficient cyclic processor or free erasure is inferred.

Q-RAIF C supplies a positive history-clock route for each specified finite decision circuit [1], with all held body content, meters, thermal cells and purification references retained. Its first-interval meter word controls actual repair and thermal exchange: failure is $.02891$ and cold-cell gain $.00761031$. The same finite Hamiltonian extends across a preallocated horizon without re-preparing the held body. The full source/clock and three-interval enumeration are recorded in the supplemental source notes. Ancillas, initialized clock, phase/timing permissions and reversible memory remain paid inputs; this does not turn the repeated simulation into a fixed-capacity cyclic thermal apparatus.

Resolution under a supplied spectrum. A factorially decaying reading spectrum resolves only $\log N/(\alpha \log\log N)$ directions to fixed tolerance. Appendix A gives the two-sided inversion and full proof; it does not derive that spectrum from finite selfhood.

8 Finite self-knowledge and the explanatory gap

8.1 Three questions

Three questions about a subject are easily merged. The question of access asks what the system can know about itself. The question of organization asks how a continuing self-model structures the world as it is presented to the system. The question of experience asks why there is something it is like to be the system at all. The results above answer the first question for any finite continuing observer in the typed model, and they constrain the second. They do not answer the third.

8.2 What the results fix

Within the typed model, continued fresh expression and sufficient growing memory make the identifiable dispositions learnable to any precision. In the example of Section 3, the finite-record excess falls as $1/(2N)$ nat. The reader's continuity controls whether those records remain available, while a fixed register has the floor of Proposition 4.6. Its occurrent state can be known only within the single-copy limit, which for a uniformly distributed qubit state in the Bloch ball leaves 94% of the no-record risk in place. Its drifting parameters can be tracked only up to a floor, with the stated square-root dependence in the slow-drift Gaussian regime. Its self-model shapes its blind spots, and its model-conditioned assessment cannot see beyond them.

These estimation statements require no brain-specific substrate. Their interpretation as self-knowledge comes from the observer's physical role in the experiment, not from an estimation law unavailable to an external reader.

8.3 A bridge hypothesis

Hypothesis 8.1 (Dispositional self-acquaintance)**.** *In a system whose continuing self-model has the structure above, what is available to reflective first-person access corresponds to the self-model's knowledge of the system's own dispositions and their reliability, which grows with its history and is refreshed at a finite rate. The occurrent state is not available to that access beyond the single-copy limit; it is lived through rather than read off. The sense that present experience is given immediately and the impossibility of describing it completely are, on this hypothesis, two aspects of one finite organization.*

The proposed mechanism is participation: readings, memory and corrections belong to the physical process whose dispositions they describe. Recurring couplings and habits of access express those dispositions again; an arbitrary unknown state supplied once does not provide a separate ensemble of copies. This distinction supports the hypothesis without making every classical present state unknowable. Habits of attention and perception can become objects of reflection while the ongoing experience remains its medium. *The World as Imprint* develops this reading of the self as a maintained way of participating [8]. The causal-return hypothesis below asks which retained distinctions actually sustain that participation.

8.4 Why propose this particular bridge?

The bridge is an inference about organization, not an inference from finite estimation error to the existence of feeling. It has four independent motivations. First, a reflective description is available to this system only if acquired distinctions can return to its own continuing operations; an external observer's accurate catalogue is not yet that return. Second, the distinction between a world change and a change of one's own access determines different possible corrections. The pointer task makes this difference causal even when content and confidence reports agree. Third, retained structure supplies a continuity of dispositions through new states and damaged components; requiring every raw episode to remain stored is unnecessary. Fourth, the current lived state and the model by which it is described occupy different roles in this account. The single-copy result constrains one declared reconstruction task; it does not show that all present experience is quantum, ineffable or inaccessible in every respect.

Together these reasons locate a candidate basis for reflective ownership: a maintained model is used by the process whose access it organizes. They do not derive valence, qualitative character or phenomenal presence. An ordinary non-experiential controller can have this organization. The experiential addition is the hypothesis that a subject's reflective continuity depends on this organization, to be tested with independently justified markers rather than defined true by controller success.

The strongest objection is that the hypothesis concerns access and organization, while the explanatory gap concerns experience. A system without experience would satisfy every theorem above, so the theorems cannot show that experience is present or what it is like. The hypothesis does not claim otherwise. It relocates the gap: the formal account fixes what a subject can know about itself and how that knowledge is organized in time, and an explanation of experience must connect at that point. It also excludes something definite. An account granting

a physical observer exact reconstruction of an arbitrary unknown occurrent state from operations on its sole preparation is incompatible with the single-copy limit. Prior knowledge of the state or fresh preparations change that premise.

The comparison of dispositions and the present state must fix the prior, target and available preparations. A common scoring rule alone does not order calibration. For the specified Bloch-ball experiments, fresh informationally complete expressions drive the dispositional risk to zero, whereas the one-preparation target has the stated positive risk. For a self-parameter with the confirmed Gaussian random-walk and readout laws, Theorem 5.3 gives the exact predictive loss and its slow-drift scaling. Those are quantitative tests of the functional model, not universal laws for all subjective judgments.

8.5 Causal return and reflective continuity

Hypothesis 8.2 (Renewable causal self-acquaintance)**.** *In an observer exhibiting reflective first-person access, its continuity depends on a maintained, causally usable distinction between changes of the represented world, its reading interface and its retained records. Selective disruption of the return from acquired distinctions to permitted correction impairs reflective causal self-attribution even when routine content reports remain matched. Retaining the sufficient causal organization can preserve that access through replacement of components or loss of redundant raw logs.*

The pointer result supplies a functional witness for this proposal. Ordinary reports stay the same while internal causes differ; a repair can normalize a report without repairing those causes. Acquired records alter which part is corrected and whether another question is worth asking. Brain Anchor supplies a different physical witness in which retained access history chooses future protection of still-held unknown content [9]. These witnesses identify a return mechanism that a theory of reflection can investigate. Their success does not make either device a demonstrated subject.

The empirical contrast should keep current content performance, routine confidence output and read/actuation budgets matched before the diagnostic intervention. Manipulate world state, sensor orientation and record reading separately. Compare own-record diagnosis with independent-trial replay, blocked record-to-operation return, and deletion of raw logs after preserving the sufficient diagnostic state. Score the inferred cause, the correction chosen and its actual effect, not merely whether the old report becomes normal. A biological application also needs independently justified markers of reflective access or ownership; the pointer's restoration score is not such a marker by definition.

A candidate marker class is error-sensitive nonverbal opt-out behavior, paired with source-ownership attribution on separate probe trials. Neither measure is the diagnostic selector or restoration score. Before using it here, independent validation must establish its relation to reflective access and control for first-order performance, motor capability and reward sensitivity. It is a proposed assay, not an established marker of experience. Preregister a minimally detectable change and an equivalence margin. If selective blocking abolishes cause-specific diagnosis while this validated marker remains equivalent to sham within that margin, with adequate power and matched controls, the claimed dependence is refuted. A replay/control contrast chosen only after seeing the outcome would not meet this test.

The hypothesis predicts that preserving the causal return, rather than the displayed autobiography alone, preserves reflective self-attribution across these changes. The preregistered refutation criterion above uses equivalence within the declared margin, not a failure to reject a difference. A model restricted to the matched routine records can reproduce current reports but

not the cause-specific repair in Proposition 7.3. An ordinary controller with the same full interventional laws can reproduce it. Distinguishing that controller from an experiencing subject needs an additional empirical bridge; Proposition 7.2 prevents hidden experiential labels from being declared observable without the corresponding permission.

8.6 Competing explanations and discriminating interventions

A report-generation account can explain confidence and autobiography without making them causally necessary for repair. A generic metacognitive-control account can explain the entire functional task, including causal diagnosis, with no commitment about experience. A broader integration account may instead attribute the change in reflective markers to a disruption of global communication or task competence. Finally, a theory on which experience survives loss of reflective access may accept all functional effects here while rejecting this dependence for phenomenal presence. These are genuine alternatives; a loss of reports alone does not decide among them.

The proposed experiment therefore separates three interventions. Block or replay the record-to-correction path while matching routine content and confidence before the diagnostic event. Erase redundant raw logs after retaining the causal decision statistic, and separately erase that statistic while keeping the logs or displayed autobiography available. Finally, restore the causal state and update program in replacement hardware. Measure cause-specific correction and independent reflective markers, with acquisition, stimulation, timing and actuation budgets declared in each condition. A matched sham disruption and a nonspecific processing-load control test whether any change is due simply to poorer global performance.

The causal-return proposal predicts selective loss when usable causal distinctions are broken, preservation when sufficient structure survives, and recovery when its permitted return is restored. A display-only account need not predict this pattern; a fully equivalent causal controller does. The marker comparison uses the preregistered equivalence criterion stated above. If all permitted interventional laws and markers are matched, this proposal has no further observable discriminator. A distinct experiential interpretation then requires an additional bridge, not a new hidden label attached to the same functional result.

8.7 Relation to modular time and to Paper G

The accessible algebra of Consciousness I carries a modular flow that the Cut correspondence reads geometrically for standard Cuts [2]. That flow is algebraic time, defined by the state and the algebra. Nothing here identifies it with the time in which a self-model is refreshed, or with experienced time. Paper G distinguishes a selected target operation from its larger executor and prices sharp readouts in a finite program register [4]. The risk floors here instead come from one copy, a finite reading rate against drift, and permissions chosen by the self-model. No fixed-point impossibility is used or claimed.

9 Conclusion

A finite observer can acquire self-knowledge, assess its reliability and use that assessment to revise its access. Failure, restricted permissions and fixed storage leave different gaps. The unknown state supplied once and drift during reading add further boundaries. T-DOME III gives histories with equal sign confidence but different maintenance values [7]; here, causes

with identical routine reports require different actual repairs. Paid internal diagnosis separates those causes, and finite physical gates preserve its task advantage. With the same body left in place through repeated damage, normalizing routine reports cannot maintain its hidden organization. A finite self-reader can instead carry its posterior into decisions about future repair. Optimizing only the present can squander that history; considering the next damaged body gives a constructive benefit. This puts a causal role for reflection beside its estimation limits.

The experiential proposal is that reflective continuity follows a maintained return from records to the organization of access, rather than a displayed self-description alone. It has selective-disruption and retention tests. Equal routine reports do not defeat those tests; equal complete interventional laws do. The remaining question is how this causally organized self-acquaintance relates to the lived state, and what independently justified evidence can decide that relation.

A The resolution order of a finite self-model

Suppose a self-model has independent directions $j = 1, 2, \ldots$, each estimated from records with information I_j per record, and call direction j resolved when $NI_j \geq c$ for a fixed tolerance constant c. This is a statement about precision, not identifiability: a direction with small positive I_j is identifiable but needs many records.

Proposition A.1 (Resolution order)**.** *If* $\log I_j = -\alpha j \log j + O(j)$ *with* $\alpha > 0$ *and* $L = \log N$*, then*

$$K(N) := \max\{j : NI_j \geq c\} = \frac{L/\alpha}{W(L/\alpha)}\big(1 + o(1)\big) \sim \frac{\log N}{\alpha \log\log N}, \tag{16}$$

with W *the Lambert function. The same asymptotic holds for the summed budget* $\max\{K : \sum_{j\leq K} I_j^{-1} \leq B\}$ *with* $L = \log B \to \infty$.

Proof. Write $-\log I_j = \alpha j \log j + \epsilon_j$ with $|\epsilon_j| \leq Cj$. If $NI_K \geq c$, then $\alpha K \log K - CK \leq L - \log c$; if $\alpha j \log j + Cj \leq L - \log c$, then $NI_j \geq c$. Since $j \log j \pm Cj/\alpha = j \log j\,(1 + O(1/\log j))$ and the inverse of $x \mapsto x \log x$ is $y \mapsto y/W(y)$, both the largest admissible K and the largest j satisfying the sufficient condition equal $(L/\alpha)/W(L/\alpha)\,(1 + o(1))$. For the summed budget, $\log \sum_{j\leq K} I_j^{-1} = \alpha K \log K + O(K)$, and the same inversion applies. □

For $\alpha = 1$ the values of $L/W(L)$ at $L = 10^2, 10^3, \ldots, 10^6$ are 29.54, 190.49, 1382.77, 10770.56 and 87847.54. The approach to $L/\log L$ is slow, with relative corrections of order $\log\log L/\log L$. The proposition turns a stipulated information spectrum into a count of resolvable self-directions; it does not supply the spectrum. A candidate signal factor with exact factorial suppression appears in the small-ball modular Hamiltonian (Appendix B), but a physical spectrum needs independently normalized directions, a fluctuation kernel and a permitted measurement, none of which is constructed here.

B The small-ball signal and its scope

The full one-dimensional CHM moment calculation is retained in `CIII_smallball.tex` in the supplemental source notes. Its factorial signal suppression alone supplies neither normalized directions nor reading noise. Consciousness II gives the ball-family multipoles and a conditional Fisher spectrum with $\log I_k = -4k \log k + O(k)$ under supplied noise [3], so Proposition A.1 then applies with $\alpha = 4$. A physical noise law remains to be derived.

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